



Commvault® Professional Advanced

Student Guide

Module 2 – Storage Advanced
Configuration

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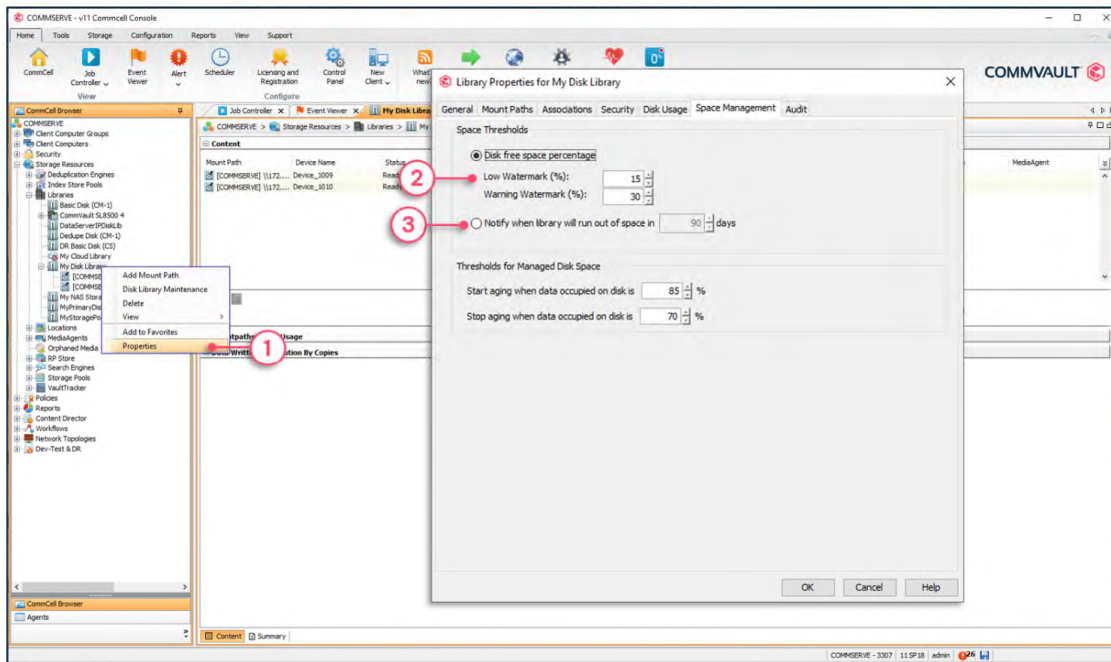
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Disk Libraries

Disk Library Advanced Options



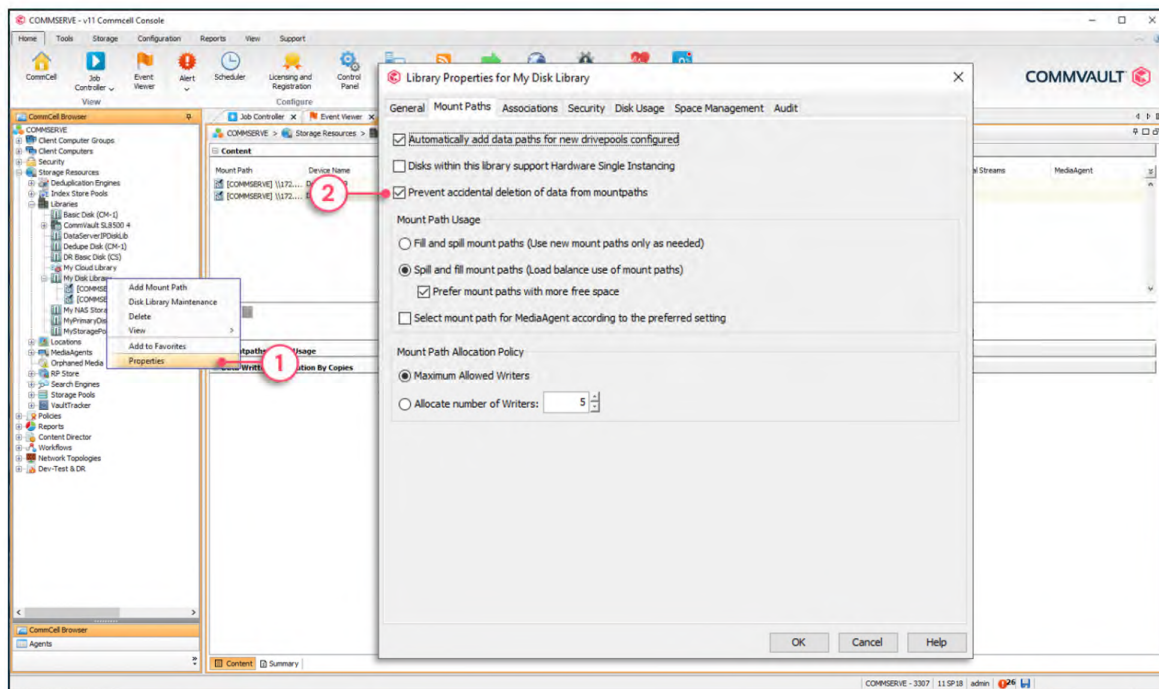
Set Low Disk Space Watermark

The 'Disk Library Low Watermark' is configured with a percentage threshold that reports an event to the Event Viewer when total disk capacity for all mount paths reaches or falls below the specified percentage. Alerts can also be configured to notify users when the threshold has been reached.

Instead of a low watermark, an estimated number of days before running out of space can be used. Based on the daily usage history, the system estimates when it expects the library to be saturated. When the number is equal or lower to the defined value, an event is sent in the Event Viewer.

1. Right-click the **library** > **Properties**
2. Set free space watermarks to send notifications in the Event Viewer
3. Alternatively, use a threshold to trigger the notification based on the number of days until the library is full

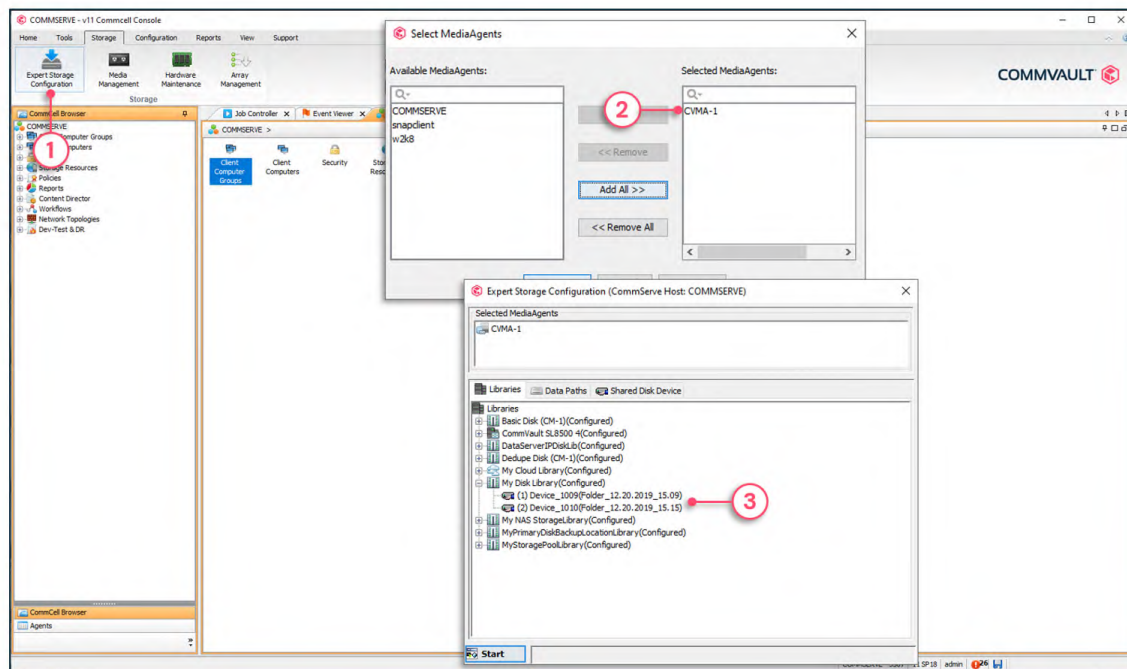
Prevent Accidental Deletion



The 'Prevent accidental deletion of data from mount paths' option locks disk mount paths content from being deleted for Windows MediaAgents. It is strongly recommended that this setting is selected. When it is enabled and a user connects to the MediaAgent to delete data using Windows Explorer, the operating system displays an access denied error message.

1. Expand Libraries > right-click the **library** > **Properties**
2. Check the box to lock down the content of mount paths

Determine the Order for Multiple Mount Paths



The 'Mount Path Usage' section determines the order in which incoming client streams are distributed across mount paths when multiple mount paths are configured for a disk library.

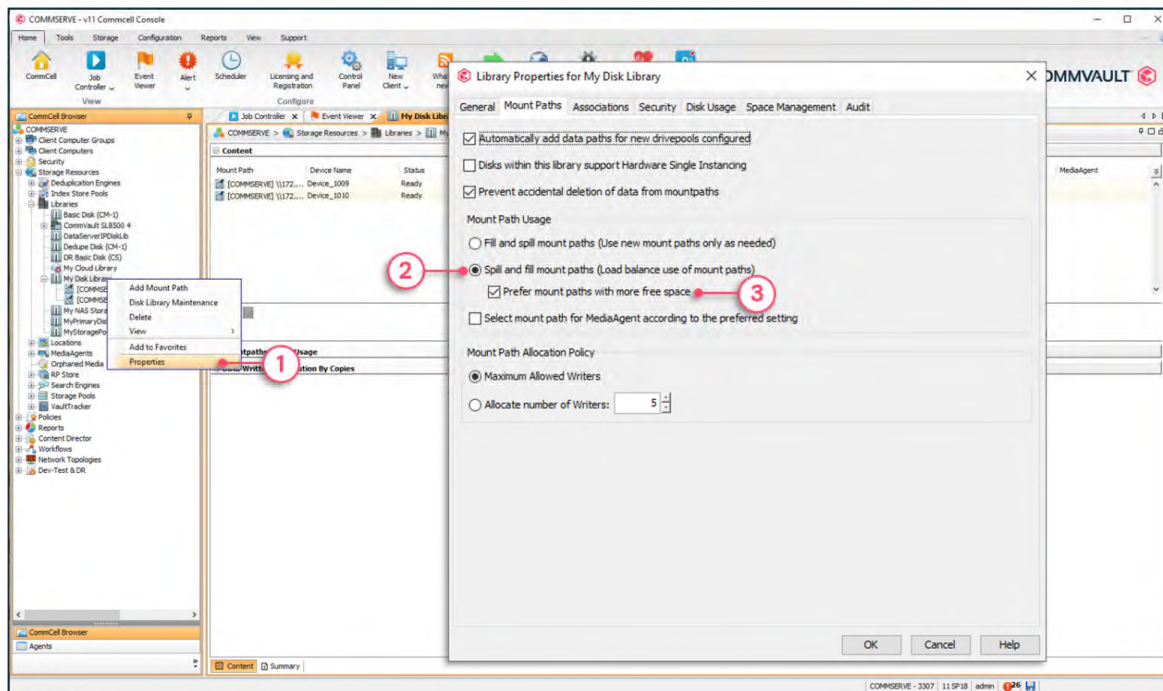
1. From the *Storage* tab > **Expert Storage Configuration**
2. Select the MediaAgent that hosts the disk library
3. Lists the mount paths in order in which they were created

Mount Path Best Practices

Which should you use for best performance? For disk libraries, part of a bottleneck are the disks themselves, but the other part of the bottleneck is the I/O path to the disks. The use of Fill & Spill or Spill & Fill (load balancing) should be determined by how the mount paths were created and the I/O path to each mount path.

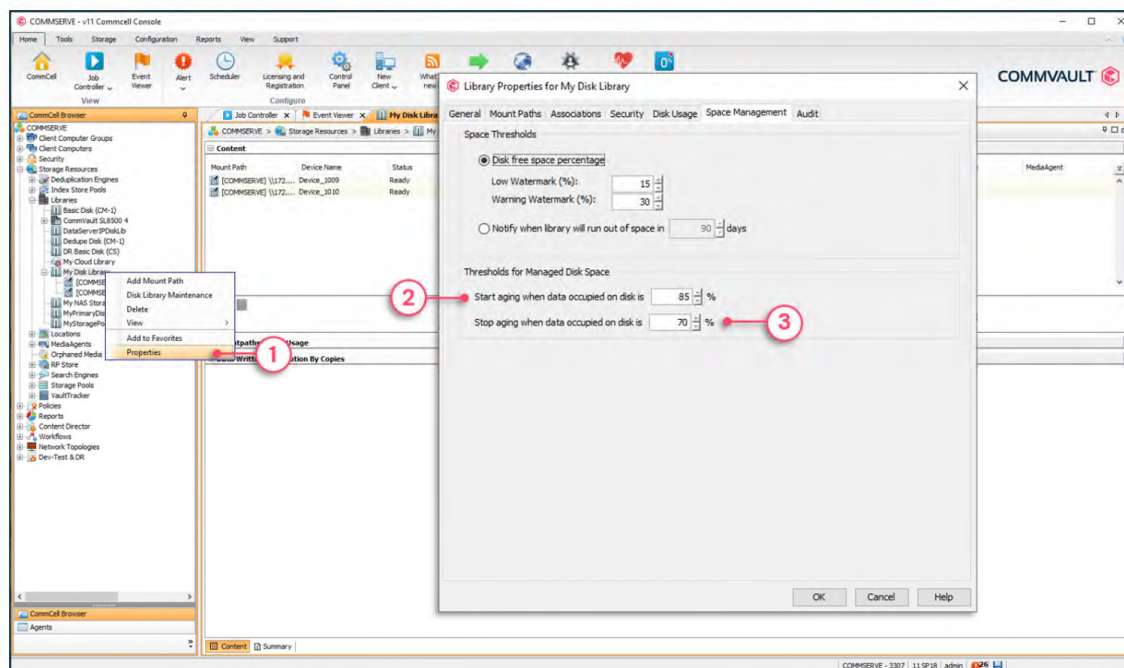
- **Fill & Spill** – The default setting for mount path usage, uses mount paths in the order in which they were created. Use Fill & Spill if the mount paths are part of the same disk array and there is a single I/O path to the mount paths.
- **Spill & Fill** (load balancing) - Load balances device streams between all mount paths in the library. Use Spill & Fill if the mount paths are on separate disks and/or if there are separate I/O paths to the mount paths. If there is only one mount path in the library, consider setting mount path usage to Spill & Fill. If a disk array is later added, the disk library will automatically use the mount paths in a load balancing manner. A common misconfiguration is when disks are added to a library and the administrator or engineer forgets to set this option.

Configure Mount Path Usage Mode



1. Expand Libraries > right-click the **library** > **Properties**
2. Define the mount path usage mode
3. Check to prioritize mount paths with more free space

Configure Disk Space Thresholds



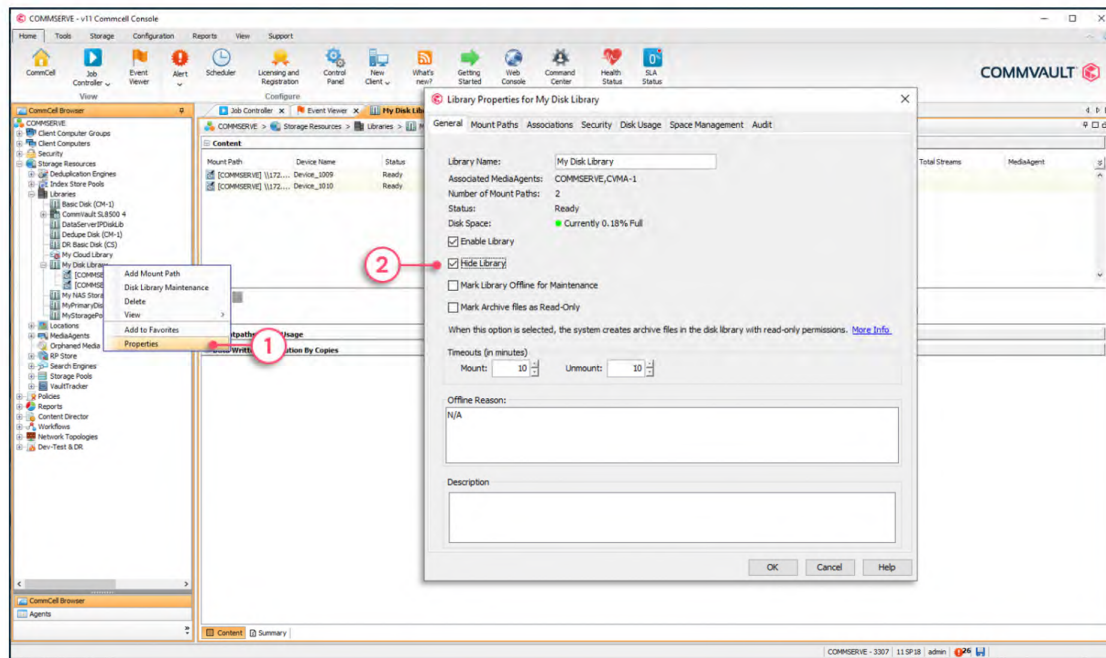
Managed disk space is a Commvault software feature that allows a job stored on disk to be retained beyond its standard retention settings if free space exists. Managed disk space thresholds do not apply to disk libraries using Commvault deduplication.

- Start aging when data occupied on disk is (High Watermark) – determines the amount of disk space consumed before aged jobs will be pruned.
- Stop aging when data occupied on disk is (Low Watermark) – stops purging jobs when amount of disk space consumed falls to the specified value.

Configuration

1. Right-click the **library > Properties**
2. Set the high threshold at which the system starts aging jobs
3. Set the low threshold at which the system stops aging jobs

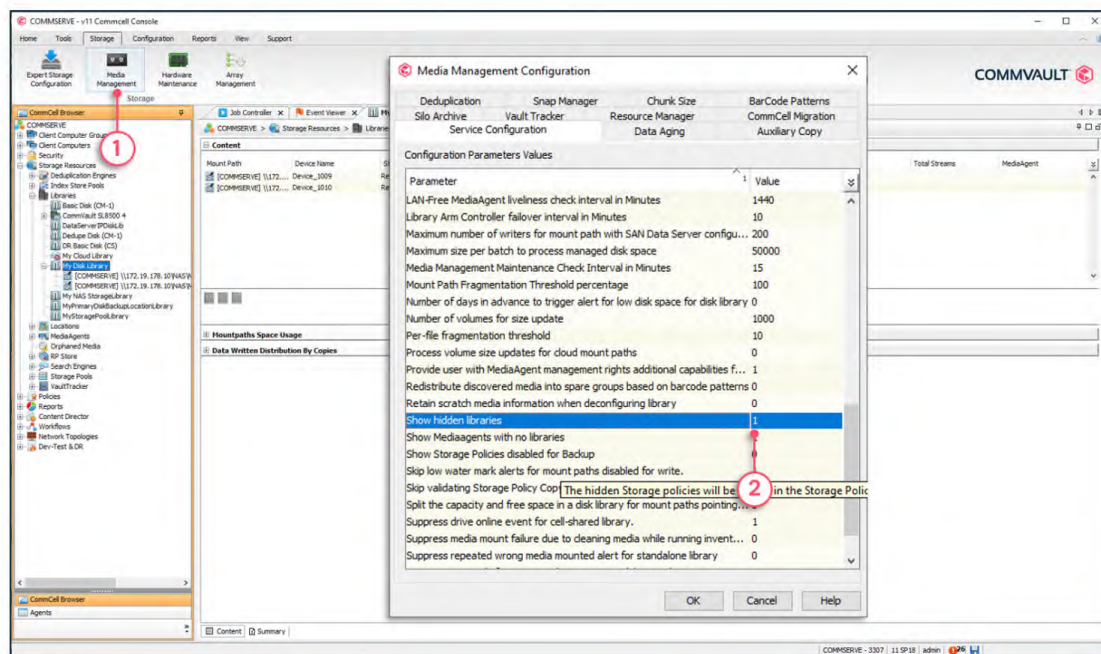
Hide a Decommissioned Disk Library



If a library is not to be used anymore for new backups, it does not mean that it can be deleted from the CommCell console. Doing so would result in destroying the data that might still be in the library and required for restores. Leaving the library in the CommCell console, however, could potentially lead to an administrator creating a new storage policy pointing to that library. If you want to prevent the administrator from using the disk library for any new storage policies and completely hide it in the console, the 'Hide Library' option can be used.

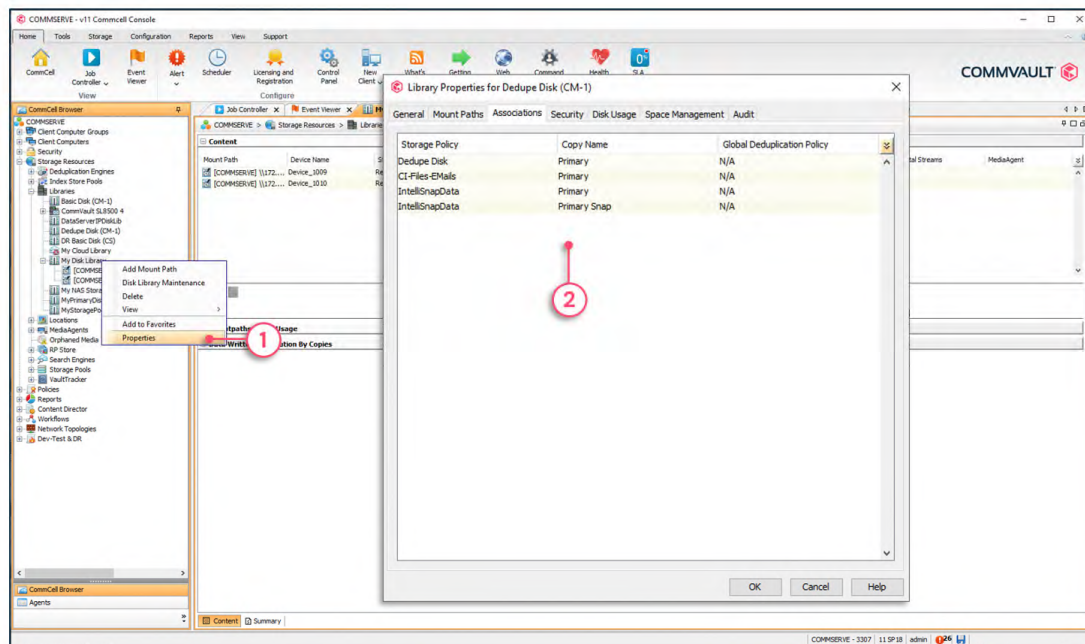
1. Right-click the **library** > **Properties**
2. Check the box for **Hide Library** to completely hide the library in the CommCell console

Unhide a Library



1. *Storage* tab > **Media Management**
2. Set the value to 1 to display hidden libraries

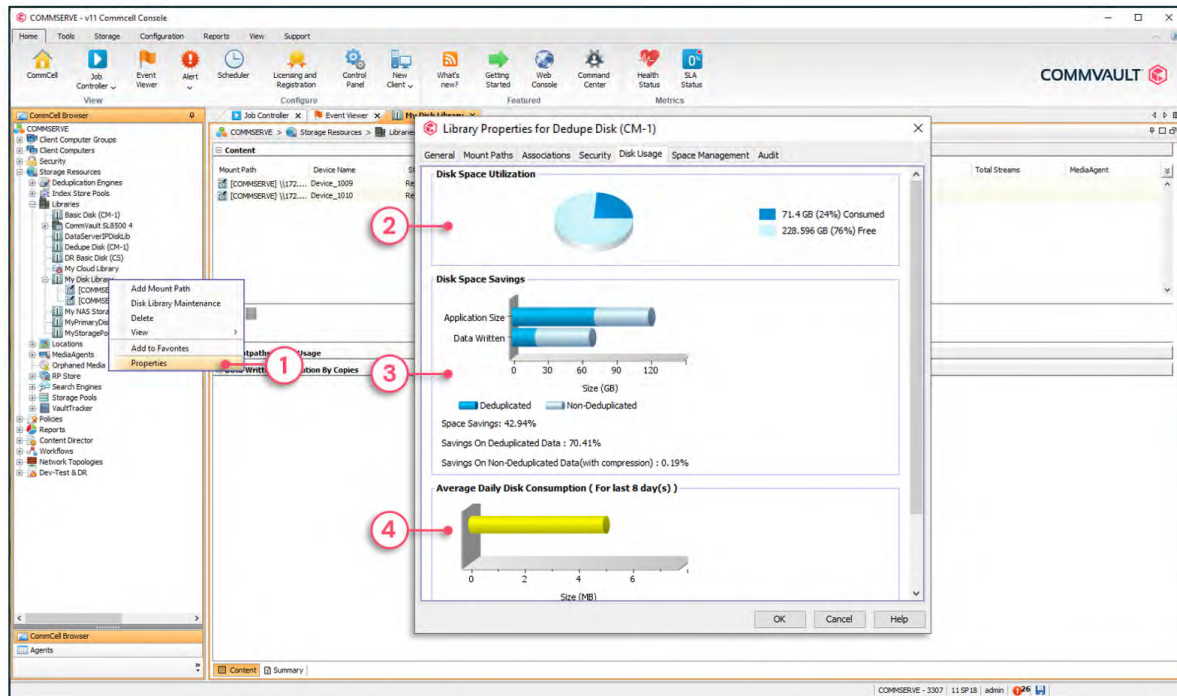
Disk Library Association Storage Policies



The Associations tab of a disk library displays all storage policies that have a copy sending data to the disk library. This is useful information when you plan on decommissioning a disk library or taking it offline for maintenance.

1. Right-click the **library > Properties**
2. Displays storage policy copies sending data to the disk library

Disk Library Usage



The disk library usage tab displays the space usage, deduplication space saving and data aging pruning information. Keeping an eye on this tab helps in avoiding filled library situations and helps in planning the acquisition or replacement of hardware. Based on the average daily disk space consumption and the average space freed up by data aging, the system can estimate a possible date on which the library is expected to be filled.

1. Right-click the **library > Properties**
2. Shows current space usage
3. Displays deduplication efficiency
4. Shows daily average for new space consumption and data aging space freeing up

Mount Path Advanced Options

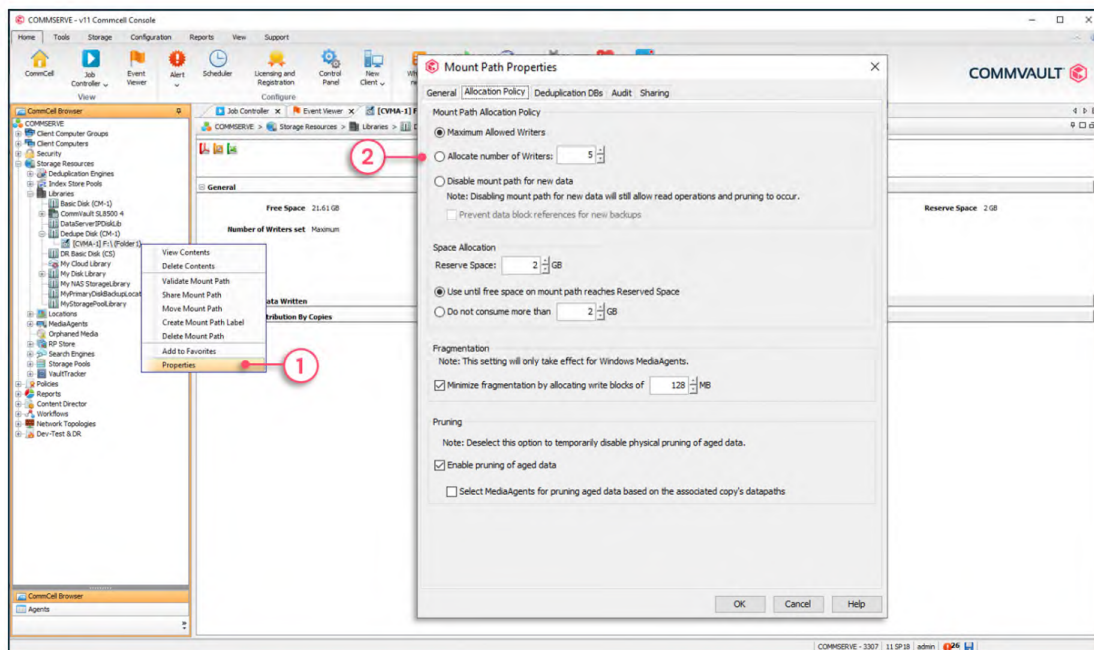
- Mount path writers
- Space allocation
- Validate a mount path
- Move a mount path
- Mount path status

Mount path allocation determines the number of concurrent write operations that are performed to a mount path. The default is set to 'Maximum allowed.' If a defined number is set, increasing this number could potentially improve overall backup performance within the CommCell environment by allowing more jobs to run concurrently. However, setting this number too high could have a negative impact on the MediaAgent, network, and disk I/O performance.

Before modifying this setting consider the following:

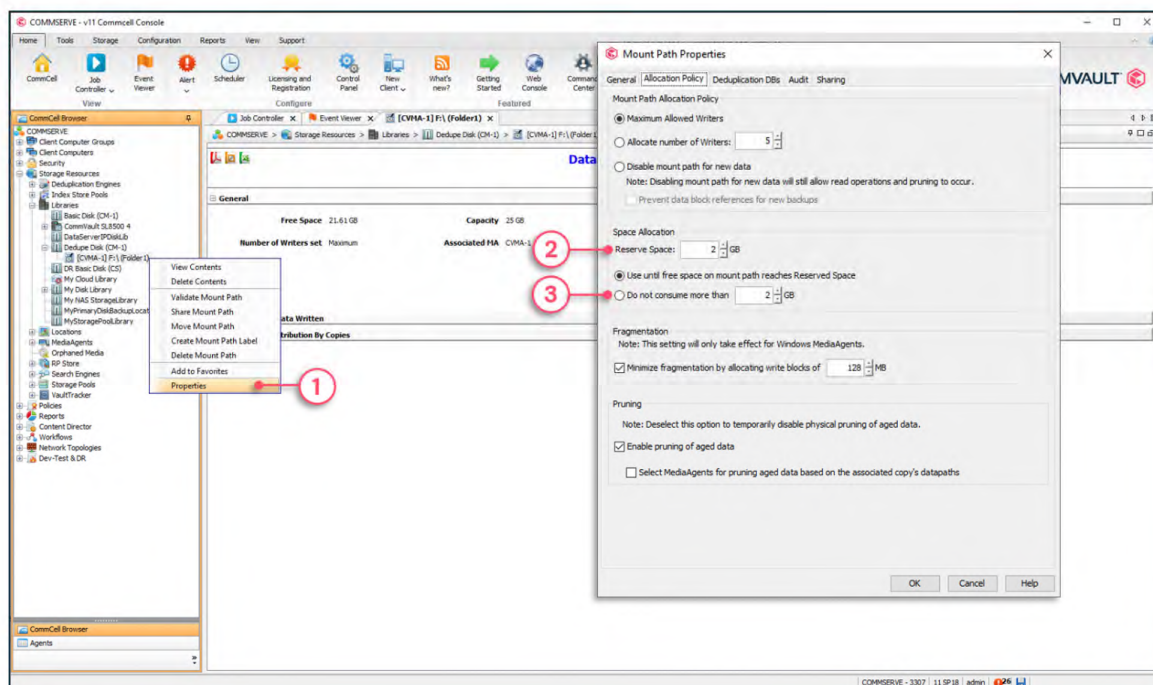
- Increasing this setting will run more streams through the network, MediaAgent, and disk. For many environments increasing this number marginally may not have a negative impact on the overall environment. It is recommended to incrementally adjust this number higher (increments of three to five at a time) and carefully monitor the network, MediaAgent, and disk performance.
- If multiple mount paths exist in the library, consider how the mount paths have been carved out of the disks. If two mount paths set to five writers are allocated from the same physical disk array then, there would be ten concurrent write operations to the disks, five from each mount path.
- Consider the I/O path from the MediaAgent to the disks as a potential bottleneck. Depending on the disk storage being used (Fibre, iSCSI or NAS) the I/O paths can become a bottleneck.
- Increasing the number of writers could have a positive impact on overall backup performance within the CommCell console by running more jobs but may slow down individual jobs due to the extra network and disk load.

Define Disk Library Mount Path Writers



1. Right-click the **mount path** > **Properties**
2. Define the number of writers for mount path or set to read only

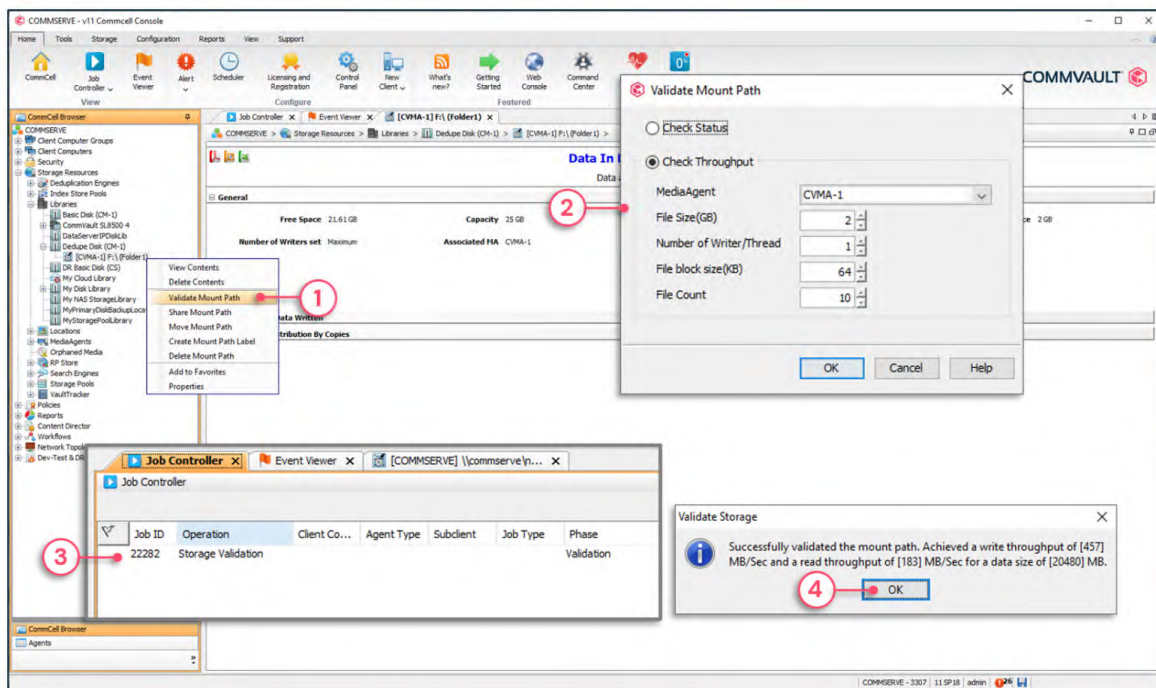
Determine the Space Allocation for a Mount Path



Space allocation specifies cut-off limits for a mount path. By default, it is set to reserve 2GB of disk space. If more reserve space is needed, increase this setting, or use the 'Do not consume more than' setting to limit how much disk space will be allocated to write operations. This can be useful if you require reserving more space for maintenance purposes, such as a defrag operation.

1. Right-click the **mount path** > **Properties**
2. Define the amount of free space to reserve on a mount path
3. Alternatively, define the maximum amount of space to use on a mount path

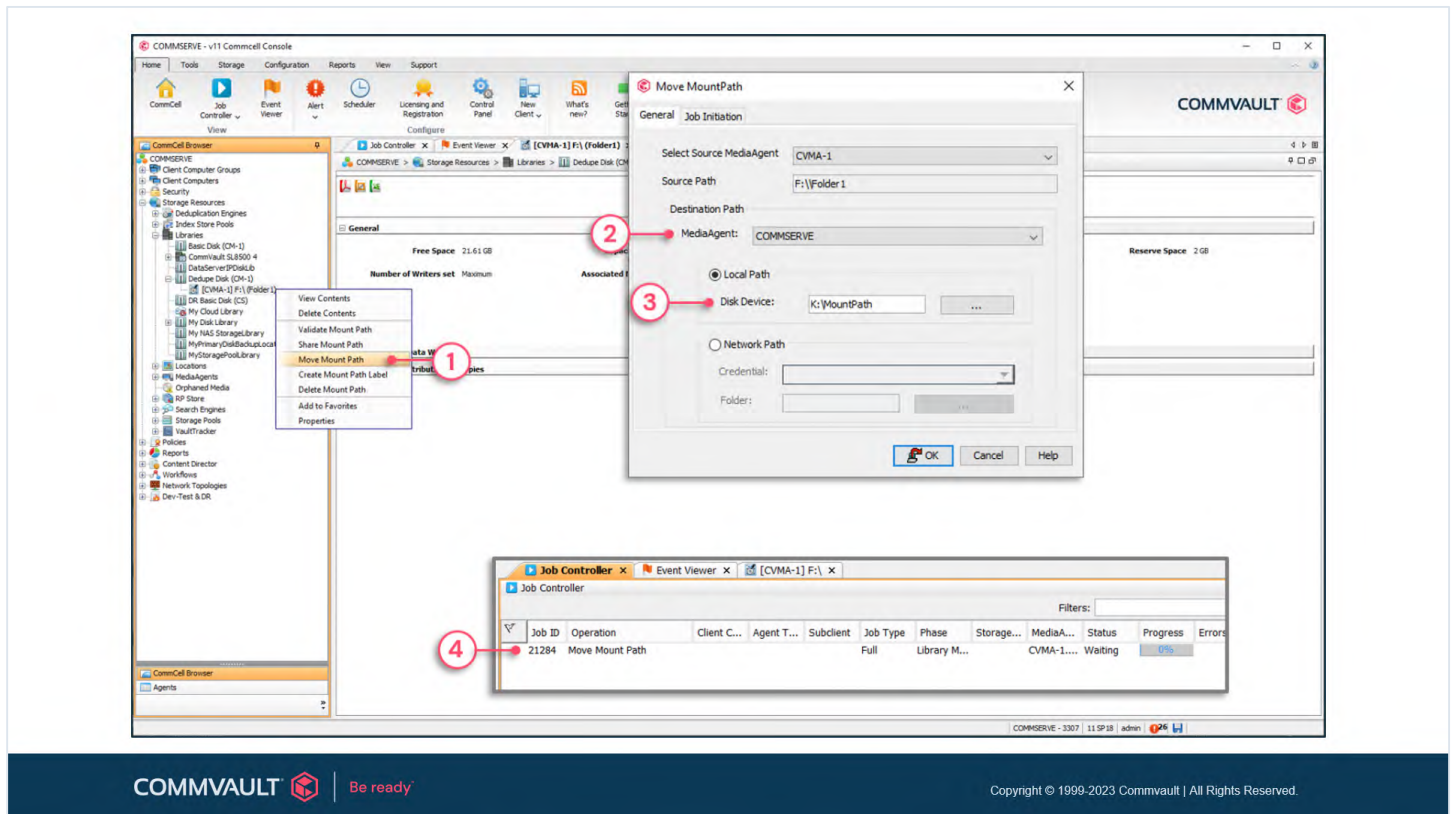
Validate a Mount Path



Mount path validation tests throughput speeds of library mount paths. The validation operation is configured using the MediaAgent, file size, number of writers, block size, and number of files.

1. Right-click the **mount path** > **Validate Mount Path**
2. Specify mount path validation settings and click **OK**
3. A storage validation job is displayed in the Job Controller
4. Results of the validation job will be displayed in a popup window

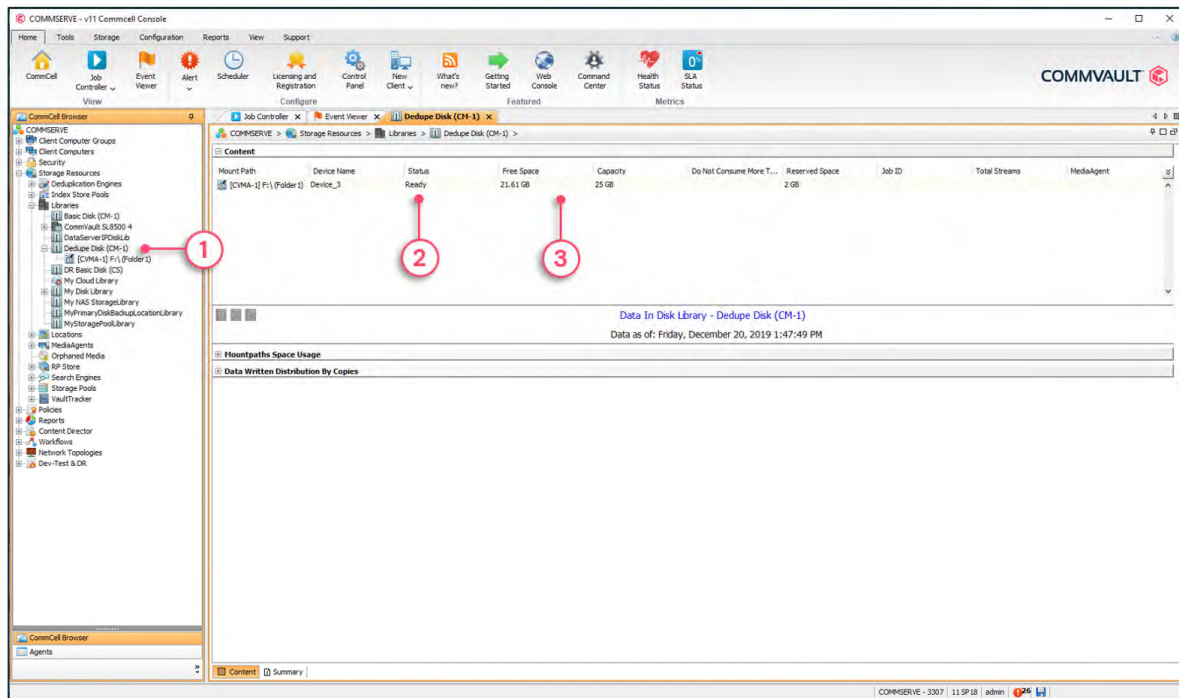
Move a Disk Library Mount Path



A disk library mount path can be moved to a different location using a 'Move Data Path' operation. This fully automated functionality is used when a disk library mount path is lacking space and needs to be moved on a bigger storage volume, or when an older storage unit is replaced. The new storage must first be presented to the MediaAgent operating system.

1. Expand the library > right-click the **mount path** > **Move Mount Path**
2. Select the MediaAgent
3. Click to browse for the new location
4. The job is displayed in the Job Controller

Disk Library Mount Path Status



1. Expand Libraries > click the **disk library**
2. Displays the mount path status
3. Displays the total capacity and free space left for each mount path

Disk Library Maintenance

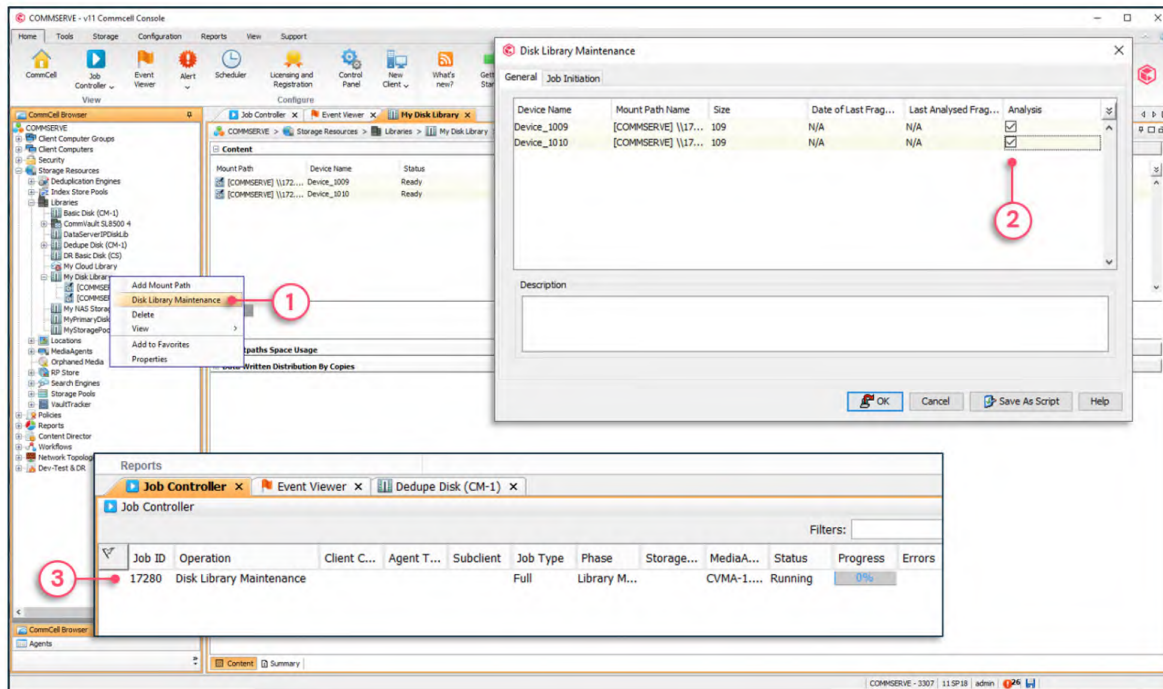
- Run a disk library maintenance analysis
- Execute the Library and Drive Configuration report
- Disable the library or mount path, or set to read-only
- Run a defragmentation tool

Disk library data is written in chunk files. When using Commvault deduplication, every written block that makes a chunk file is unique. When subsequently encountered in backup jobs, similar blocks are discarded, and references are made to the same block. When all jobs age and this block is no longer referenced to by any other jobs, it is marked to be purged. If the block is at the trail of a chunk file, it is deleted, and the chunk file is truncated. If it is in the middle of a chunk, and the library is hosted on DAS or SAN storage or NAS storage supporting the sparse file attribute, the block is deleted, leaving a 'hole' in the chunk file.

This can lead to fragmentation and decreased performance over time. To avoid this, the mount paths can be defragmented using any defragmentation tool supporting the sparse file attribute. Windows built-in defragmentation tool does not support the sparse file attribute and therefore, cannot be used. Commvault does not support calls on any third-party defragmentation tools. However, Disk keeper has certified its 2011 (and higher) release version for online volume defragmentation.

Before running any defragmentation jobs on a mount path, a disk library maintenance job and a report can be generated to provide information about fragmentation. Based on the result displayed in the report, the Commvault administrator can easily establish if maintenance is required.

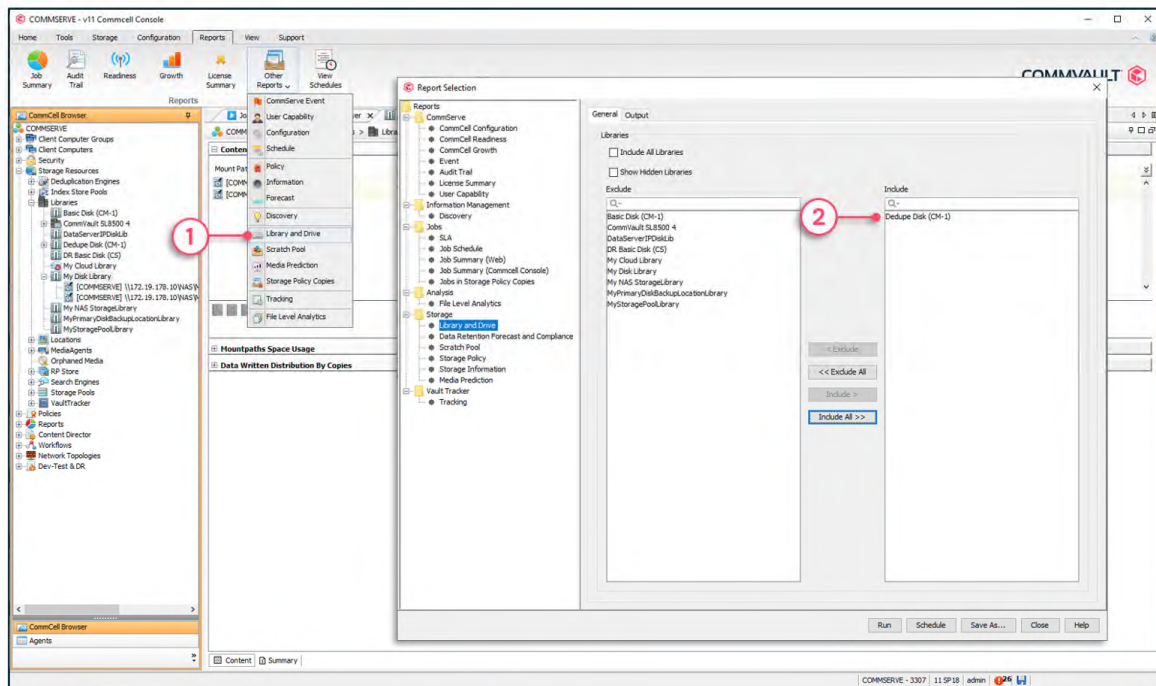
Disk Library Maintenance Job



Before defragmenting mount paths, you need to establish if the mount paths require defragmentation by running a disk library maintenance job that captures the status of fragmentation for every mount path of a disk library. Commvault recommends running the Disk Library Maintenance Job and the Library and Drive Configuration report twice yearly.

1. Right-click the **disk library** > **Disk Library Maintenance**
2. Select the mount paths to analyze
3. Job appears in the controller

Library and Drive Configuration Report



Once the Disk Library Maintenance job is completed, the Library and Drive Configuration report summarizes the result. The report can be scheduled to run after the Disk Library Maintenance job.

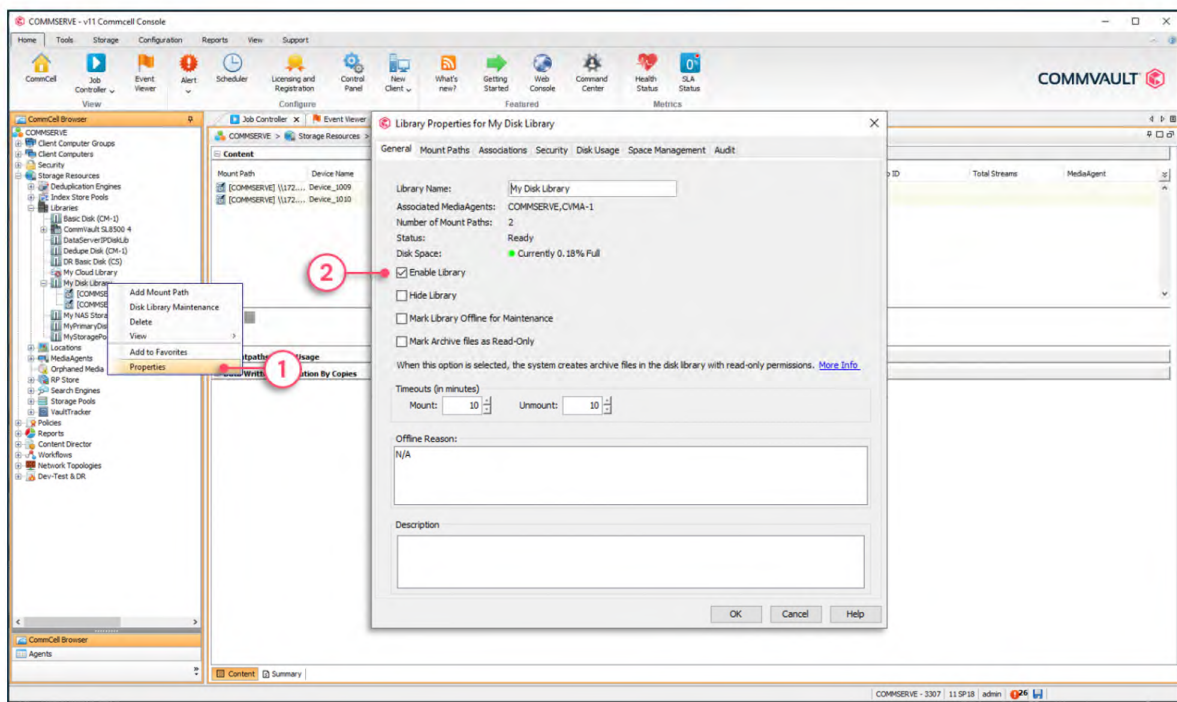
1. **Reports** tab > **Other Reports** > **Library and Drive**
2. Select the libraries to include in the report

The screenshot displays the COMMVAULT v11 CommCell Console interface. A 'Report Selection' dialog box is open, showing various report categories on the left and configuration options on the right. Red circles 3 through 7 indicate the steps for running a report. Below the dialog box, a report table is visible, showing 'Mount Path Information' and 'File Fragmentation Analysis' data.

Mount Path Information										File Fragmentation Analysis			
Mount Path	Capacity (TB)	Space Left (TB)	Space Occupied (TB)	Size of Application (TB)	Space Savings (%)	Allocated Writer	Offline Reason	Files Scanned	Fragmentation Per File	Fragmentation Threshold (in %)	Files Fragmented	Fragmentation Percentage	Last Successful Analysis Run
[CVMA-1] F:\	0.024	0.021	0.003	0.008	59.89%	Maximum	N/A	382	10	100	3	0.40	04/17/2017 13:50:11
[CVMA-1] H:\MP2	0.059	0.058	0	0	0%	Maximum	N/A	382	10	100	3	0.40	04/17/2017 13:50:11

3. Select the report output
4. Choose where to save the report
5. Define report settings
6. Click **Run** to run the report immediately or click **Schedule** or run later
7. In the report, locate the section referring to mount path fragmentation

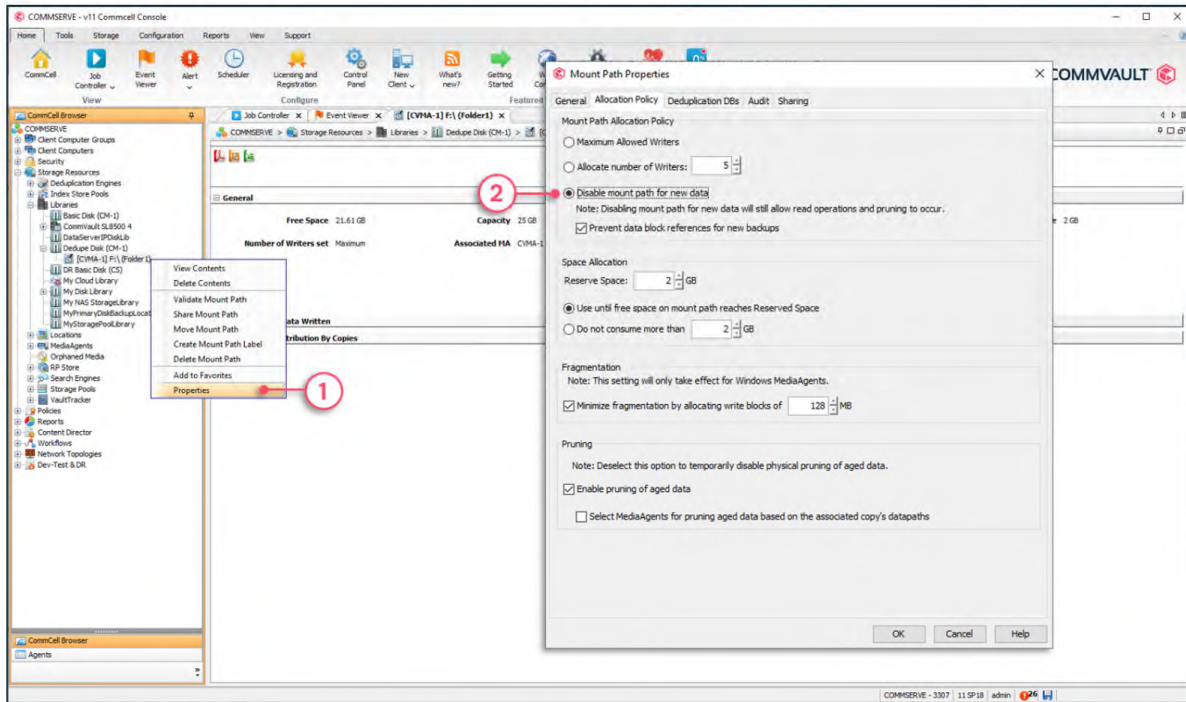
Disable Disk Library or Mount Path



If the report indicates that a high fragmentation level has been reached, run a defragmentation tool. It is important that no backups are running. Setting a mount path as read-only has an advantage over disabling the entire library. Restores can still be achieved and if there are enough additional mount paths, backup jobs can still run using the remaining mount paths.

1. Expand Storage Resources > right-click the **disk library** > **Properties**
2. Uncheck the box for **Enable Library** to disable the disk library

Set a Mount Path as Read-Only



1. Right-click the **mount path** > **Properties**
2. Select **Disable mount path for new data** to mark the mount path as read-only

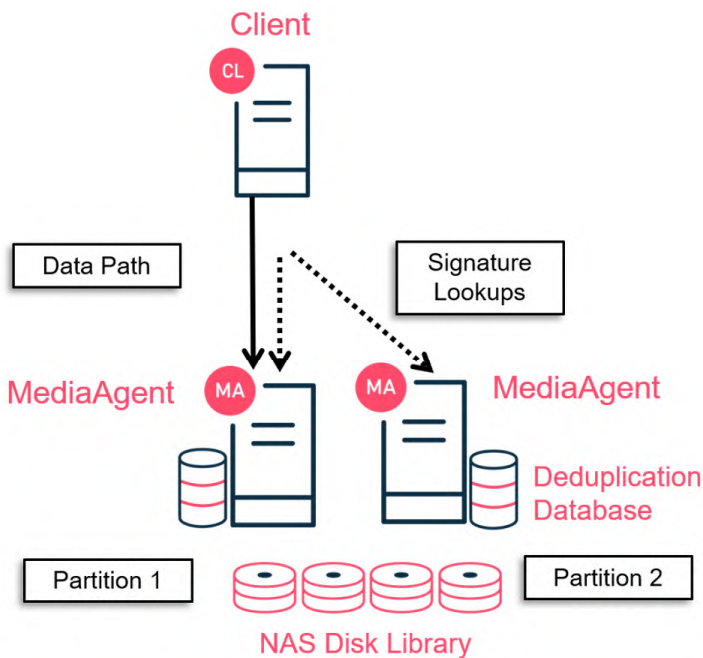
Run the Defragmentation Tool

Run the defragmentation tool on the mount path. Consult your software vendor documentation to ensure it supports online defragmentation the sparse file attribute. Once the defragmentation job completes, put the library back online or to set the mount path to read-write.

Deduplication

Partitioned Deduplication Database Overview

- Used for scalability
- Recommended when using a shared NAS library
- Up to four partitions (six for HyperScale)
- Client connectivity to all MediaAgents



Partitioned deduplication provides higher scalability and deduplication efficiency by allowing more than one Deduplication Database (DDB) partition to exist within a single deduplication engine. It works by logically dividing signatures between multiple databases. If two deduplication partitions are used, it effectively doubles the size of the deduplication store. Commvault software supports up to four database partitions.

During data protection jobs, partitioned DDBs and the data protection operation work using the following logic:

1. The signature is generated at the source. For primary data protection jobs using client-side deduplication, the source location is the client. For auxiliary DASH copy jobs, the source MediaAgent generates signatures.
2. Based on the generated signature it is sent to its respective database. The database compares the signature to determine if the block is duplicate or unique.
3. The defined storage policy data path is used to protect data. Regardless of which database the signature is compared in, the data path remains consistent throughout the job. If GridStor Round-Robin has been enabled for the storage policy primary copy, jobs will load balance across MediaAgents.

Partitioned Databases and Network-Attached Storage (NAS)

If partitioned deduplication is going to be implemented using two MediaAgents, use a shared disk library with a Network-Attached Storage (NAS) device. The NAS storage allows either MediaAgent to recover data even if the other MediaAgent is not available.

Partitioned Database for Scalability

The primary purpose for partitioned DDBs is to provide higher scalability. By balancing signatures between database partitions, you can scale up the size of a single deduplication store. If you have two partitions, the size of the store doubles; four partitions quadruples its size.

Partitioned Database for Resiliency

Using partitioned databases ensures resiliency. For instance, if one MediaAgent hosting a DDB goes offline, the other MediaAgent continues data protection jobs as the available DDB continues signature lookups. However, with the loss of one database, all signatures previously managed by the offline database would now be looked up in the remaining online database. This causes existing signatures managed in the offline database to be compared in the online database, which results in the signatures being treated as unique and additional data being written to the library.

Storage Pool Wizard

The Storage Pool wizard is used to create a global deduplication policy and to associate it to storage. Once the Storage Pool is configured, it is displayed under the Storage Pool entity of the CommCell Browser. It can be expanded to provide a centralized view of the deduplication engine it created as well as the storage it is associated to. The storage pool components can also be viewed from their respective location. The Global Deduplication Policy created by the wizard is displayed under the Storage Policies entity and is named after the Storage Pool name.

Display the Storage Pool

COMMVAULT COMMCELL CONSOLE

Home | Tools | Storage | Configuration | Reports | View | Support

COMMVAULT

CommCell Browser

Client Computer Groups
Client Computers
Security
Storage Resources
Index Store Pools
Libraries
Locations
Media Agents
Orphaned Media
RP Store
Search Engines
Storage Pools
Commvault ML6030 8
My Disk Storage
Deduplication Engines
My Disk Storage/My Disk Storage_Primary
My Disk Storage_1003 [ID: 1003]
[CVMA-1] E:\DDB [ID: 1004]
[CVMA-1] E:\DDB [ID: 1004]
VaultTracker
Policies
Reports
Content Director
Workflows
Network Topologies
Dev-Test & DR

My Disk Storage_1003 [ID: 1003]

Storage Resources > Storage Pools > My Disk Storage > Deduplication Engines > My Disk Storage/My Disk Storage_Primary > My Disk Storage_1003 [ID: 1003]

Deduplication Database - My Disk Storage_1003 [ID: 1003]

Data as of: Thursday, February 18, 2021 9:34:50 AM

Properties

Deduplication Engine Name: My Disk Storage_1003
Creation Time: 02/11/2021 09:12 AM
Version: N/A
Number of Partitions: 2
Total Application Size: 0 Bytes
Total Data Size on Disk: 0 Bytes
Deduplication Savings: 0%
Total Number of Unique Blocks: 0 MB
Total Size of Unique Blocks: 0 MB
Number of Pending Prunable Records: 0

Number of Jobs: 0
Baseline Size (GB): 0 Bytes [Application Size: 0 Bytes]
Average Q&I Time across Partitions in last 3 days: 0 ms (0%)
Average Q&I Time across Partitions in last 14 days: 0 ms (0%)
Free Space On Library: 21.62 GB
Number of Active Streams: 0
Pruning Status: Enabled
Maximum Allowed Connections: 200
DDB last verified on: N/A
Existing jobs on Disk and DDB last verified on: N/A

DDB Statistics (Across Partitions)

No chartable data

Chart Options
Display Options

DDB Size

Size of DDB path
Free space on DDB Disk

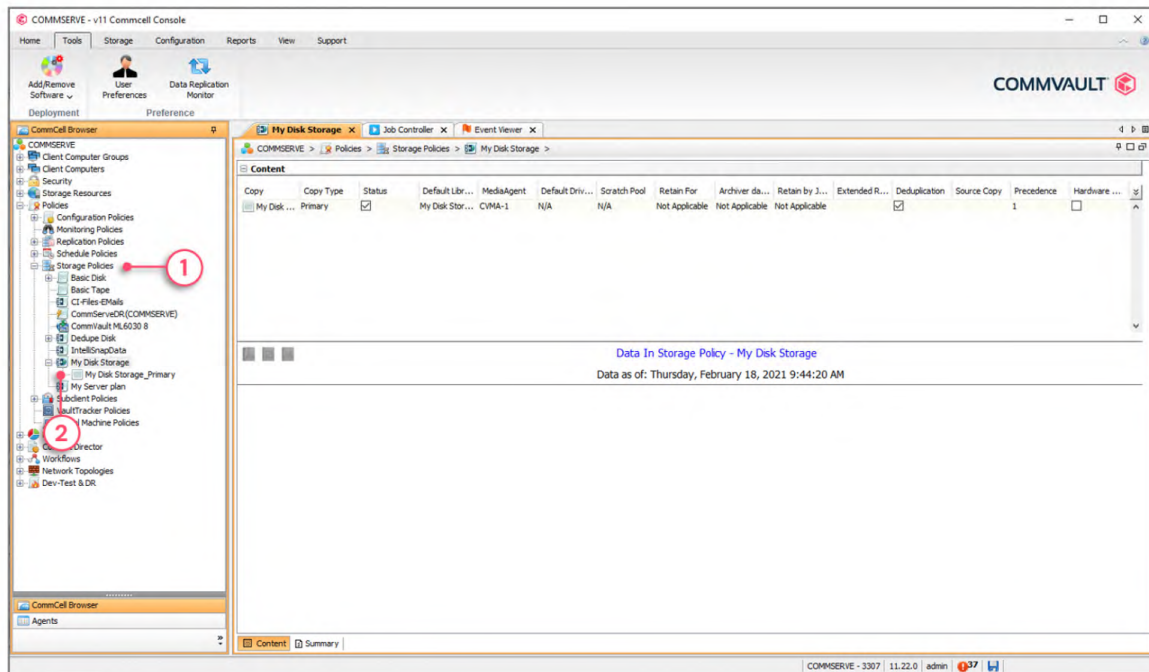
[CVMA-1] E:\DDB
[CVMA-1] E:\DDB

0 6 12 18 24 30

COMMVAULT - 3307 | 11.22.0 | admin | 37

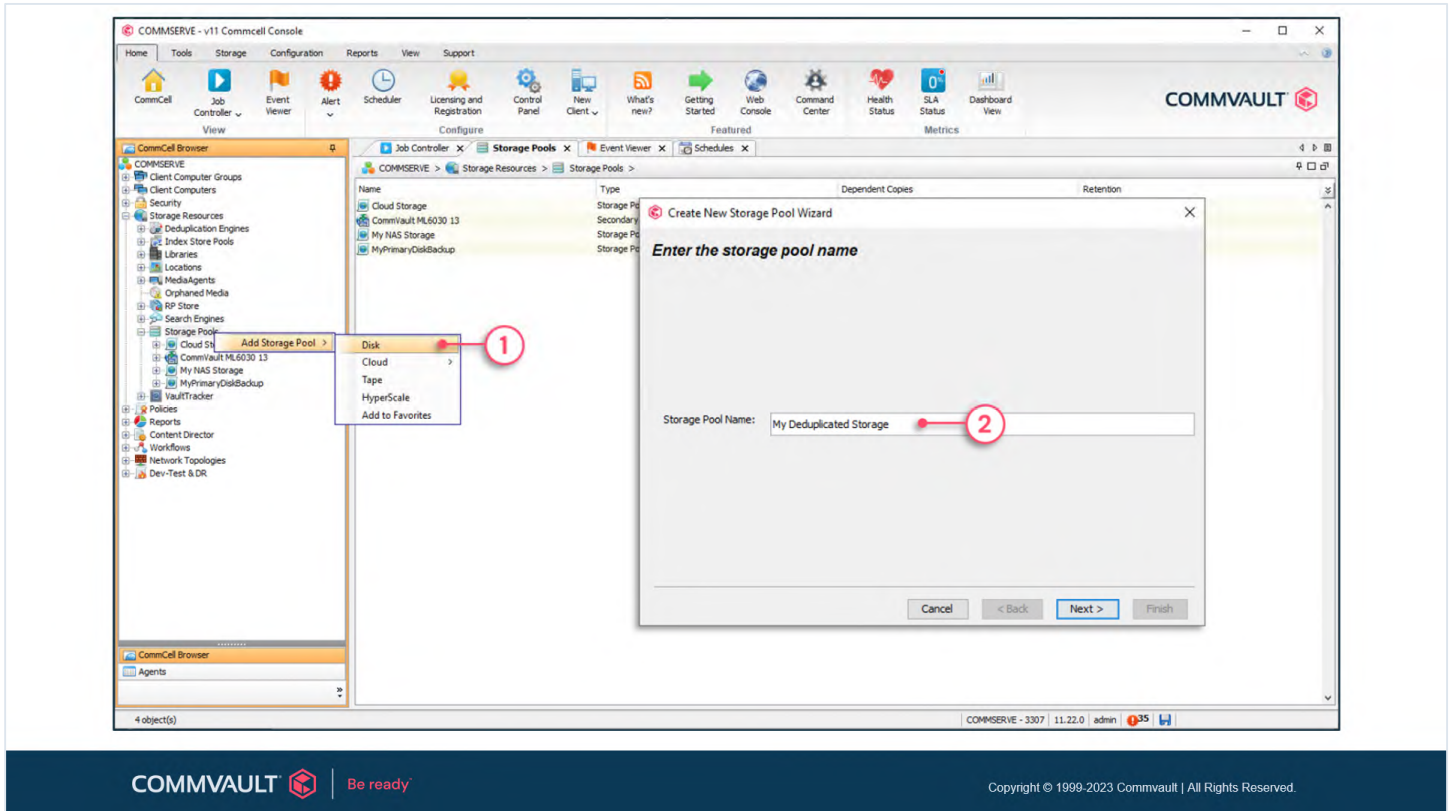
1. Expand Storage Resources > Storage > **Storage Pool name**
2. The deduplication engine is displayed and can be expanded to display its DDB partitions
3. The storage is displayed and can be expanded to display the mount paths

Display the Global Deduplication Policy

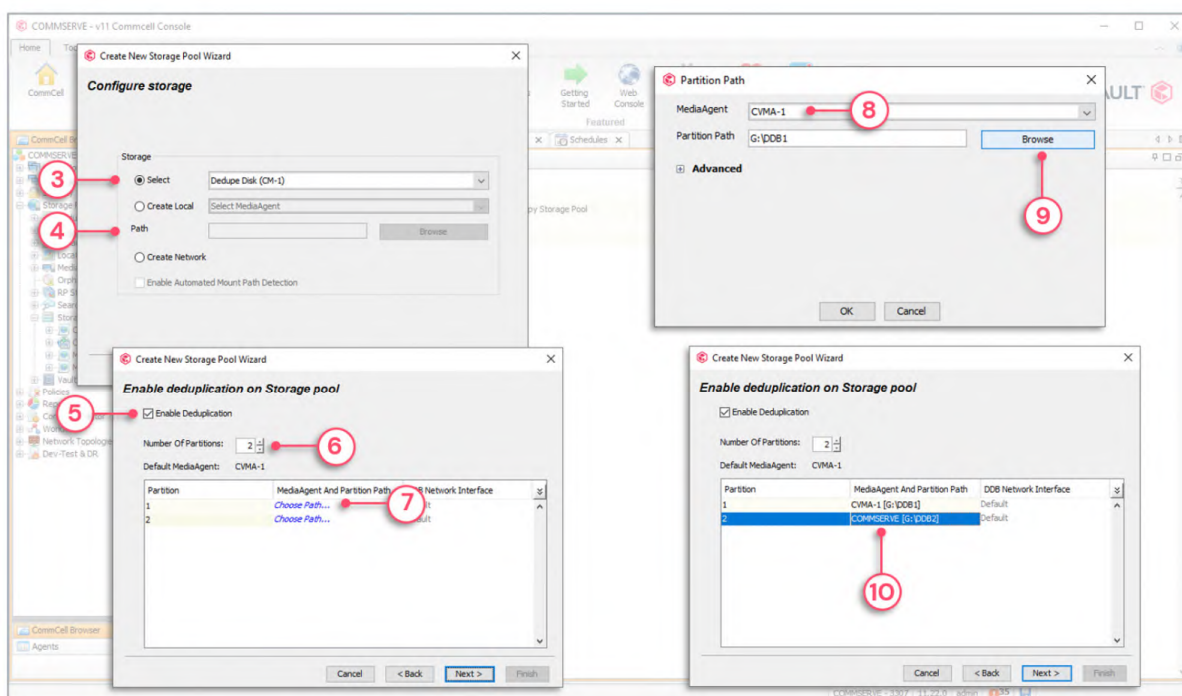


1. Expand Policies > Storage Policies
2. Global Deduplication Policies are shown here

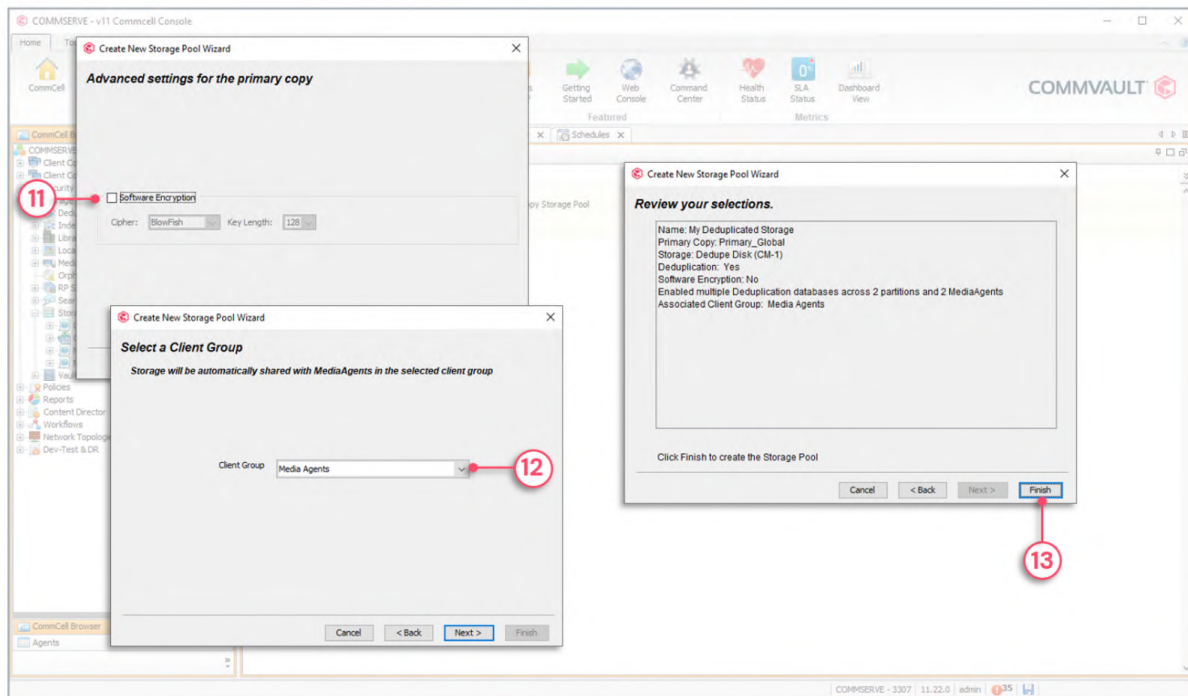
Create a Deduplicated Storage Pool



1. Right-click **Storage Resources** > right click **Storage Pools** > **Add Storage Pool** > **Disk**
2. Enter a name for the storage pool

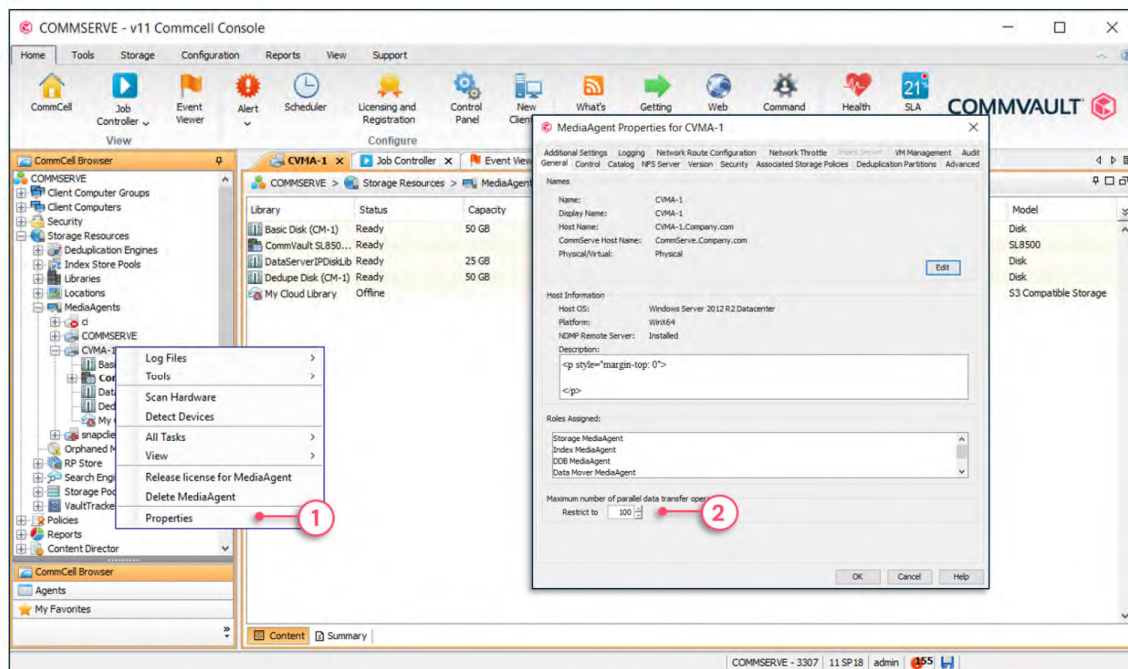


3. Select an existing disk library from the list
4. Alternatively, create a new local or network-based library
5. Check the box to enable deduplication
6. Define the number of partitions to use; if the disk storage is shared among multiple MediaAgents, several deduplication database partitions can be used
7. Click the **Choose Path** link to define the location of the first database partition
8. Select the MediaAgent hosting the first DDB partition
9. Click **Browse** to browse to the dedicated high-performance volume
10. Repeat the same procedure to define the location of the additional partitions



11. Check the box for **Software Encryption** if encryption is required
12. If more than one DDB partition was defined, the library is automatically share among the MediaAgent hosting a partition; if the library also needs to be shared with additional MediaAgents not hosting any partition, create and select a client computer group containing these MediaAgents
13. Click **Finish**

Deduplication Advanced Options



Streams and Deduplication

It is important to properly configure the streams when implementing multiple MediaAgents and DDBs of different sizes. The number of streams that a MediaAgent or a DDB can receive is based on the hardware of the MediaAgent and the speed of the DDB volume.

There are two options to consider for stream parameters with deduplication:

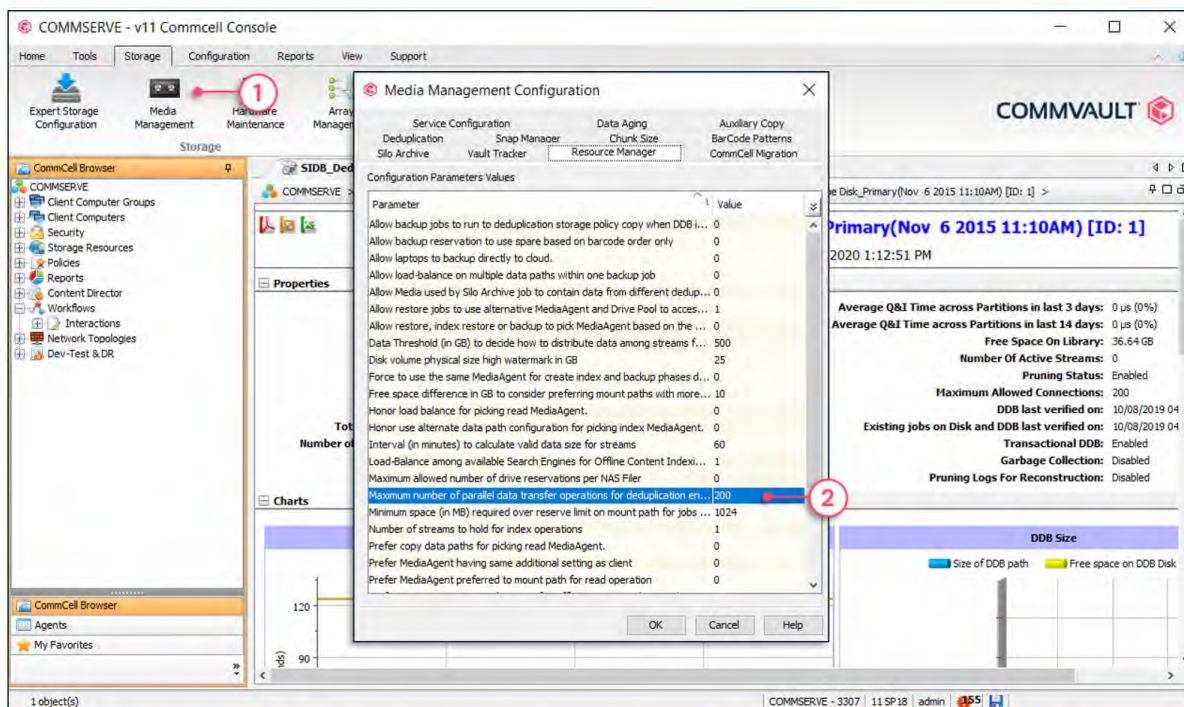
- MediaAgent maximum number of parallel data transfer operations – This is the maximum number of streams that the MediaAgent can receive. It is configured in the General tab of the MediaAgent Properties page.
- DDB maximum number of parallel data transfer operations – This is the maximum number of streams that any DDB of the CommCell environment can receive. This is applied to all DDBs. It is configured in the Resource Manager of the Media Management applet.

Example: If there are three MediaAgents; one is extra-large and two are medium, how can streams be adjusted? In this scenario, the DDB value should be set to the number of streams that the largest MediaAgent can receive, which in this case is 300 streams. This means that the extra-large MediaAgent maximum number of streams must also be set to 300 streams. Lastly, the two medium MediaAgents can be set to 100, preventing them to receive more streams than their hardware can handle.

MediaAgent Stream Options

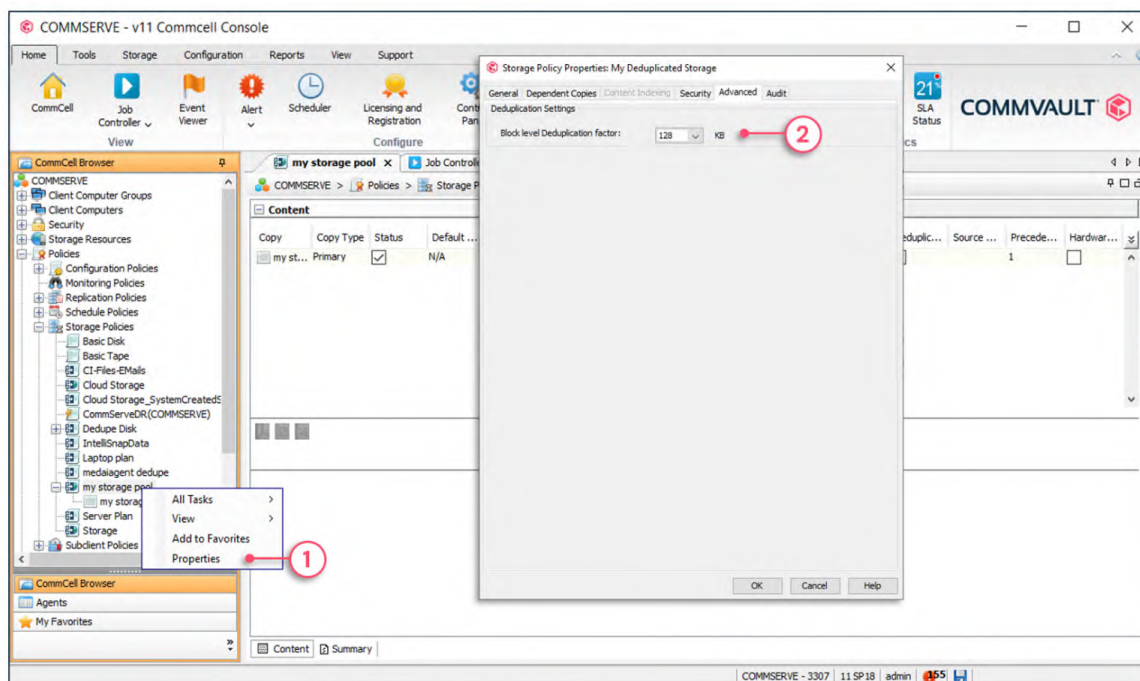
1. Right-click the **MediaAgent > Properties**
2. Define the maximum number of parallel streams that can be received by the MediaAgent

Configure the Maximum Number of Streams for DDB



1. *Storage* tab > **Media Management**
2. Define the maximum number of streams for deduplication databases

Deduplication Block Size



The deduplication block size determines the size of a data block that is hashed during data protection operations. The default block size is 128KB and is configurable from 32 – 512 KB. A smaller block size provides a marginally better deduplication ratio but reduces the maximum size of the deduplication store. A higher block size may reduce deduplication ratios but allows the deduplication store to hold more data. The block size recommendation is 128KB, which provides the best balance for deduplication ratio, performance, and scalability.

Deduplication block size is configured in the global deduplication policy properties. All storage policy copies associated with the global deduplication policy always use the block size defined in the global deduplication policy.

1. Expand Storage Policies > right-click the **Global Deduplication Policy** > **Properties**
2. Select the required block size from the list

Block Size for Large Databases

For databases, the recommended block size is from 128KB to 512KB depending on the size of the data, number of full backups, and retention requirements. Initial protection of the database does not yield high deduplication ratios due to the uniqueness of block data, so block size does not have a significant impact on deduplication ratios. However, protecting the database redundantly over time can yield very high deduplication ratios, depending on the application and compression usage.

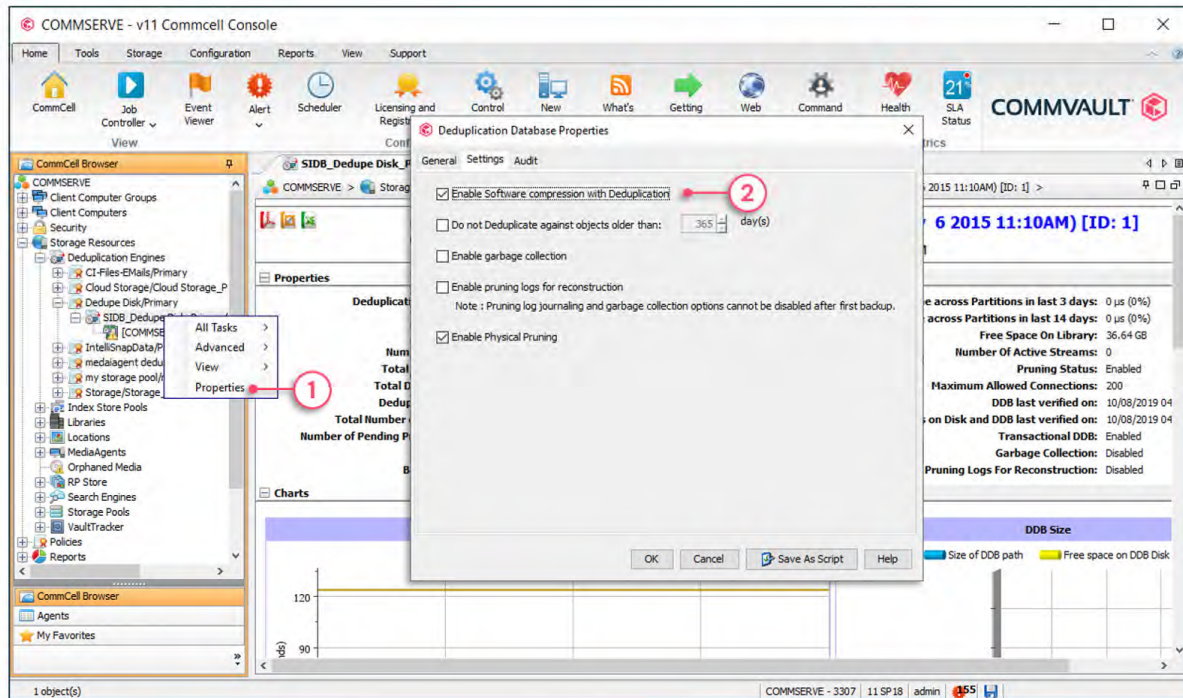
Block Size for Large Static Repositories of Binary Data

For large data repositories managing binary data types such as static media repositories, larger deduplication block sizes can provide more scalability because the potential for duplicate data blocks is minimal. Using a higher block size allows more data to be stored with a smaller DDB size. In this scenario, the advantage is using client-side deduplication and DASH full backups results in processing and backing up the block data only once. This solution works very well for static data types, but if processing such as video editing will be performed on the data, deduplication may not be the best method to protect the data.

Changing Deduplication Block Size

If the block size is changed on a deduplication store containing data, it results in a resetting the baseline of all data in the store since new signatures being generated do not match existing signatures. If the block size needs to be changed, ensure no jobs are currently running, seal the current DDB, and then change the block size.

Compression

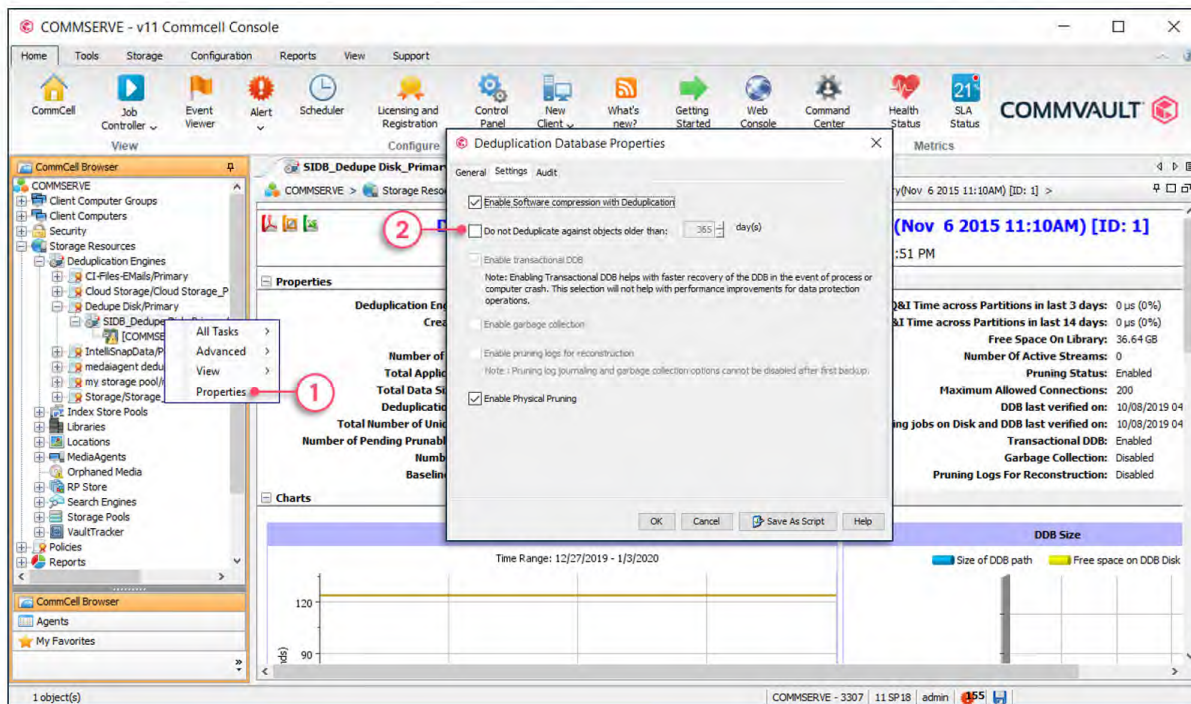


By default, compression is enabled in a global deduplication policy. Data is compressed prior to signature generation, which is optimal for files and virtual machines. For database backups, signatures are generated first and then compression takes place providing better deduplication ratios for large databases. When running auxiliary copy jobs to tape, compression is enabled by default for the tape data path. Most modern tape drives are advanced enough to attempt compression on the block and determine if it is beneficial. If the block compressed well, the block remains compressed. If not, it will un-compress the data on the fly.

In some cases, where application data is being compressed prior to backup, such as in certain Oracle RMAN (Recovery Manager) configurations, it is best not to use Commvault compression. Use only one compression; never use both. For most data, Commvault compression results in the best deduplication ratios.

1. Expand the Storage Resources > Deduplication Engine > right-click the **primary copy** > **Properties**
2. Check the box to enable compression when using deduplication

'Do Not Deduplicate Against Objects Older Than' Option



This option should only be used when data verification jobs frequently identify bad blocks in the store. This is indicative of issues with the disks themselves, which may need to be replaced. Note that when this option is enabled, the baseline of specific blocks is reset once they exceed the 'days' threshold.

1. Expand the Storage Resources > Deduplication Engine > right-click the **primary copy** > **Properties**
2. Check the option and set a maximum number of days for deduplication

Data Verification

- Levels
 - Full verification
 - Incremental verification
- Types
 - Jobs on disks and database
 - Database only
 - Quick verification
- Incremental verification schedule

A corrupt block in the deduplication store can result in data from multiple jobs not being recoverable. Commvault conducts live data verification operations while data protection jobs are running.

Verification of Existing Jobs on Disk and Deduplication Database

This verification method uses checksum data to verify block integrity by reading data chunks (Sfiles), uncompressing and decrypting, and using Cyclic Redundancy Check (CRC) information to validate block integrity. This option also verifies chunk metadata using CRC checks. Any blocks failing the check will be marked in the DDB. New blocks generating the same signature as a block marked bad are rewritten to disk and a new signature entry is written to the DDB. This verification method also verifies chunk integrity between the DDB and disk library.

Verification of Deduplication Database

This verification method performs all the same tasks as the 'Verification of Existing Jobs on Disk and the Deduplication Database' except metadata chunk validation.

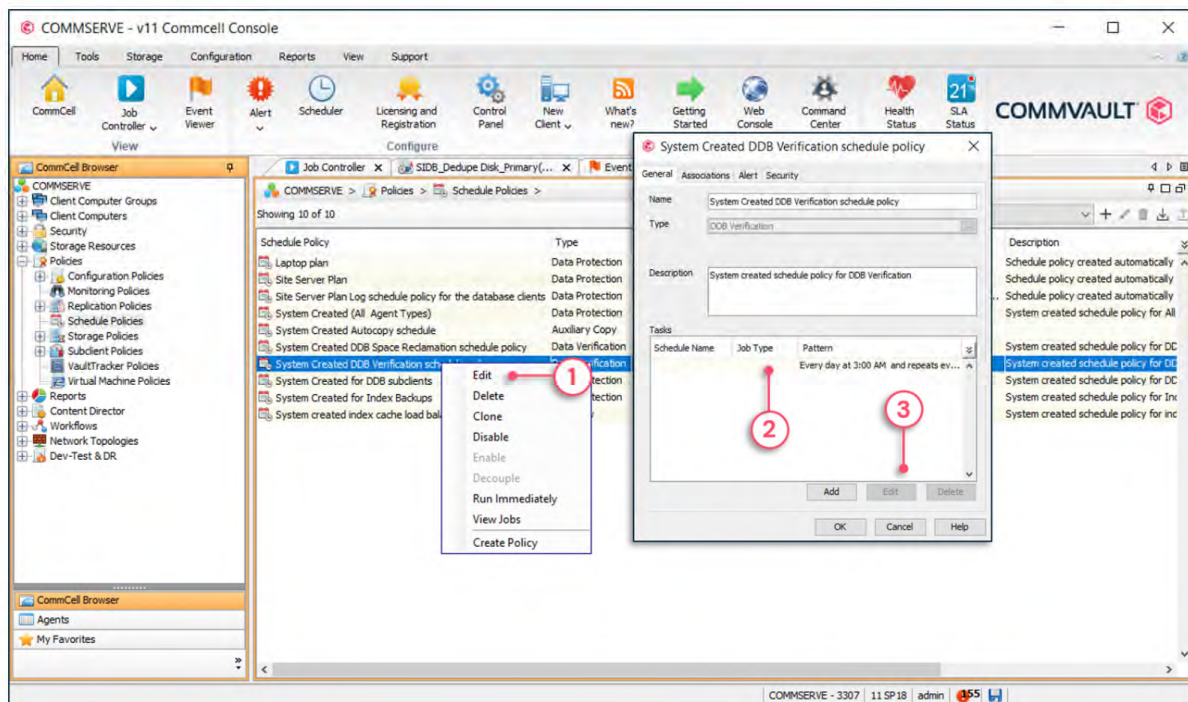
Quick Verification of Deduplication Database

The verification method quickly verifies chunk integrity between DDB and disk library.

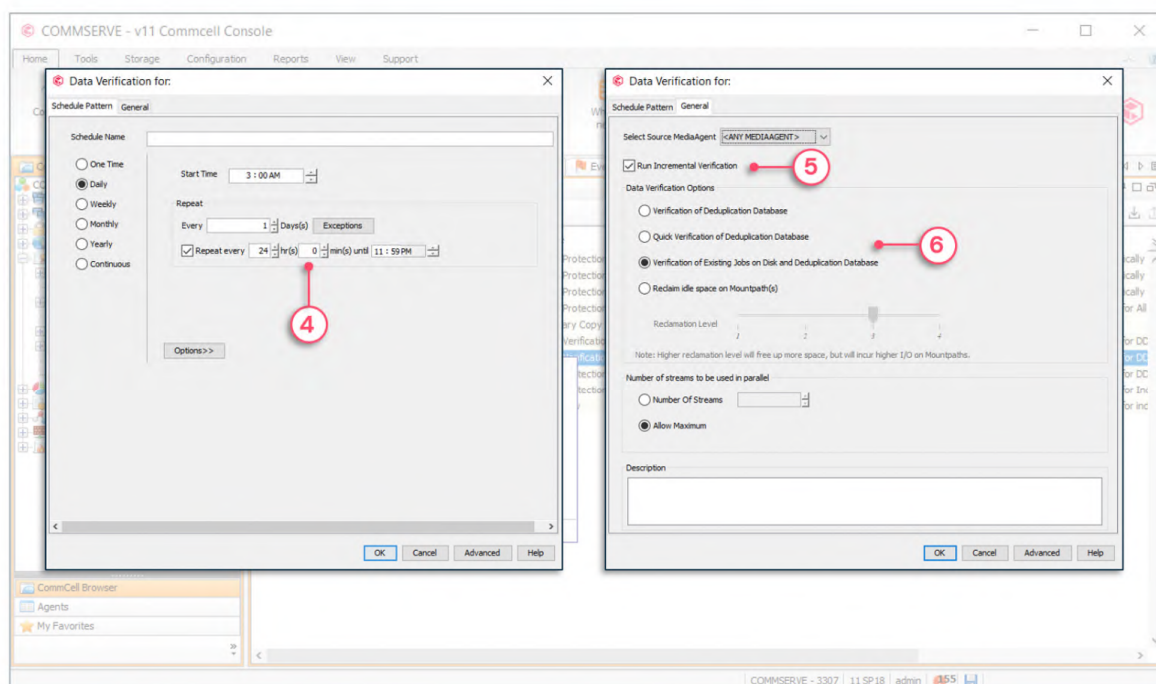
Incremental Verification

This method verifies data integrity for new jobs added since the last verification job. This option is available when running 'Verification of Deduplication Database' or 'Verification of Existing Jobs on Disk and the Deduplication Database' options. Commvault executes an incremental verification every day, at 11:00 AM. Since this method only verifies new jobs, full verification jobs should periodically be executed (e.g., monthly or quarterly).

Edit the DDB Verification Schedule

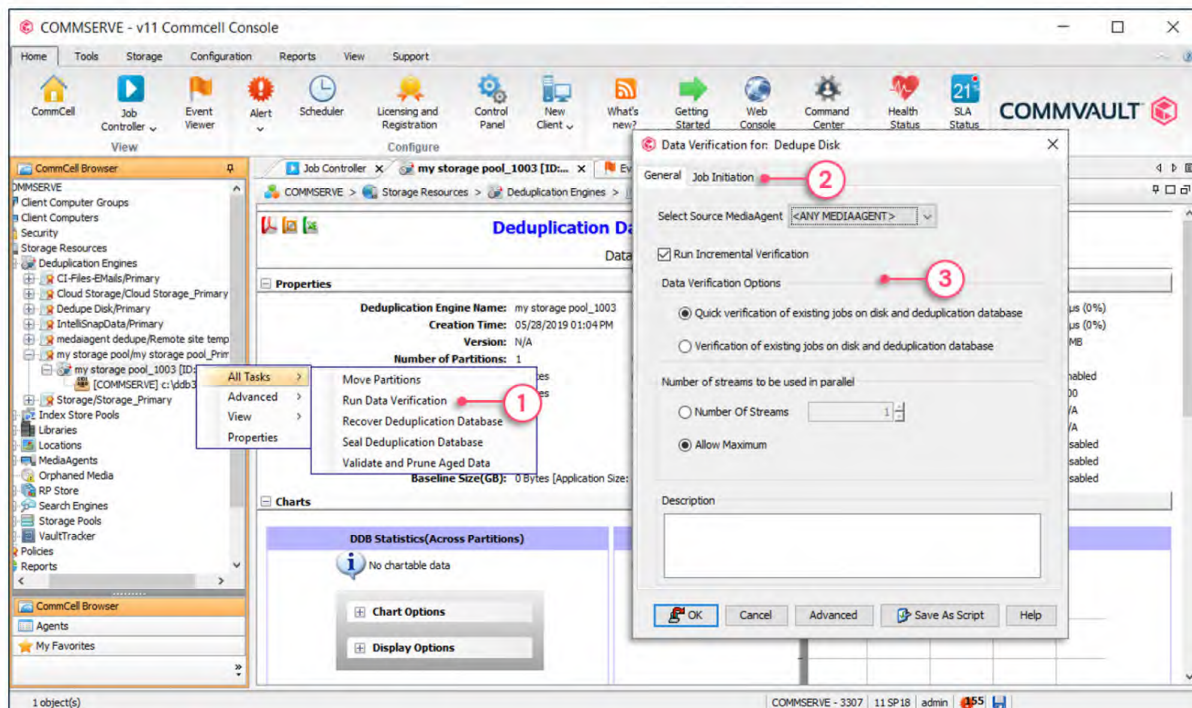


1. In the CommCell browser > **Schedule Policies** > right-click the **System-created DDB verification schedule policy** > **Edit**
2. Select the schedule
3. Click **Edit**



4. Modify schedule options
5. Check the box to run an incremental verification or uncheck to run a full verification
6. Select the verification options

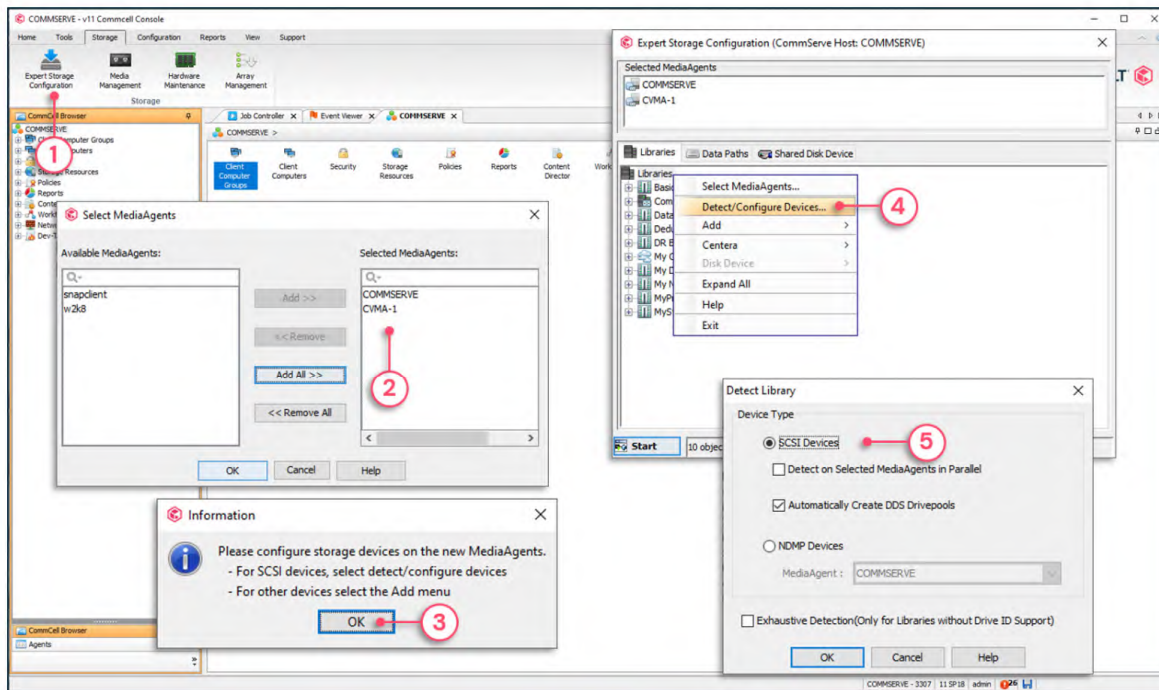
Run a Data Verification Job



1. Right-click DDB partition > **All Tasks** > **Run Data Verification**
2. Data verification job can be scheduled, run immediately, or saved as a script
3. Configure data verification options

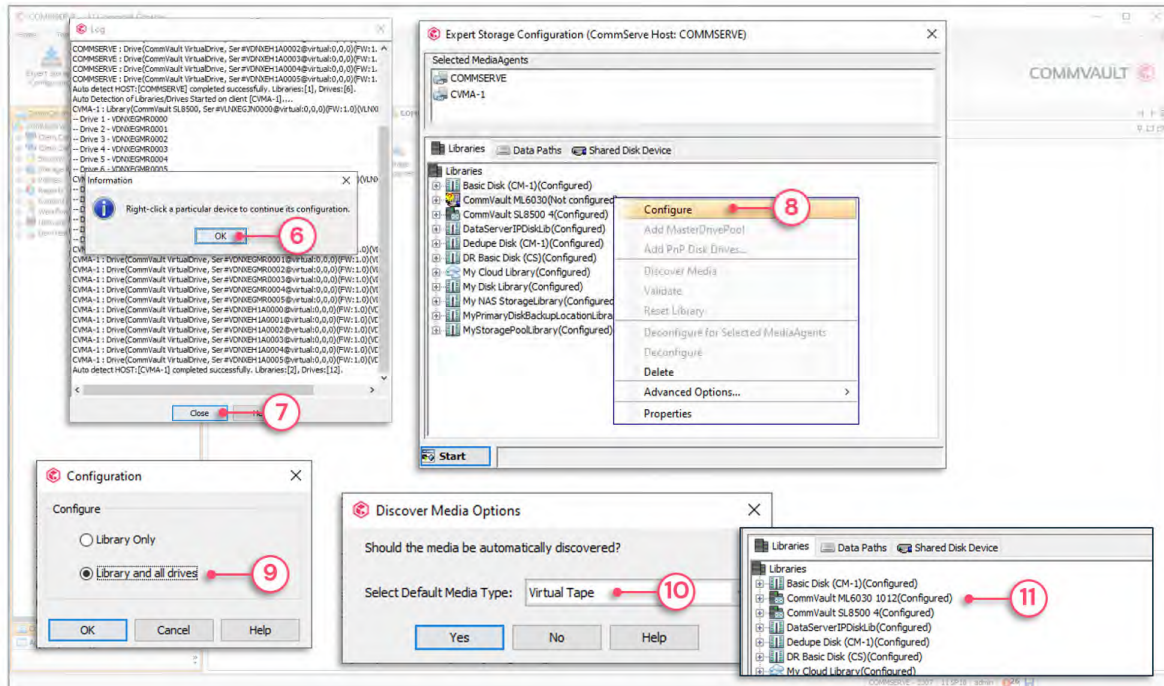
Tape Libraries

Share a Tape Library



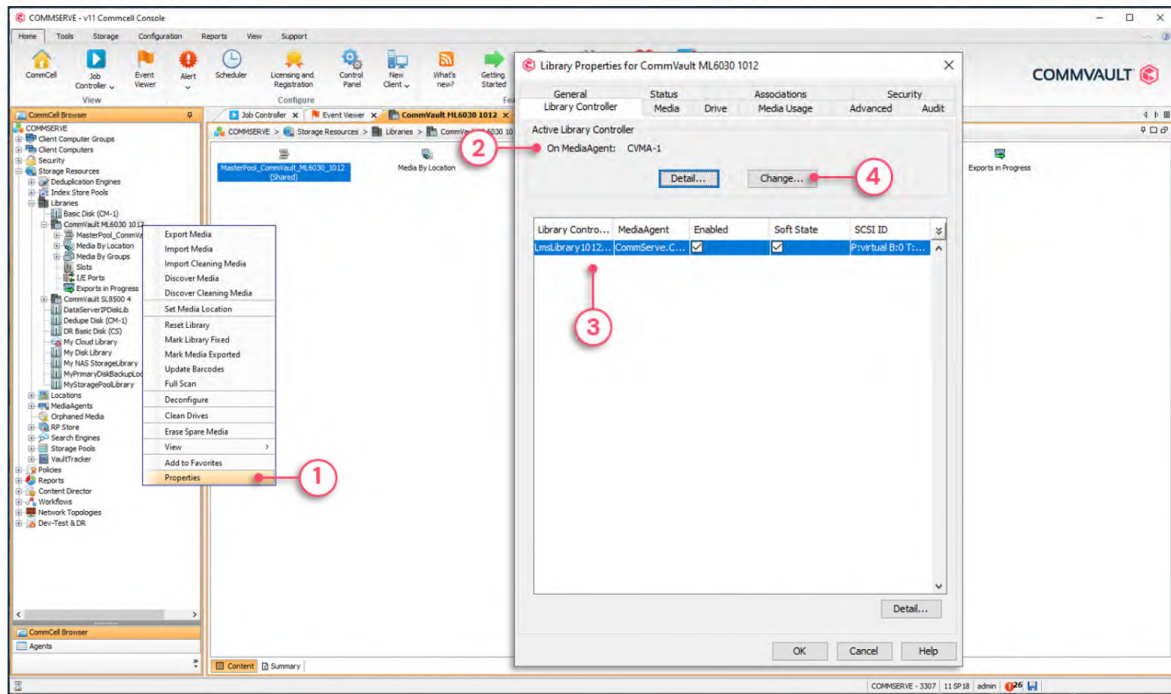
Before you configure a tape library in Commvault software, it must be attached or zoned to be seen by the MediaAgent operating system. The device manager should detect one medium changer and one or many drives. Install the vendor drivers' specific to the library. If the library is dedicated to a single MediaAgent, it is configured directly in the CommCell browser. If it is shared, it must be configured using the Expert Storage Configuration wizard.

1. **Storage** tab > **Expert Storage Configuration**
2. Select all MediaAgents sharing the library and then click **OK**
3. Click **OK** on the information message
4. Click **Detect/Configure Devices**
5. Click **SCSI Devices** and click **OK**



6. Click **OK** to close the information message
7. Click **Close** to go back to the Expert Storage Configuration screen
8. Right-click the **unconfigured library** (red question mark on the icon) > **Configure**
9. Choose **Library and all drives** and click **OK**
10. Choose the default media type from the list
11. The library is shown as configured for both MediaAgents

Active and Failover Library Controllers

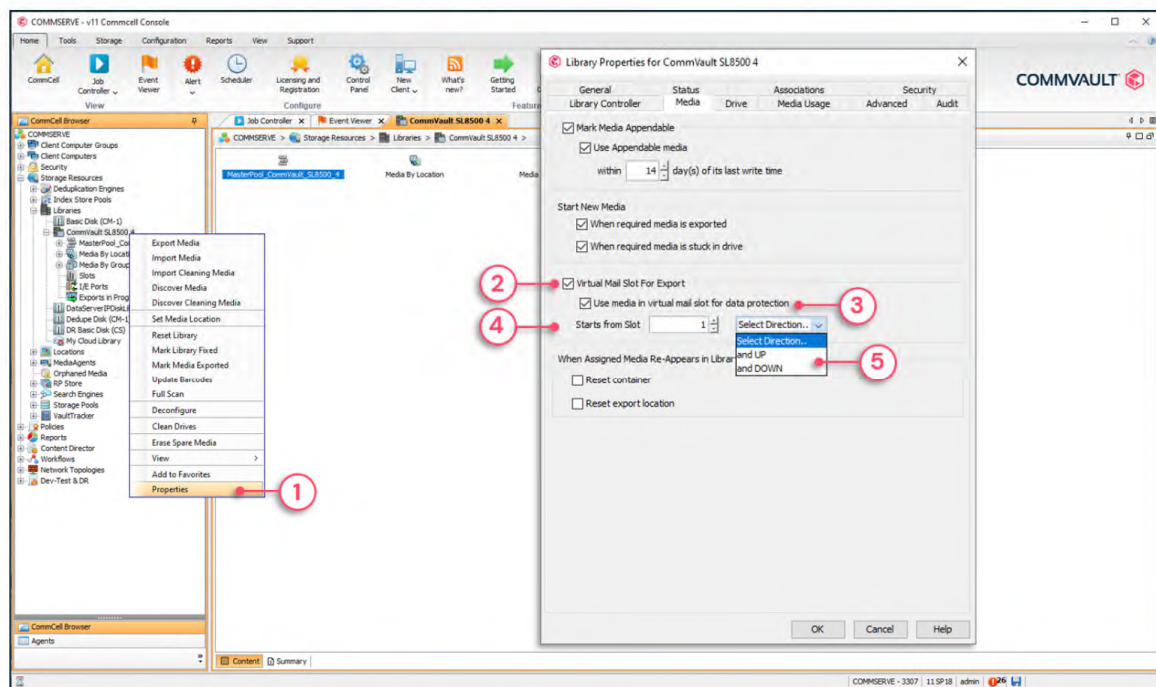


When a tape library is shared among multiple MediaAgents, only one of the MediaAgent acts as the active library controller. The active controller is responsible for sending SCSI commands to the device such as mounting/unmounting a tape, cleaning a drive, resetting a drive, exporting/importing media, etc.

Even if another MediaAgent requires writing data, the library operation is always conducted by the active controller. Failover candidates can be defined to replace the active controller if the active controller become unavailable.

1. Right-click the **tape library** > **Properties**
2. Displays the current active controller
3. Displays failover candidates for this shared library
4. Click **Change** to manually switch another controller as the Active Controller

Tape Library Advanced Options



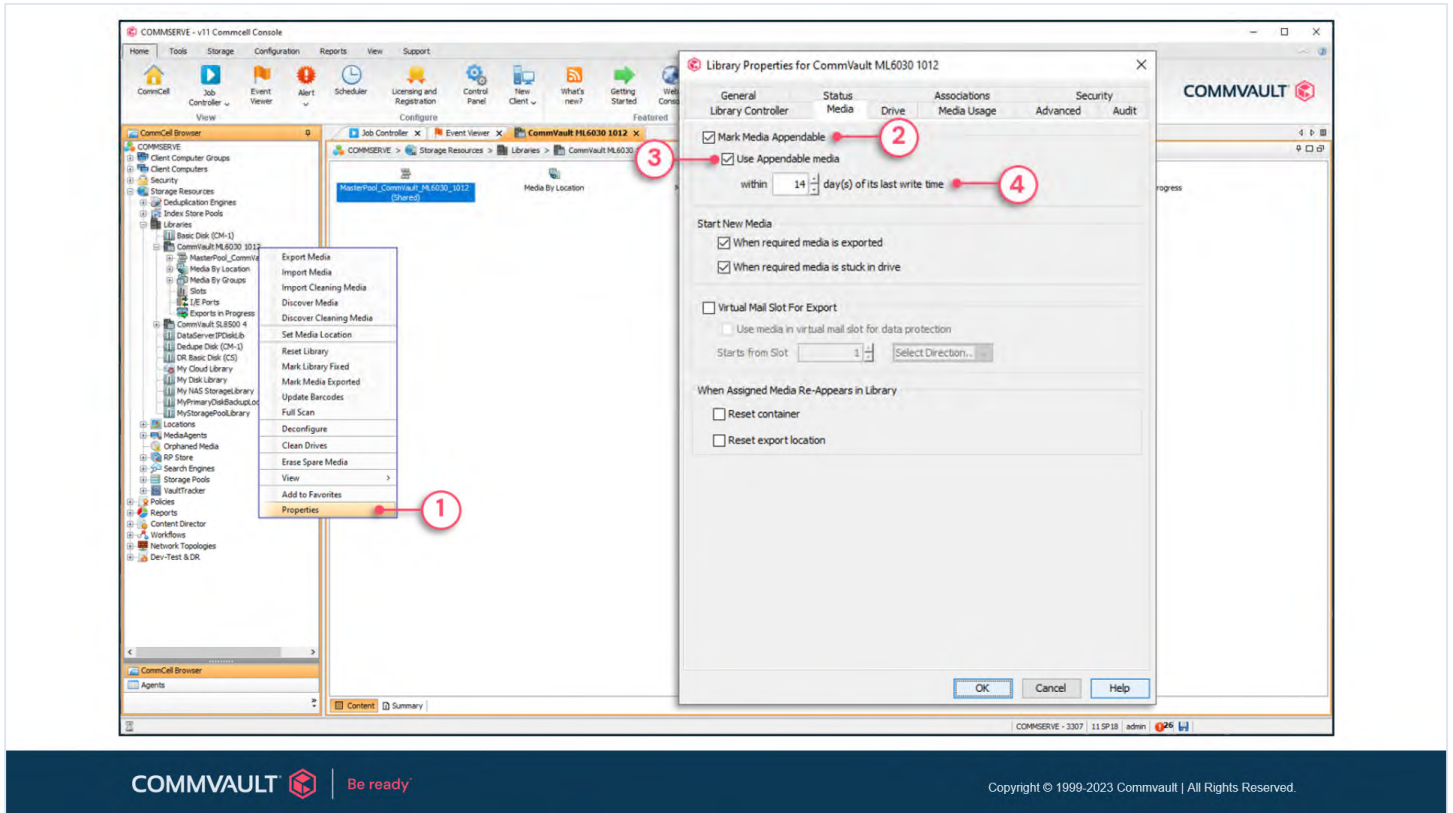
Virtual Mail Slots

When exporting media, the library uses I/E ports referred to as mail slots. If several tapes require exporting and the number of I/E ports are limited, the tape operator must open the library multiple times, which triggers a time-consuming library scan each time. To avoid this situation, virtual mail slots are configured to allow the entire magazine or library door to be open to access the media.

When defining virtual mail slots, a starting port number must be defined, as well as the order for additional media. When using virtual mail slots, leave at least one empty slot in the library, as it is required by the system when re-ordering tapes.

1. Right-click the **tape library** > **Properties**
2. Check the box for **Virtual Mail Slot For Export**
3. Check the box to use media in a virtual spot for data protection
4. Define the starting port to use as virtual mail slot
5. Define the order to use for additional virtual mail slots

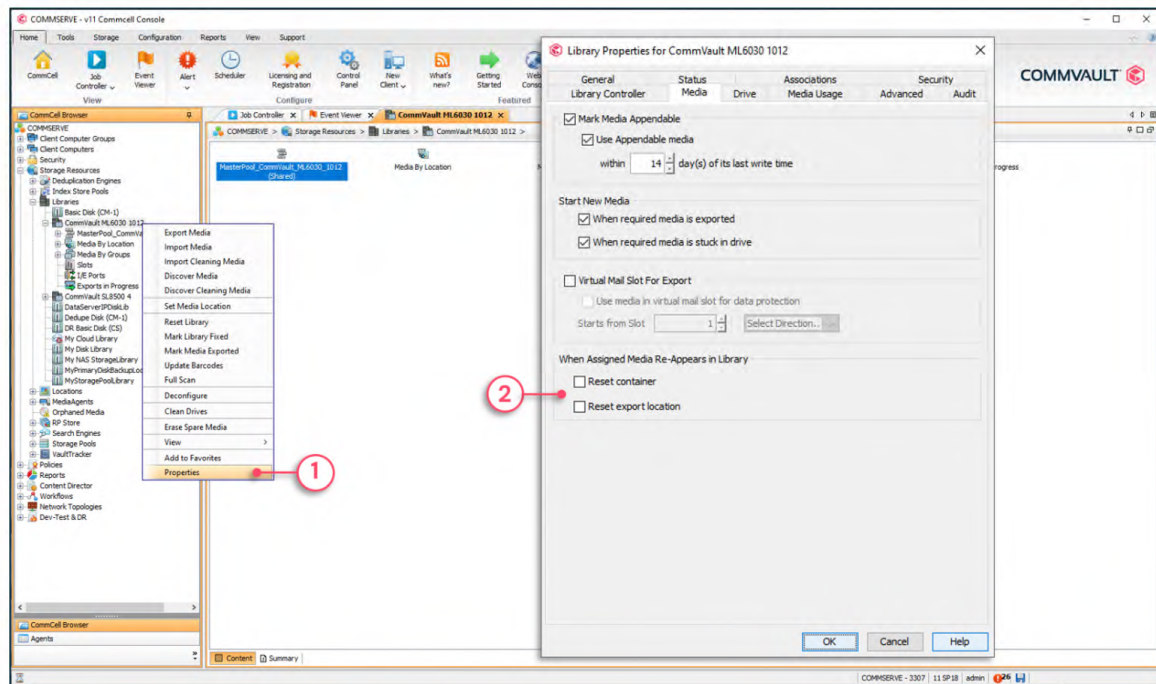
Appendable Media



By default, Commvault software tries to fill a tape completely before exporting it. This default behavior is overridden by either manually exporting the unfilled tapes or by scheduling an export media job using the 'Export Active Media' option. In such cases, the exported media is assigned a status of 'appendable.' If the media becomes appendable but gets reinserted into the library within 14 days, it is used by the system for subsequent jobs. If it is reinserted after 14 days, the system does not use it anymore. The 14-day threshold can be modified.

1. Right-click the **tape library** > **Properties**
2. Check the box to mark media as appendable when exported before being filled
3. Check the box to use appendable media for subsequent jobs when reinserted in the tape library
4. Define the threshold in days to reinsert a media and append data to it

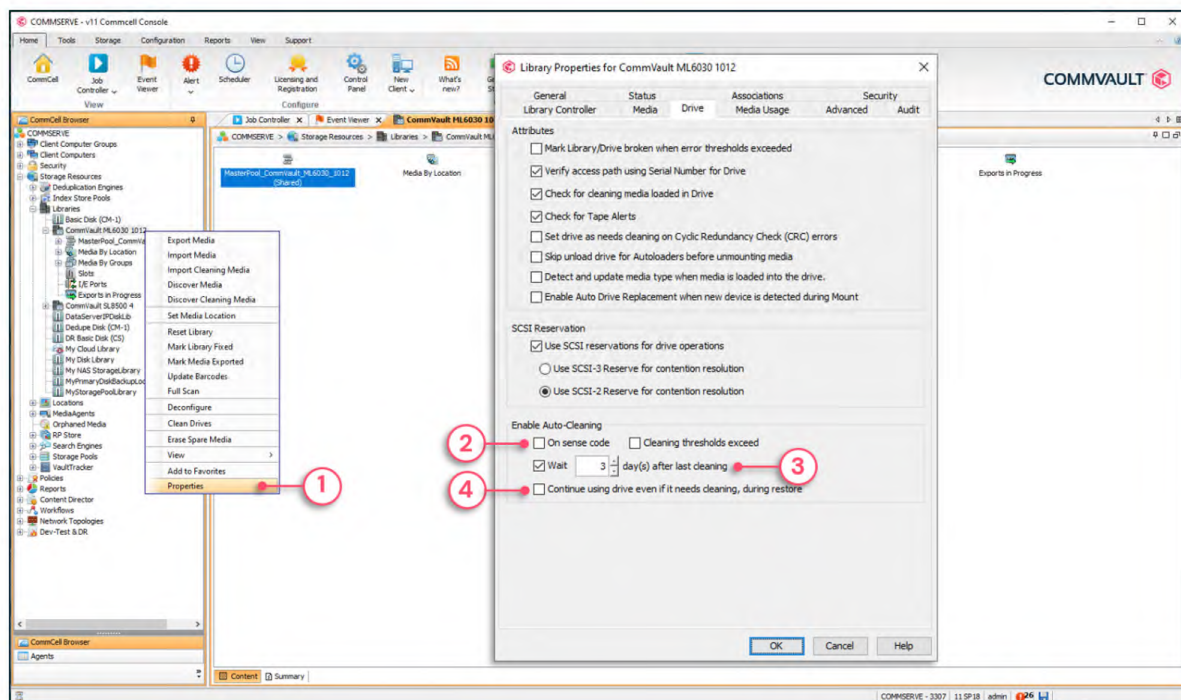
Media Container and Export Location Reset



When a tape is exported, its container and export location can be manually defined or defined by the VaultTracker 'Export Media' job. When a tape is reinserted into the library, this information can be reset by checking the 'Reset container' and 'Reset export location' options in the library properties. These options can be enabled without any adverse effects and is useful when using VaultTracker.

1. Right-click the **tape library** > **Properties**
2. Check to reset container and export location when a media is reinserted into the tape library

Clean Tape Drives



Dirt can accumulate on the tape drive read/write heads. Clean the drives using a cleaning media. This process is usually automated but can be executed manually. Note that only one method can be used. Both cleaning methods are equally effective since both use hardware sense code and/or cleaning thresholds. The preferred method should be determined based on the manufacturer's recommendations.

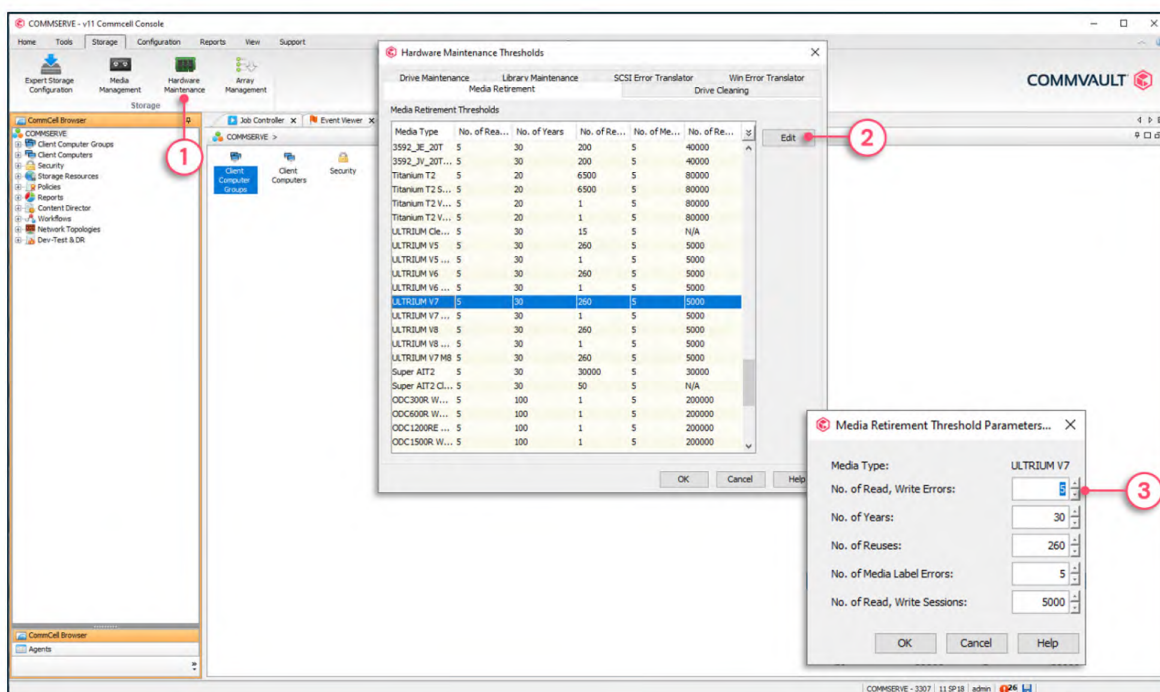
- Hardware-controlled cleaning – The cleaning of the drives is handled by the library. Auto-cleaning must be disabled in Commvault software and must be enabled on the library by using either the administration web portal or its control panel. The library configures dedicated cleaning slots where cleaning media are stored. The dedicated cleaning slots and cleaning media are not visible in Commvault software and cannot be used. Manual cleaning operations must be initiated from the library web page.
- Software-controlled cleaning – The cleaning of drives is handled by Commvault software. Auto-cleaning must be disabled on the library and must be enabled in Commvault software. The library does not reserve any dedicated cleaning slots and Commvault software is aware of the cleaning media. Cleaning must be initiated from Commvault software.

If the software receives a sense code and cleans the tape drive heads but the drive still encounters errors, a hardware malfunction should be investigated. In this situation, to avoid having the system trying to clean the drive again, a minimal number of days since the last cleaning can be set before a new cleaning attempt is conducted.

When a drive status is 'dirty' and the system cannot clean the drive, Commvault software stops using that drive completely for both backups and restores. This prevents damaging the media by using dirty tape drive. If resources are limited and a restore requires a tape drive, the 'Continue using drive even if it needs cleaning, during the restore' option is used.

1. Right-click the **tape library > Properties**
2. Choose to enable software cleaning 'on sense code' and/or cleaning thresholds
3. Check the box and define a minimal number of days between cleaning to avoid multiple cleanings on hardware malfunction
4. When resources are limited, check the box to continue using a dirty drive only on restores

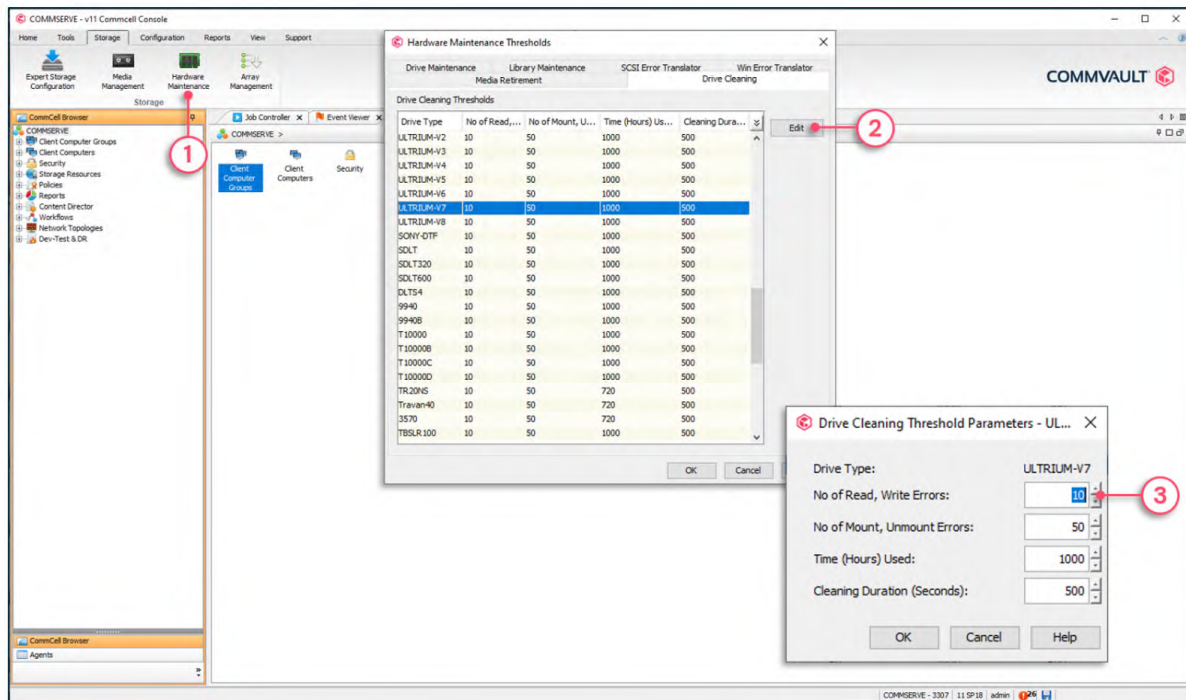
Configure Media Retirement Threshold



By default, the threshold to retire a bad media is five read/write errors, while the threshold to clean a dirty drive is ten read/write errors. This means that up to two tapes could be retired before the drive gets cleaned, even though the tapes may still be usable. To avoid this situation, you can increase the tape threshold and/or decrease the drive threshold to ensure that the drive is cleaned before the media is retired.

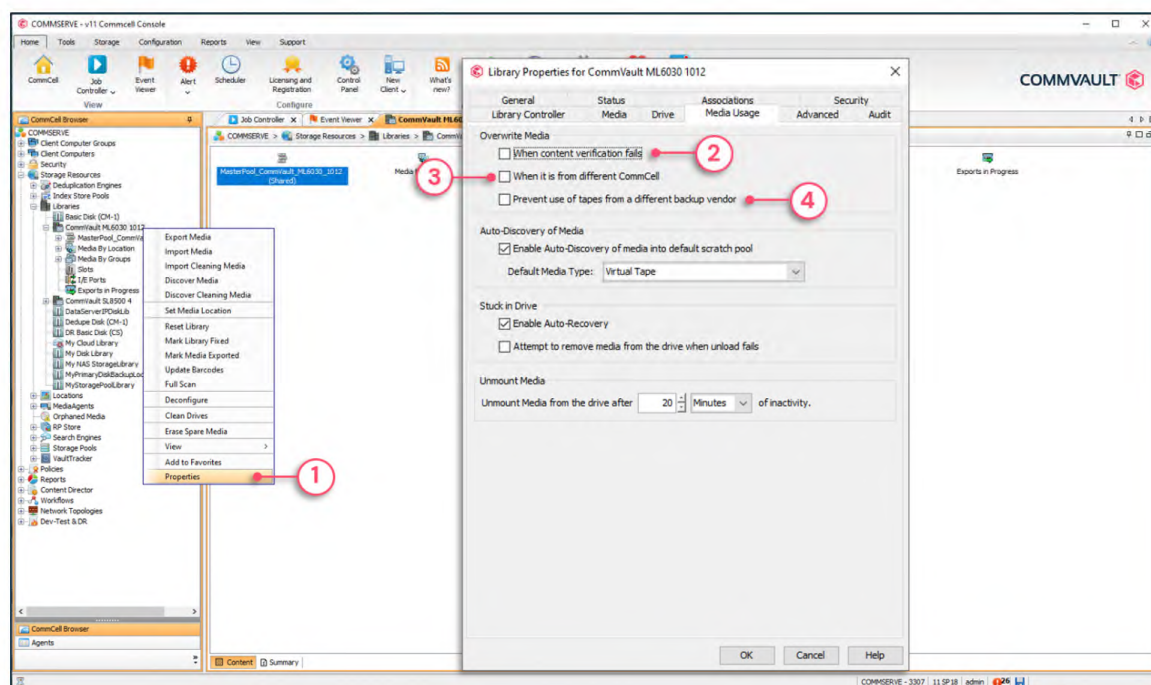
1. *Storage* tab > **Hardware Maintenance**
2. *Media Retirement* tab > select the tape technology and click **Edit**
3. Adjust the values

Configure Drive Cleaning Threshold



1. *Storage* tab > **Hardware Maintenance**
2. *Drive Cleaning* tab > select the drive technology and click **Edit**
3. Adjust the values

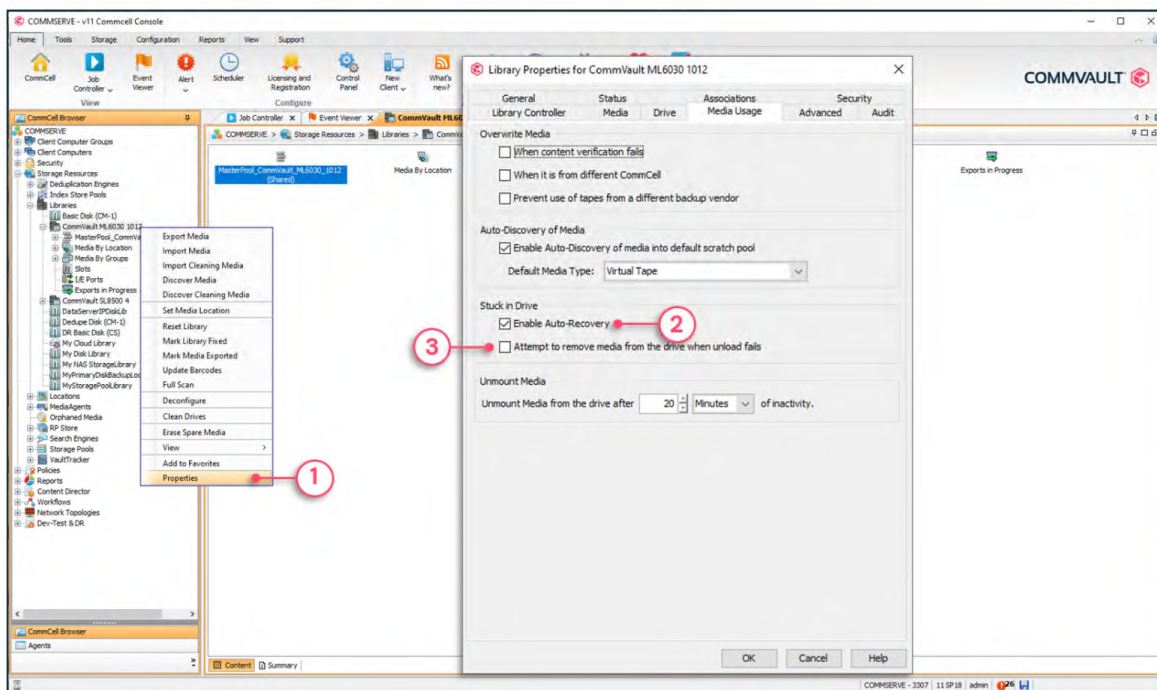
Overwrite Media Options



Overwrite media options are set when configuring a tape library. These options apply to specific scenarios and should be carefully evaluated before enabling them.

1. Right-click the **tape library** > **Properties**
2. Check the box for **When content verification fails** to automatically overwrite the tape if the MediaAgent encounters media that it cannot read, such as a 'bad On Media Label (OML)'; this option should be carefully planned as it could result in overwriting valid data
3. Check the box for **When it comes from a different CommCell environment** to use media that came from a different CommCell as scratch tapes
4. Check the box for **Prevent use of tapes from a different backup vendor** to enable Commvault to detect and avoid writing to this media; this option is useful when the library is shared with another backup software

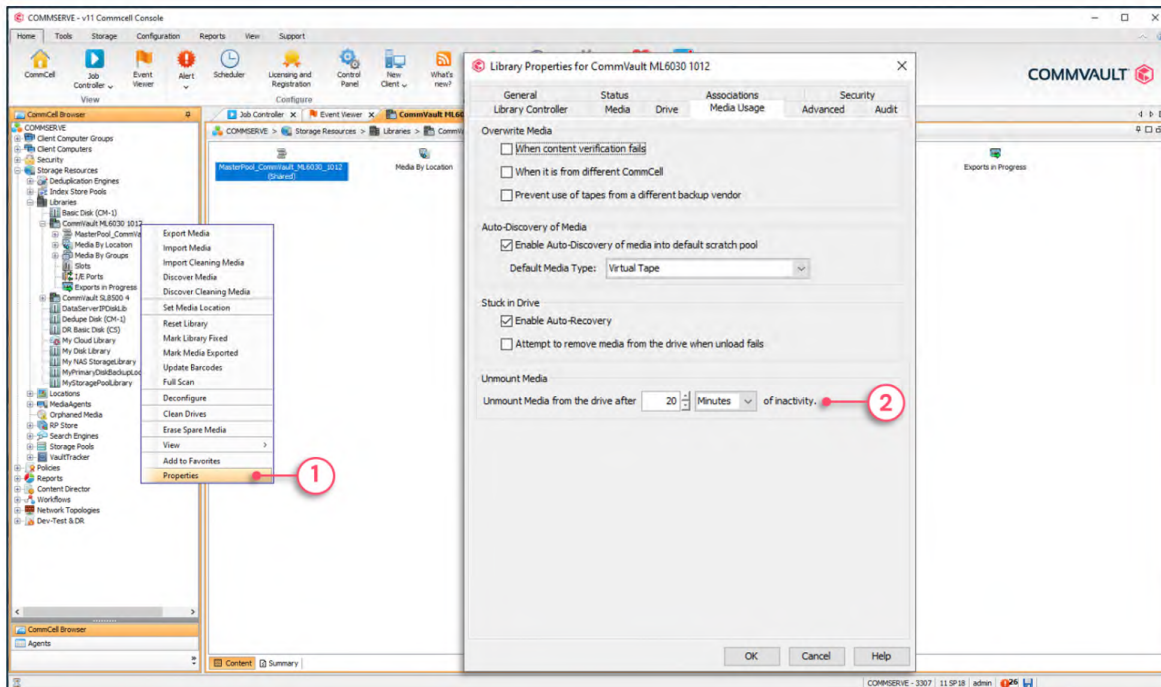
Media Stuck in Drive



If media is stuck in a drive for any reason, Commvault software options to help resolving the situation include:

1. Right-click the **tape library > Properties**
2. Check the box for **Enable Auto-Recovery** to configure the MediaAgent to attempt recovery every 20 minutes
3. Check the box for **Attempt to remove media from the drive when unload fails** to configure the MediaAgent to attempt to move stuck media out of the drive

Unmount Idle Media Timeout



When a job completes, the idle media stays mounted for an additional 20 minutes by default. By doing so, when a new job requiring the same tape starts within 20 minutes, the robotic arm does not attempt to unmount the media. After 20 minutes of idle time, if the tape is not required by any job, it is unmounted. During the 20 minutes of idle time, if another job requiring a different tape does not have any resource available, the idle tape is unmounted to accommodate the active job even if the time limit is not reached.

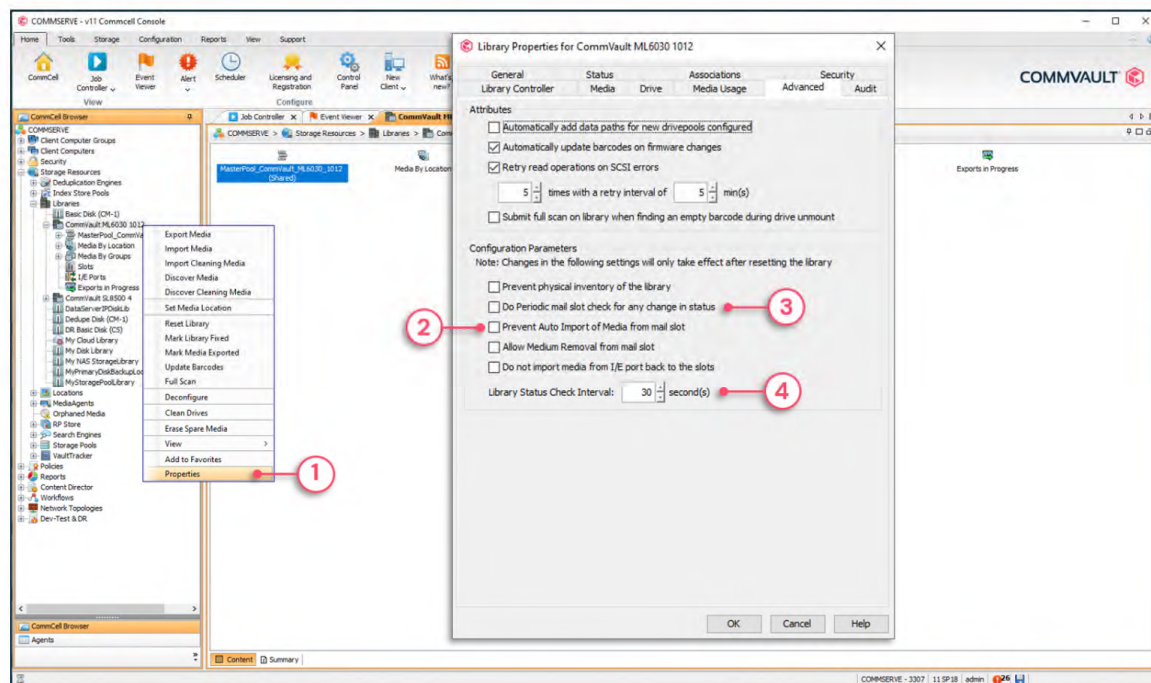
1. Right-click the **tape library** > **Properties**
2. Set the time value at which the idle media will be unmounted

Automatic Auxiliary Copy Jobs to Tapes Scenario

Auxiliary copy jobs are scheduled to run automatically every 30 minutes. If you conduct frequent backups during the day, such as hourly transaction log backups, these jobs will be copied to tapes every 30 minutes. If the default value of 20 minutes is used to unmount idle media, tapes will constantly be mounted and unmounted throughout the day, which is not recommended for the robotic arm.

Solution: Increase the 'Unmount Media from the drive after' value to 35 minutes. The first auxiliary copy job of the day then mounts the tape, which stays mounted all day long if no other jobs are requiring the drive.

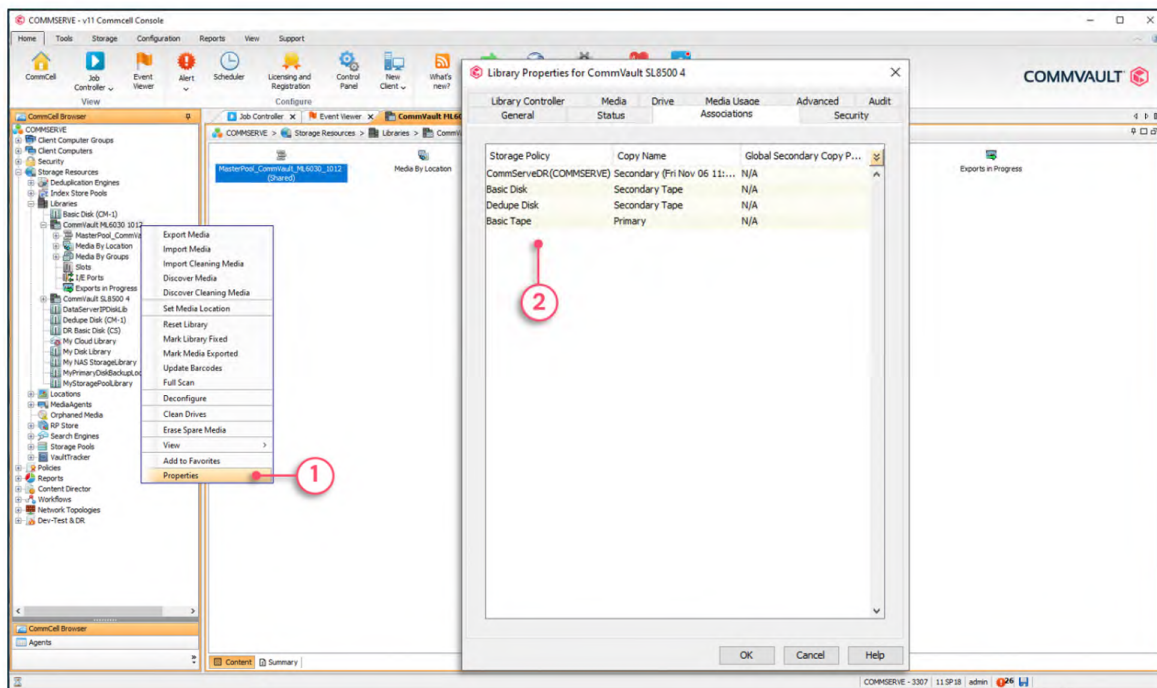
Partitioned Library Recommended Options



When a tape library is partitioned and shared with another CommCell environment or another backup software, some options should be considered and could be beneficial as it ensures that the media are imported in the proper library partition.

1. Right-click the **tape library** > **Properties**
2. Check the box for **Prevent Auto Import of Media from mail slot** to force the use of manual imports and prevent the MediaAgent to automatically import tapes
3. Check the box for **Do Periodic mail slot check for any change in status** to periodically poll the library I/E ports for changes
4. Set the frequency for periodic mail slot checks

Tape Library Association Storage Policies



The Associations tab of a tape library displays all storage policies that have a copy sending data to the tape library. This is useful information when you plan on decommissioning a library or taking it offline for maintenance.

1. Right-click the **tape library** > **Properties**
2. Displays storage policy copies sending data to this tape library

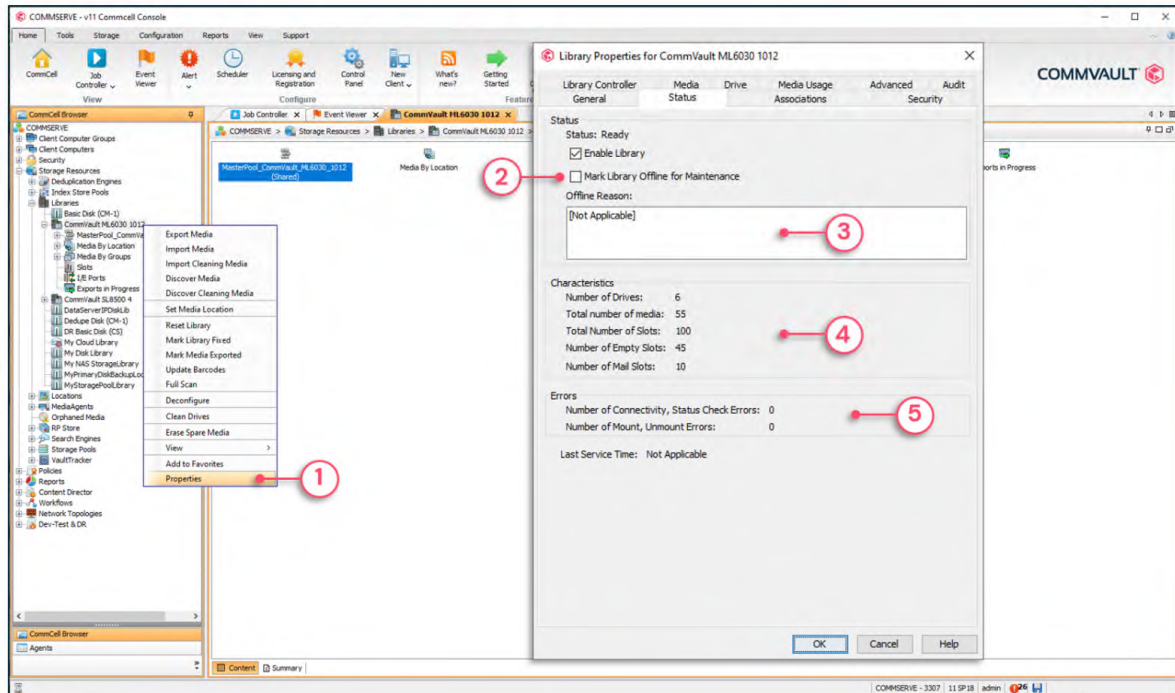
Tape Library and Drive Status Errors

Drive No.	Drive	Drive Type	Status	Barcode	Job ID	Mount Status	Operation Phase	MediaAgent	Storage Policy
1	CommVault VirtualDrive_1	VIRTUAL	Drive Fully Accessible				None	COMMVAULT	N/A
2	CommVault VirtualDrive_2	VIRTUAL	Drive Fully Accessible				None	COMMVAULT	N/A
3	CommVault VirtualDrive_3	VIRTUAL	Drive Fully Accessible				None	COMMVAULT	N/A
4	CommVault VirtualDrive_4	VIRTUAL	Drive Fully Accessible				None	COMMVAULT	N/A
5	CommVault VirtualDrive_5	VIRTUAL	Drive Fully Accessible				None	COMMVAULT	N/A
6	CommVault VirtualDrive_6	VIRTUAL	Drive Fully Accessible				None	COMMVAULT	N/A

View the status for all drives of the library. If a drive is displayed as offline, it can be a defective drive. If all drives are displayed as offline, it could indicate a communication or failure with the entire library.

1. Expand the tape library and click the master drive pool
2. The status of each drive is displayed

Access the Tape Library Status Tab



If a library is offline, the library's Status tab provides an explanation, statistics, and error counts. If counters start to increase, this could be an indication of connection errors or that a drive is about to fail.

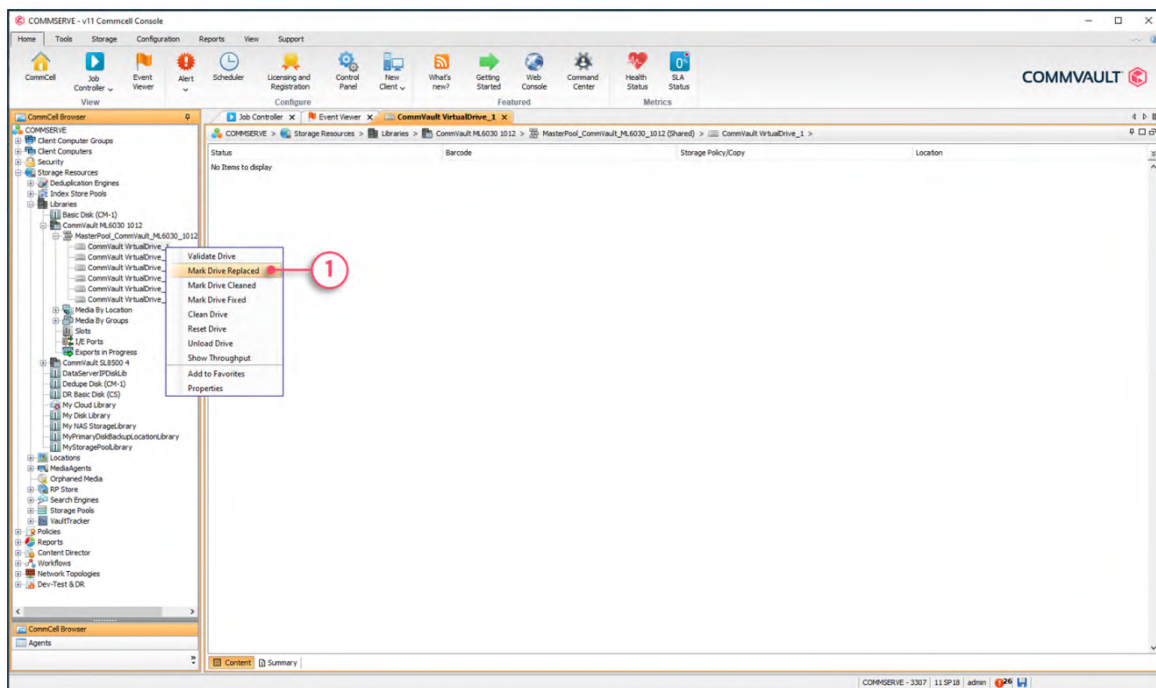
1. Right-click the **tape library > Properties**
2. Check the box to take library offline for maintenance
3. If library is offline the reason will display here
4. Displays statistics and error counts for this library
5. Displays the number of errors encountered since counters were last reset

To monitor progress, right-click the library and choose 'Mark Library Fixed' to reset counters. If the problem persists, a library can be disabled or taken offline for maintenance.

The screenshot displays the COMMVAULT v11 CommCell Console interface. The left-hand navigation pane shows the hierarchy: CommCell Browser > Agents > CommCell VirtualDrive. A right-click context menu is open over a virtual drive, with 'Validate Drive' selected (indicated by red circle 1). The 'Validate Drives' dialog box is open, showing 'COMMVAULT' as the MediaAgent (red circle 2) and 'Any Spare Media' as the Media (red circle 3). The 'Job Controller' window shows a list of jobs, with 'Storage Validation' selected (red circle 4). A 'Validate Storage' confirmation dialog is displayed, showing the success message and the 'OK' button (red circle 5).

1. Expand the master pool > right-click the **drive > Validate Drive**
2. Select the MediaAgent to use for the validation
3. Allow Commvault to choose to write to any spare media or select specific media
4. The validation job is displayed in the Job Controller
5. Throughput results are displayed in the Validate Storage window

Tape Drive Replacement

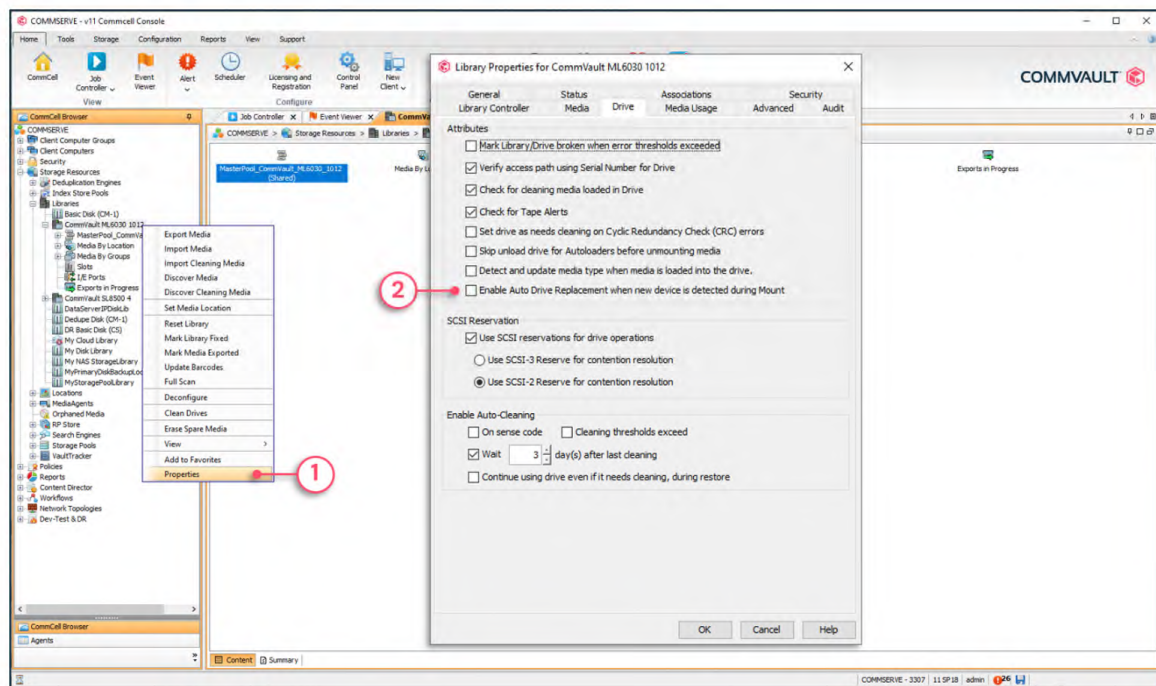


Replacing a defective drive has an impact on backup software. Tape drives are mapped to their serial number in Commvault software. Some drive models now allow the serial number of the drive to be configured, which alleviates any impact on the backup system. But if the drive is not replaced with the same serial number, it must be addressed in Commvault software.

1. Expand the master pool > right-click the **defective drive** > **Mark Drive Replaced**

After a few minutes, the system maps the drive to the new serial number and the drive becomes available.

Enable Automatic Drive Replacement

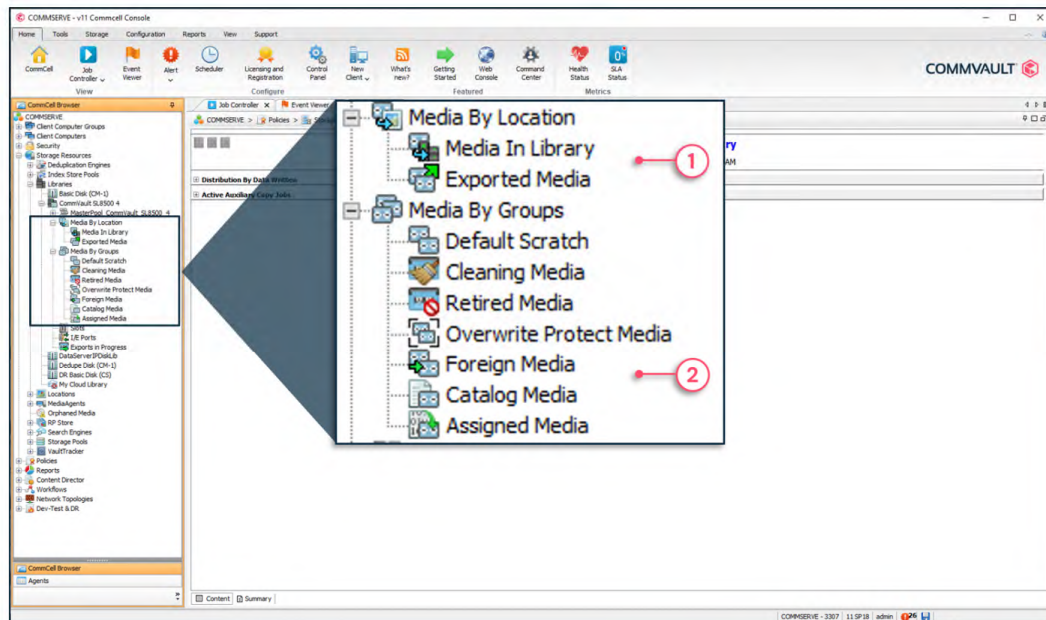


It is also possible to configure the system to attempt to map replaced drives automatically.

1. Check the box to enable Commvault to map the serial numbers of the replaced drives automatically

Tape Media Management

Tape Media Overview



1. Physical media management is tracked by its location. A tape can be inside the library or exported. Physical media actions take place within the library, such as:
 - Export tapes from the library
 - Import and discover tapes added to the library
 - Physically erase tapes
 - Load and unload tapes into drives
2. Logical management of media focuses on the media group and the state of tape. Tracking is managed within media groups. Logical actions include:
 - Load and unload tapes into drives
 - Move a tape to another media group
 - Delete tape or delete contents of a tape
 - View contents of a tape
 - Retire a tape

Media Icons

All tapes within a library display a status icon. Go to documentation.commvault.com and search for “media icons” to view the full list of icons.

Media Lifecycle

Commvault tracks the entire lifecycle of a tape. The logical lifecycle of a tape is different than the physical life. Logical management of tapes are managed with tapes in or outside the library.

Media Groups

Commvault software logically organizes all tapes into media groups. Each media group identifies all tapes associated with the group. Since all tapes are logically managed in the Commvault environment it is important to understand each media group and group management.

- Scratch Group – manages new or recycled media
- Cleaning – is reserved for library cleaning tapes
- Retired – manages tapes that exceed usage thresholds or are manually marked bad
- Overwrite Protect – manages tapes that are manually prevented from aging
- Foreign – manages tapes from a different CommCell® environment or a different backup vendor
- Catalog – manages tapes that are marked to be cataloged
- Assigned – manages tapes with active jobs currently retained and managed by storage policy copies

The diagram illustrates the Commvault media lifecycle, showing the progression of media from discovery to retirement. The lifecycle is organized into four main groups:

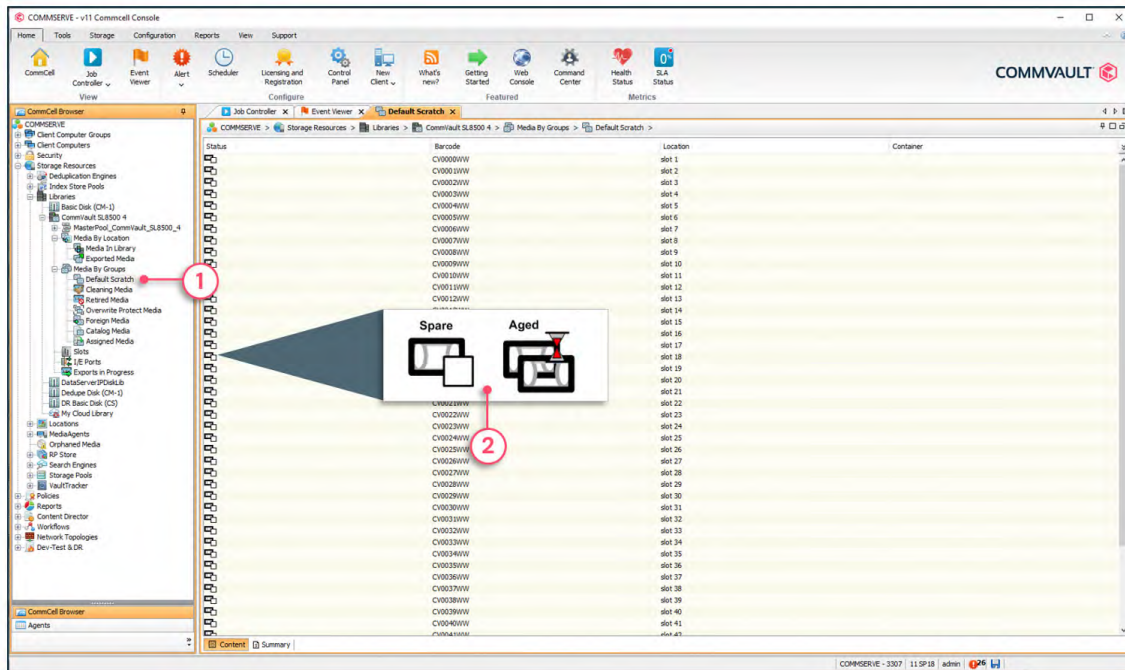
- Scratch Group:** Contains the initial states:
 - Undiscovered:** Represented by a tape icon with a question mark.
 - Spare:** Represented by a tape icon with a white square.
 - Aged:** Represented by a tape icon with a red hourglass.
- Assigned Media Group:** Contains states where media is assigned to a job:
 - Assigned (Active):** Represented by a tape icon with a blue square.
 - Assigned (Appendable):** Represented by a tape icon with a yellow square.
 - Assigned (Full):** Represented by a tape icon with a blue square.
 - Assigned (Bad):** Represented by a tape icon with a red border.
- Cleaning Media Group:** Contains the state:
 - Cleaning:** Represented by a tape icon with a blue square.
- Retired Media Group:** Contains the final state:
 - Retired:** Represented by a tape icon with a red 'X' over it.
- Overwrite Protect Media Group:** Contains states for media under overwrite protection:
 - Aged:** Represented by a tape icon with a red hourglass.
 - Assigned (Full):** Represented by a tape icon with a blue square.

The flow of the lifecycle is indicated by red arrows:

- Undiscovered** transitions to **Spare**.
- Spare** transitions to **Assigned (Active)**.
- Assigned (Active)** can transition to **Assigned (Appendable)**, **Assigned (Full)**, **Assigned (Bad)**, or **Cleaning**.
- Assigned (Appendable)** transitions to **Assigned (Full)**.
- Assigned (Full)** transitions to **Assigned (Bad)** or **Overwrite Protect Media Group**.
- Assigned (Bad)** transitions to **Retired**.
- Cleaning** transitions to **Retired**.
- Overwrite Protect Media Group** (both Aged and Assigned (Full)) transitions back to **Spare**.
- Retired** is the final state of the media lifecycle.

1. New tapes are placed in the library as undiscovered.
2. The library scans the slots, discovers the tape, and marks it as spare media in the Scratch group.
3. The tape is selected for a job where it becomes assigned media in the Active media group.
4. The tape is then appendable; so that additional jobs can be written to the media until the tape is marked full.
5. Once all jobs are aged from the tape, the tape is marked as aged and placed back into the Scratch group.
6. Tapes can also be marked as locked down where they will be placed in the Overwrite Protect group.
7. Once a tape exceeds its usage thresholds, it is marked bad and placed in the Retired Media group.
8. Cleaning media is designated as such and placed in the Cleaning Media group.

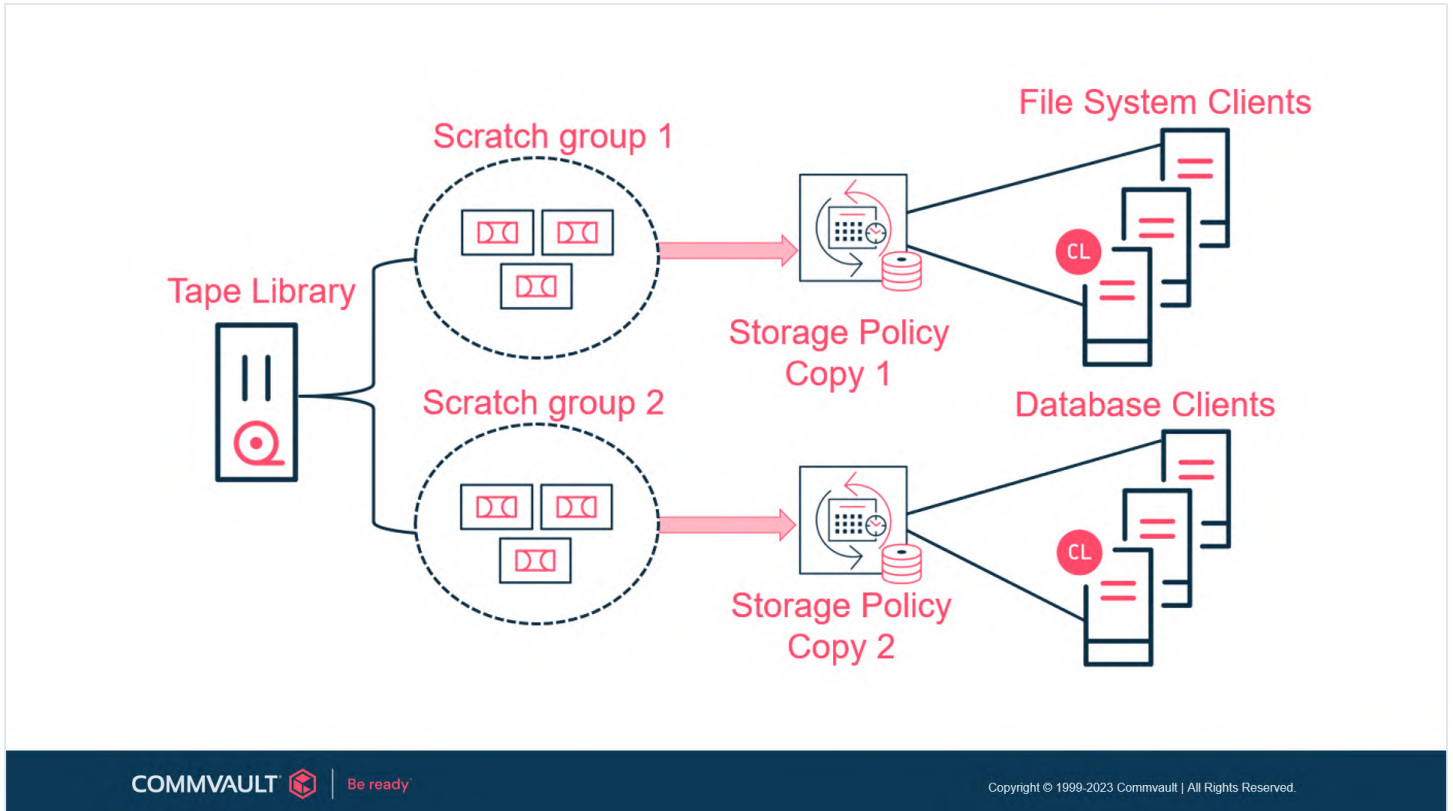
Scratch Groups



Scratch groups hold all new or recycled media. Multiple scratch groups can be used to define which tapes a job uses when it executes. When a job requires a spare tape, the tape is pulled from a defined scratch group. The storage policy copy data path is used to determine which scratch group the tape is selected from.

- These terms are used interchangeably: Scratch Pool, Scratch Group, Spare Media Group, Spare Media Pool
- All new and recycled tapes are placed in scratch groups
- Once a job is written to a tape it is moved out of the scratch group and into the assigned media group
- Multiple scratch groups can be created and assigned to storage policy copies

Manage Scratch Groups



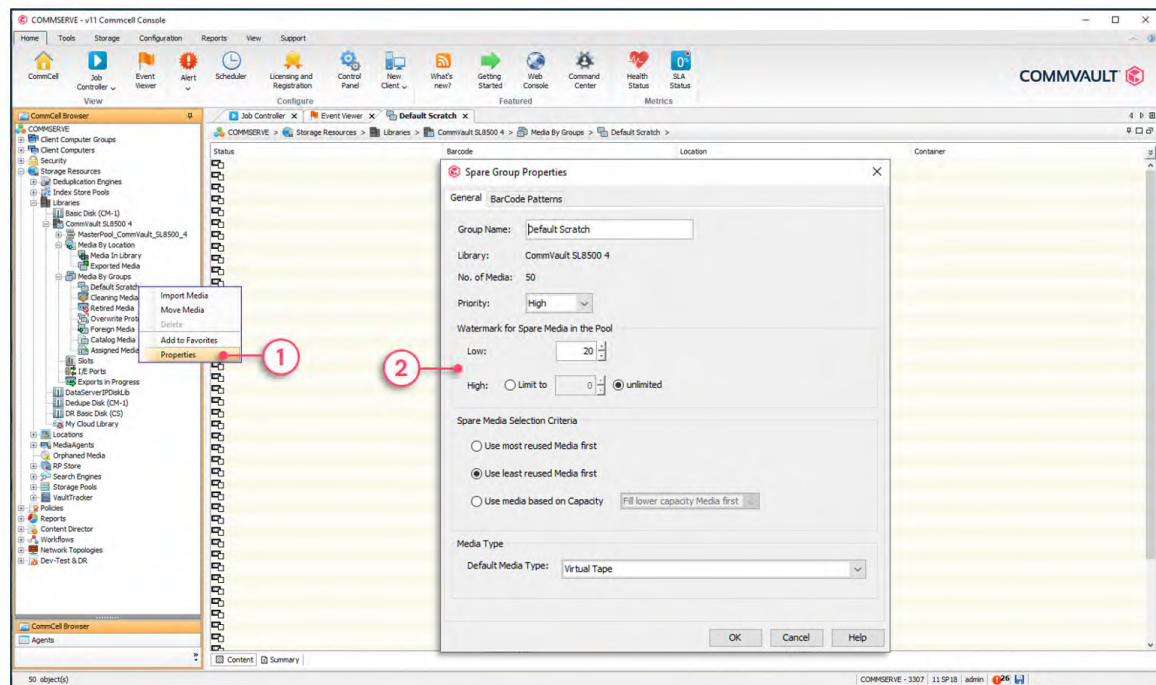
A default scratch group is created when the library is initially detected. Additional scratch groups can be created, and tapes can manually or automatically be assigned to the group. When jobs run, tapes are pulled from scratch groups and used for the job. Once data is written to the tape, it is moved out of the scratch group and into the assigned media pool.

When using multiple scratch groups, different storage policy copies define which scratch group tapes are pulled from. Also, tapes can be manually moved to other scratch groups or automatically assigned to different groups. Assigning tapes is based on barcode patterns or high watermark thresholds and scratch group priority. This allows different job types to be placed on specific media, which simplifies tape management outside the library.

Storage Policy Copy Relationship to Scratch Groups

A storage policy copy data path includes the selection of a scratch group. You can determine which tapes will be used to write certain jobs by configuring multiple scratch groups and assigning them to storage policy copies. Once data managed by the storage policy copy is written to the tape, the policy copy will own the tape until all jobs have aged, and no other storage policy copies can write data to the tape.

Scratch Group Properties



1. Right-click the **scratch group** > **Properties**

Set the low watermark to alert administrators when the library is running out of spare tapes; set the high watermark to limit the number of tapes placed in a scratch group

Use multiple scratch groups to ensure available media for backup operations

You are managing a CommCell environment and running backup jobs directly to tape each night. During the day, you run auxiliary copy jobs to tape to be sent off site. You are concerned that auxiliary copy jobs may use too many tapes and there will not be enough media for backup operations.

Solution: Create an additional scratch group and name it *Auxiliary Copies*. Set the priority to Medium. Configure a High Watermark in the Default Scratch group to be a greater number than the number of tapes required to perform nightly backups. Set the Default Scratch group priority to high.

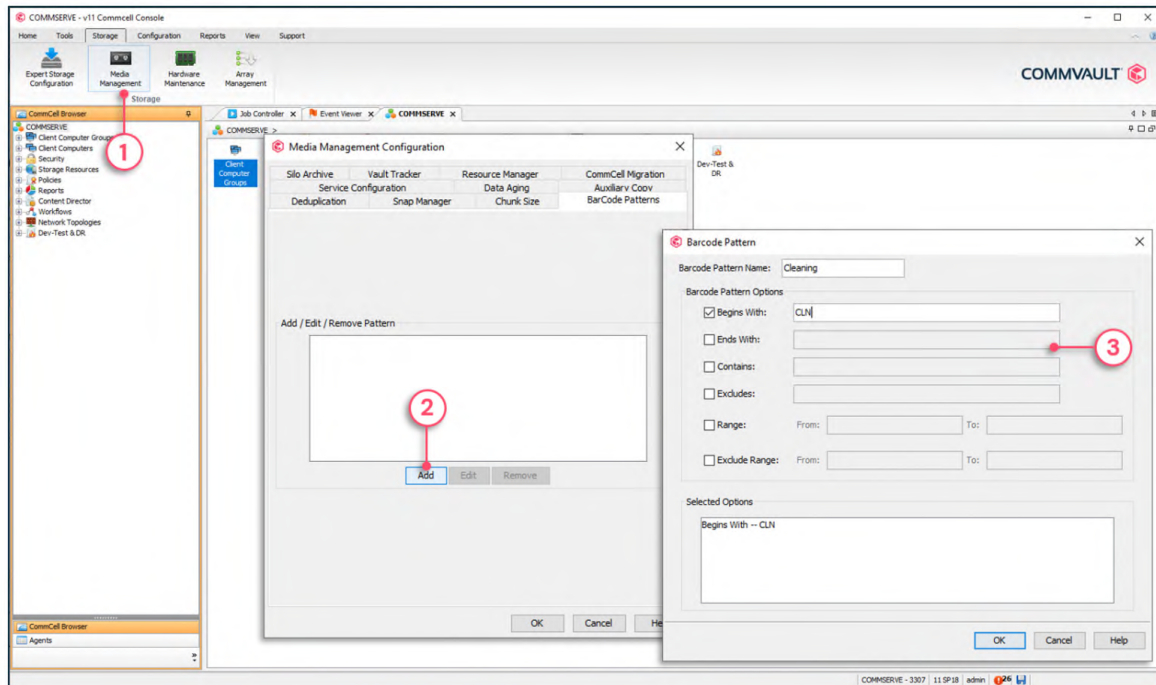
In the secondary copy of each storage policy, use the Scratch Pool drop-down box to assign the *Auxiliary Copies* scratch pool to the copy.

Cleaning Media Group

The Cleaning Media group manages all cleaning tapes for a library. Tape drives are cleaned based on drive counter usage tracked by Commvault software and/or sense codes reported from the library. Configure drive cleaning based on the library manufacturer's recommendations. Drive cleaning settings are configured in the library properties under the Drive tab.

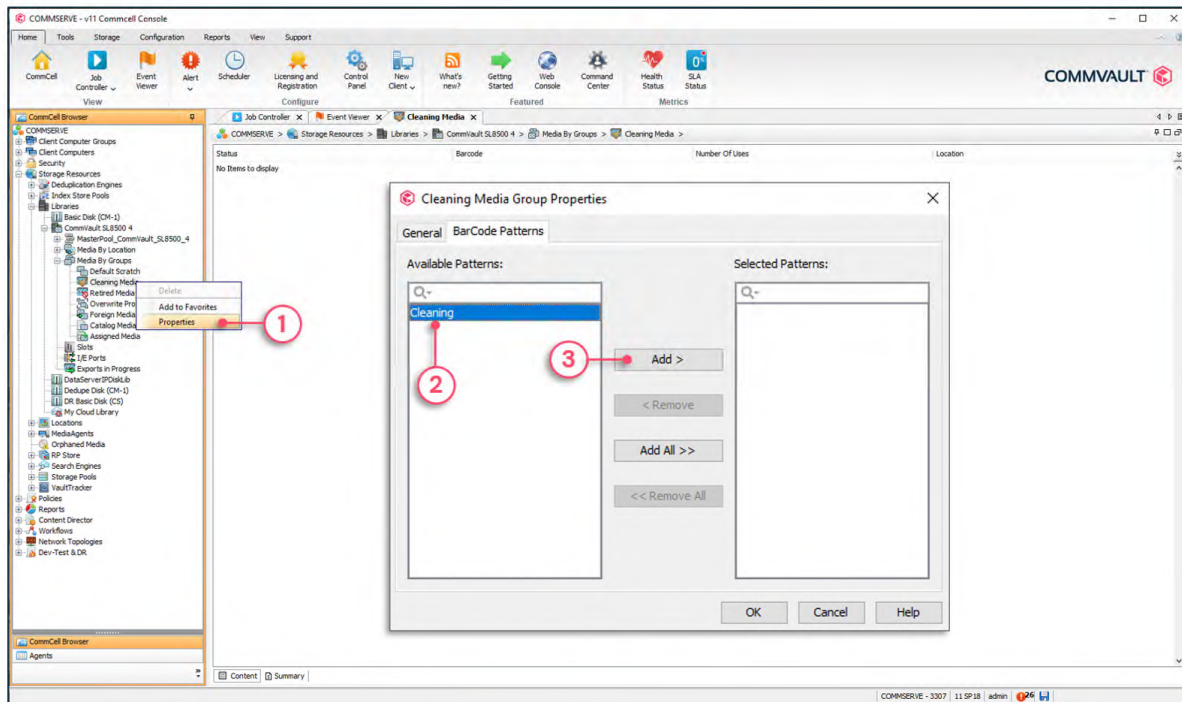
- Commvault software should automatically detect and move cleaning tapes to the cleaning media group when the tapes are discovered.
- If cleaning tapes are incorrectly identified and moved to a scratch pool, you can manually move the tapes or use custom barcode definitions to associate cleaning tapes with the cleaning media pool.
- Low watermarks can be defined to trigger events and optional alerts when the number of spare cleaning media reaches the low threshold.

Configure Barcode Patterns



1. *Storage* tab > **Media Management**
2. *Barcode Patterns* > **Add**
3. Barcode patterns can be defined based on barcode characters or ranges

Assign Barcode Patterns to a Cleaning Media Pool



1. Right-click the **cleaning group** > **Properties**
2. Select the pattern
3. Click **Add** to assign it

Retired Media Group

The Retired Media group is a holding area for all tapes that have exceeded tape error thresholds or are manually marked bad. Tapes in the Retired Media group will remain in the group until they are manually marked good or deleted. Any tapes in the Retired Media group will NOT be written to. If a tape marked as bad is in the Assigned Media group, it will NOT be moved to the Retired Media group until all jobs have aged from the tape.

- Only tapes that are not currently retaining job data are placed in the retired media group. If a tape is marked bad but is currently retaining data, it will still appear in the Assigned Media group. Once all jobs have aged from the tape it is moved to the Retired Media group.
- Tape counters are tracked for the life of a tape.
- By default, manufacturer-recommended thresholds are used for all tapes.
- Tapes can be moved out of the Retired Media group using the following methods:
 - o **Delete** – Deletes the existence of the tape from the CommServe server database. The tape can then be rediscovered and reused. The tape is treated as a brand-new tape and all counters are reset. If there are any existing aged jobs on the tape, they will not be recoverable.
 - o **Mark Media Good** – Is recommended if the tape has existing jobs that have aged but may still need to be retained. If this is the case after marking the tape good, move it to the Overwrite Protect Media group.

Tapes should be left in the Retired Media group until they are physically disposed of. This prevents a bad tape from accidentally being discovered and reused. If a bad tape is disposed of and is replaced with a new tape with the same barcode, delete the tape from the Retired Media group before putting the new tape in the library.

Tapes Incorrectly Marked Bad

Sometimes tapes can be incorrectly marked bad due to drive problems that result in tape errors. If there is a sudden increase in bad tapes this may be an indication of drive problems. However, do NOT discount the possibility that the tapes are bad. There have been situations where bulk orders of brand-new tapes are legitimately bad. If you do not know what manufacturing, delivery, or storage methods are being used; then it is critical to act on the side of caution.

Foreign Media Group

The Foreign Media group manages all media from different CommCell environments or tapes from a different backup vendor.

- Tapes from one CommCell environment cannot be directly restored into another. When a tape is loaded and the OML (On Media Label) is read, if the CommCell ID is different than the CommCell environment reading the tape, the tape is moved to the Foreign Media group.
- Commvault software will not write to tapes when the OML header is not recognized as a Commvault header, and the tape is moved to the Foreign Media group.

Overwrite Protect Media Group

The Overwrite Protect Media group logically locks down a tape, so it will NOT be written to or recycled. Tapes must be manually moved to the Overwrite Protect Media group and remain there indefinitely until they are moved out of the group.

- By default, an Overwrite Protect Media group is automatically created. Additional overwrite protect media groups can be added.
- Tapes are moved to the Overwrite Protect Media group using the following methods:
 - o For active tapes in the Assigned Media group – Right-click on the tape and select Prevent Reuse. The tape appears in the Assigned Media and the Overwrite Protect Media groups.
 - o For tapes in scratch groups – Right-click on the tape and select Move. For Media Group Type select Overwrite Protect Media group and then select the overwrite group.

Other Methods to Prevent Tapes from Being Overwritten

Moving a tape to the Overwrite Protect Media group is just one-way Commvault software can prevent data from being overwritten. Data can also be locked down at the job level.

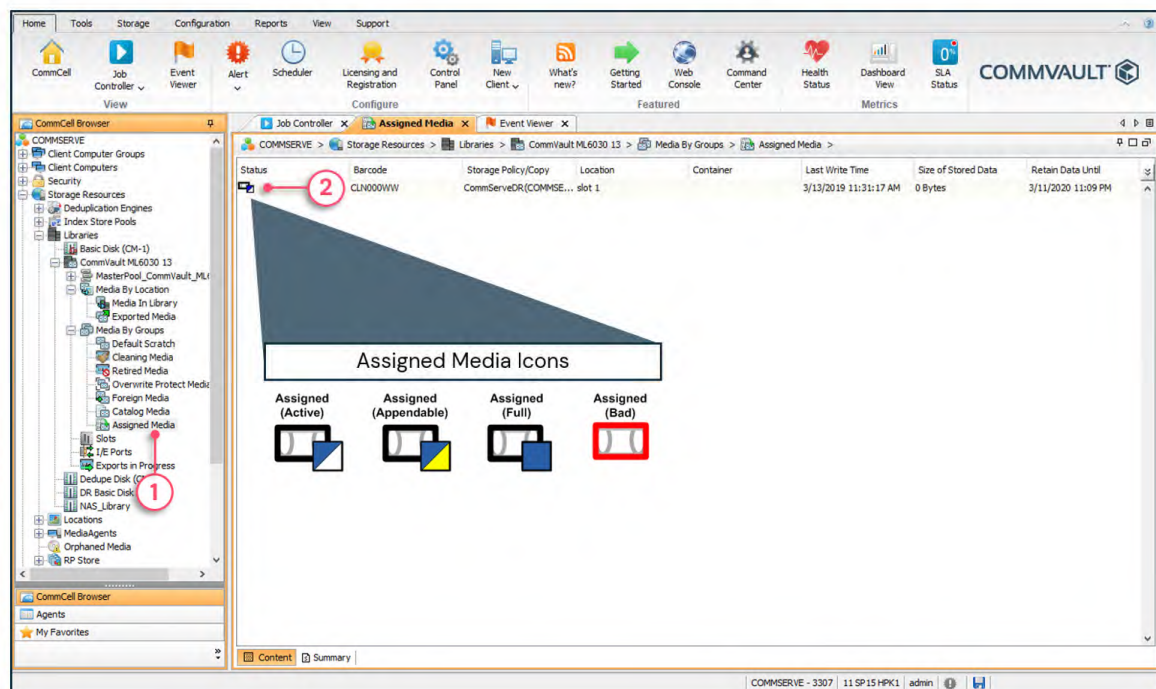
Consider a job that spanned multiple tapes. Manually moving tapes to the Overwrite Protect Media group requires you to know every tape the job was written to. The job and all tapes can also be locked down through the storage policy copy. In the job history of the policy copy, right-click on the job and select Retain Job. You are then prompted to select a date to hold the job, or infinitely retain the job.

Catalog Media Group

The Catalog Media group is used to hold all tapes that are actively being cataloged or are marked for catalog. A tape can be individually picked for catalog or multiple tapes can be picked and marked for catalog. When tapes are picked for catalog, they are moved to the Catalog Media group.

A catalog operation is used to catalog job metadata from a tape and enter the metadata back into the CommServe server database. You can perform this operation if the CommServe server database had to be restored to a point-in-time prior to the jobs on a tape finishing. This situation can result from a disaster, database corruption, or improper management of the CommServe server metadata backups.

Assigned Media Group



All tapes that are actively retaining data are in the Assigned Media group. Within a library, there is only one assigned media group. Tapes remain in the group until all jobs on the tape have exceeded retention and are marked as aged. During the data aging operation, the tape is then recycled back into a scratch pool.

- Tapes in the Assigned Media group cannot be deleted. Delete is considered a non-destructive operation.
- Delete Contents can be performed on a tape which is considered a destructive operation. To delete the contents of multiple tapes, use the Shift or Ctrl keys to select multiple media. Note that this recycles the tape, and the jobs are marked aged.

Assigned Media Relationship to Storage Policy Copies

When a spare tape is picked for a backup job, the tape becomes associated with the storage policy copy that is managing the job. Once the tape is associated with the policy copy, no jobs from other policy copies can be written to the tape. This is done intentionally to avoid mixed retention on media. Since each storage policy copy can have different retention configured, it is important to separate jobs based on policy copy ownership.

Since tapes are associated with storage policy copies it is important to properly configure and manage storage policies.

Consequences of having too many storage policies using tape media

Your environment has 25 storage policies. This results in at least 25 storage policy primary copies. If all primary copies are defining tape library data paths, then at least 25 tapes must be in the library to accommodate all potential jobs. If the storage policies also have secondary copies using a tape data path, then additional tapes must also be present to meet the media needs of the secondary copies.

In some cases, backup configurations such as Start New Media and Mark Media Full can complicate things more. It is important to understand your environment's needs and how Commvault software manages media.

Use global secondary copies to consolidate data from multiple storage policy secondary copies to the same tape sets.

Tape Media Relationship to a Storage Policy Copy

Assigned Media

commserve > Storage Resources > Libraries > LTO4-4-50 Shared TAPE (CS OR SP) > Media By Groups > Assigned Media

Status	Barcode	Storage Policy/Copy	Location	Last Write Time	Size of
	OR0001WW	Database (EX OR SP) / Local Tape	slot 17	2011/10/12 16:41:23	721.00 MB
	OR0002WW	spool test / sync	slot 18	2011/11/29 11:47:55	293.00 MB
	OR0003WW	Dedupe File System(CS EX) / Offsit...	slot 19	2011/12/06 13:12:40	1.28 GB
	OR0004WW	Backup to Tape / Primary	CommVault VirtualDrive_3	2011/12/13 20:57:05	2.00 MB
	OR0005WW	Backup to Tape / Primary	CommVault VirtualDrive_4	2011/12/13 20:59:16	9.00 MB

Content

Summary

1

COMMVAULT

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- 1. Media in Assigned Media group are associated with a specific Storage Policy Copy

Tape Media Operations

- Exporting media
- Moving media
- Deleting media
- Deleting content
- Erasing media
- Verifying media
- Marking media full / bad / good
- Viewing contents
- Using media refresh
- Using tape-to-tape copy

Managing tapes in different media groups provide the Commvault administrator with greater flexibility and simplifies the management of media. Actions can be performed on tapes in any media group.

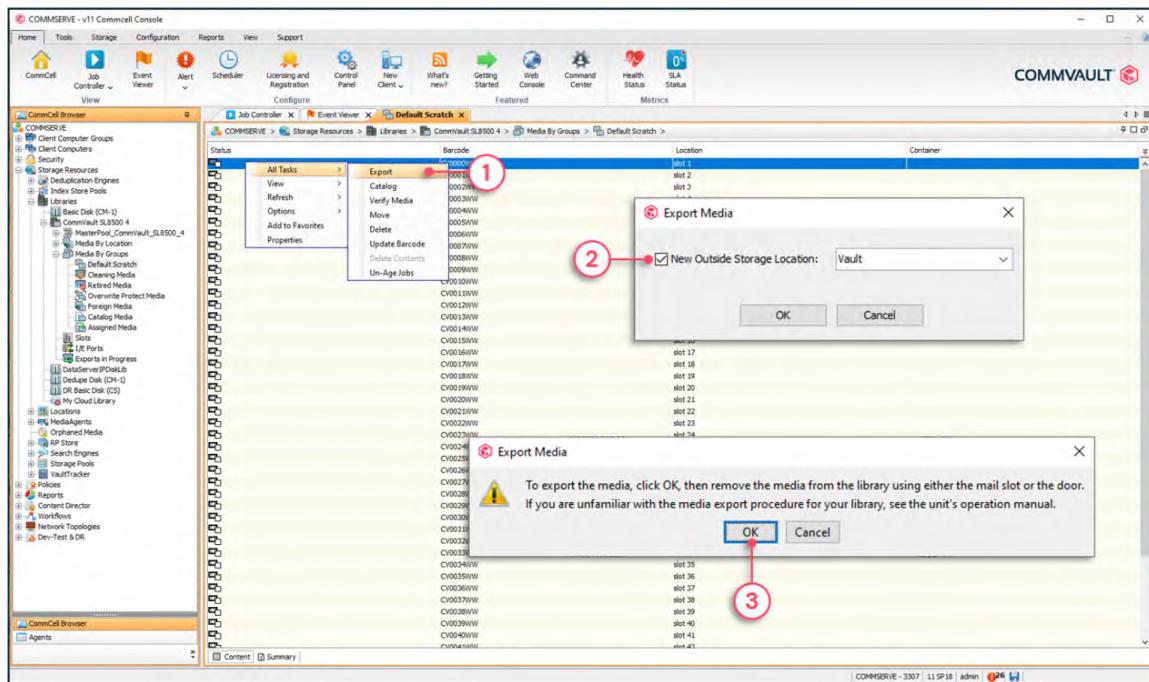
Export Media

Exporting tapes is a physical operation that sends commands to the library to eject tapes. There are several methods to export tapes from a library.

- Tape export – export individual tapes
- Library export – bulk export of multiple tapes
- VaultTracker export – automatic export of tapes based on criteria

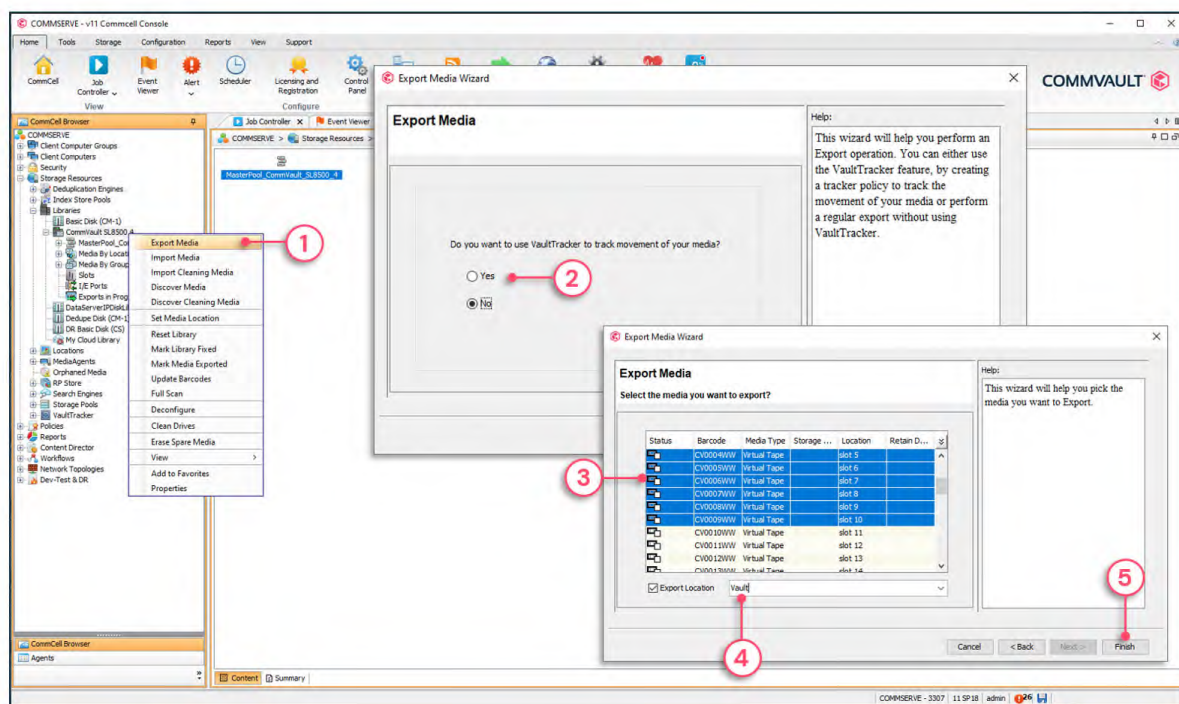
To view the progress of export operations, use the 'Exports in Progress' view in the library tree. To view tapes in the import/export slots, use the 'I/E Ports' view in the library tree.

Individual Tape Export



1. Right-click on tape > **All Tasks > Export**
2. Optionally select the location to track the tape
3. Pop-up message instructing administrator to remove tape from library's export location

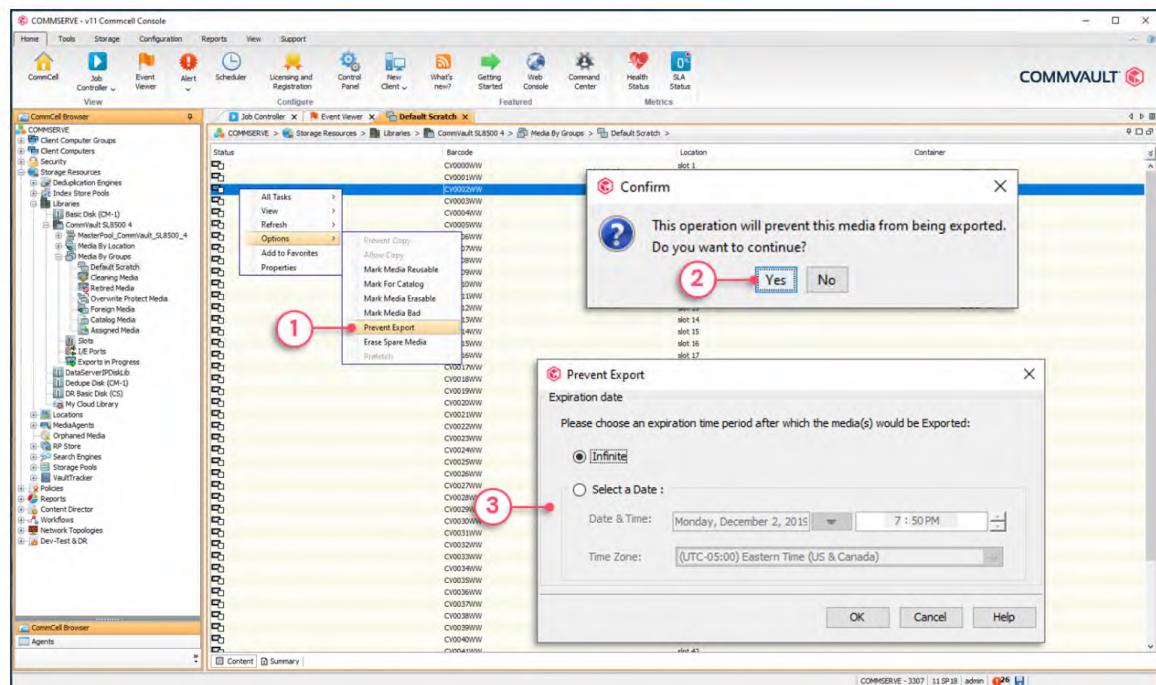
Bulk Library Export and VaultTracker® Export



Multiple tapes can be selected together to perform a bulk export. A bulk export moves tapes until the import/export slots are full. Once tapes are removed from the slots, the export operation continues until all tapes have been exported.

1. Right-click the **library > Export Media**
2. Optionally use VaultTracker to manage and track tape export
3. Select the tapes to export
4. Select an export location from the list
5. Click **Finish**

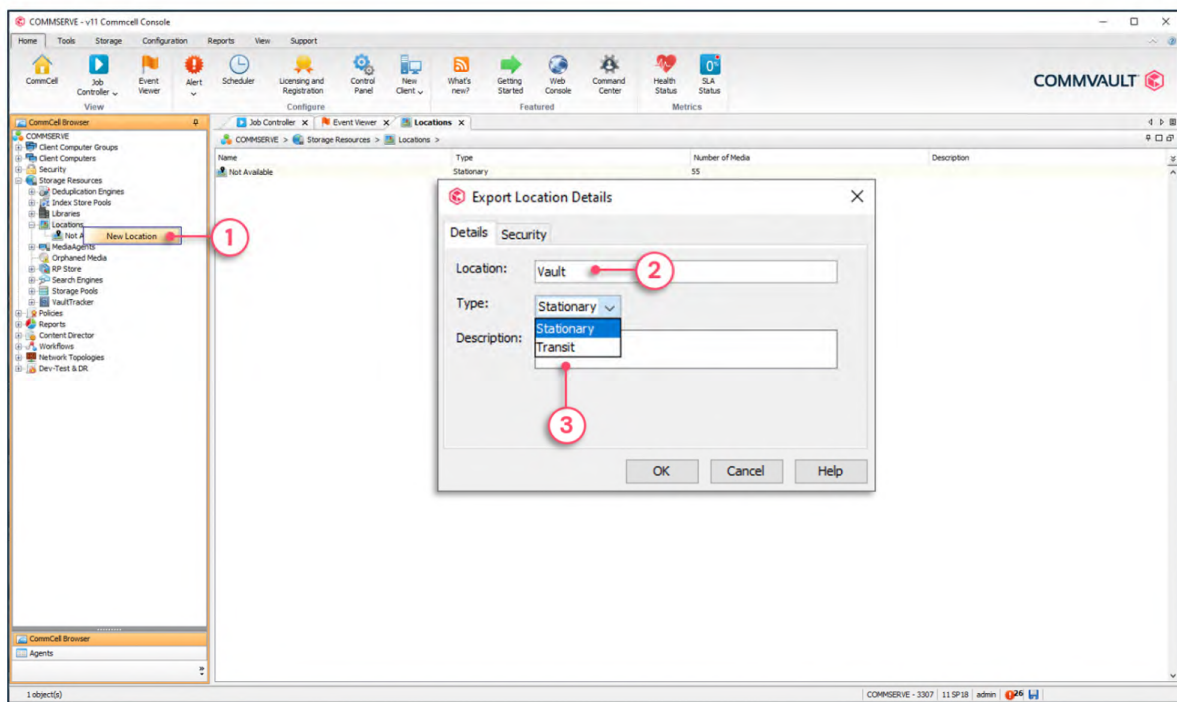
Prevent Tape Export



At times a tape must remain inside the library. To prevent a tape from being exported indefinitely or until a specific date, use the Prevent Export option. The Prevent Export option is available within any media group.

1. Right-click **tape** > **Options** > **Prevent Export**
2. Click **Yes** to confirm
3. Expiration dates can be defined for how long the tape cannot be exported

Allow Export

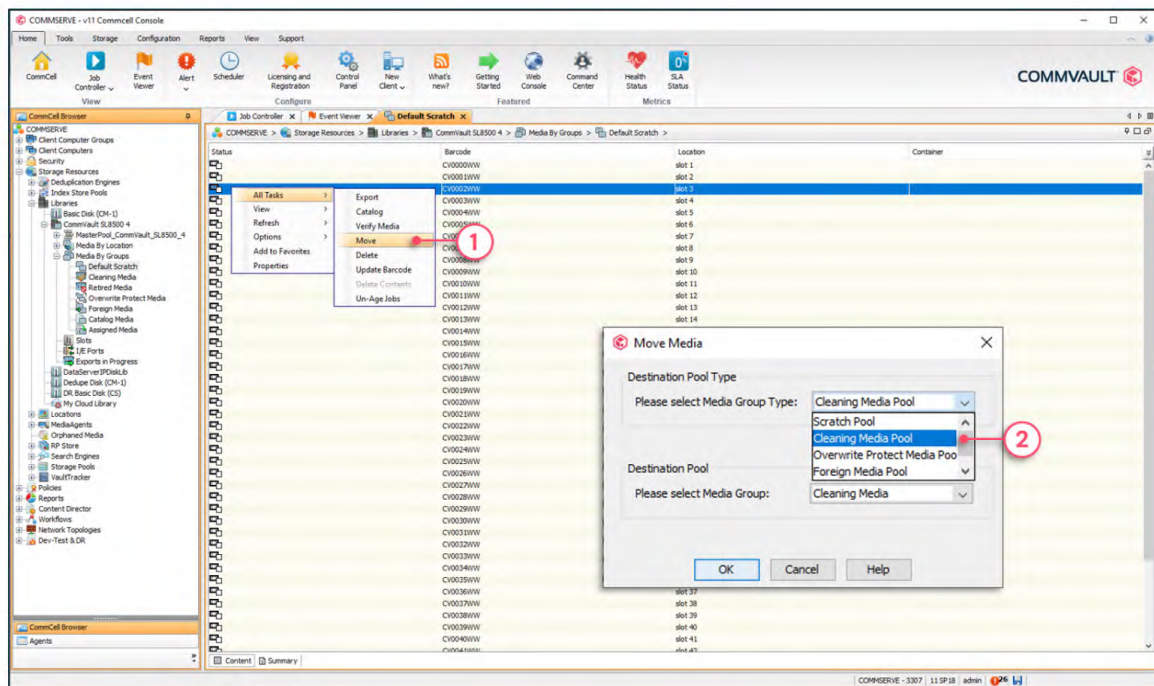


A tape that has been previously marked to prevent export can be set by the administrator to allow export. This operation overrides the previous setting specified during the 'Prevent Export' operation.

When tapes are exported out of the library, they are logically associated with an export location. There are two types of locations that can be defined: Stationary or Transit.

1. Right-click **Locations > New Location**
2. Enter a name for the location and then choose the type

Logically Move Tapes



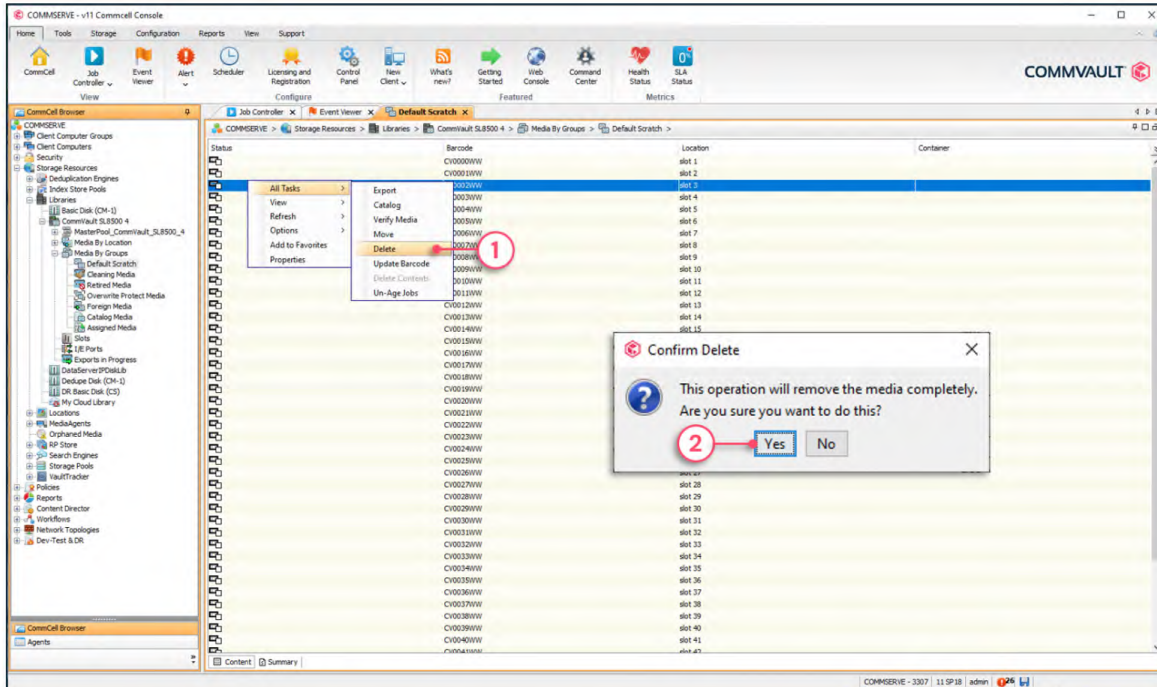
Moving tapes between groups is a logical operation that provides the administrator with a great deal of flexibility for media management. No physical media movement will be conducted in the library.

- Tapes can be moved between Scratch, Cleaning, Overwrite Protect, Foreign and Catalog media groups.
- Tapes can be moved to the Retired Media group by manually marking the tape bad. Tapes can be moved out of the retired media pool by using the option Mark Media Good or by deleting and rediscovering the tape.
- Tapes moved to the Overwrite Protect Media group will remain there indefinitely until they are manually moved out of the group.
- Tapes cannot be manually moved to the Assigned Media group.
- Tapes can logically be moved to the Overwrite Protect Media group by selecting Prevent Reuse. The tape will exist in both the Assigned Media group and the Overwrite Protect Media group until all jobs age from the tape, where it will then only exist in the overwrite protect media group.
- When a tape is cataloged or marked for catalog, it logically appears in the Catalog Media group.
- There are two methods for using the Move operation:
 - Specific tapes can be moved between media groups.
 - Groups of tapes can be moved between scratch pools.

Move Tapes

1. Right-click on **tape** > **All Tasks** > **Move**
2. Select the group type and group to move the tape and then click **OK**

Delete Tapes



A Delete operation is a logical operation that deletes the existence of the tape from the CommCell environment and CommServe server database. The delete operation does not physically erase any data and the tape does not need to be in the library when it is deleted.

This is a non-destructive operation which means a tape can only be deleted if no active jobs are currently retained on the tape. Thus, the Delete option is not available for tapes in the Assigned Media group.

When a tape is deleted, the tape records are removed from the CommServe server database, and that data is not recoverable. If data on a deleted tape needs to be recovered, the Media Explorer tool is used to catalog and recover data.

If a tape has been permanently lost or destroyed and a replacement tape with the same barcode label will replace the original, delete the original tape before adding the replacement tape to the library. Duplicate barcode patterns are not allowed and will result in an error.

The Delete operation is useful in the following situations:

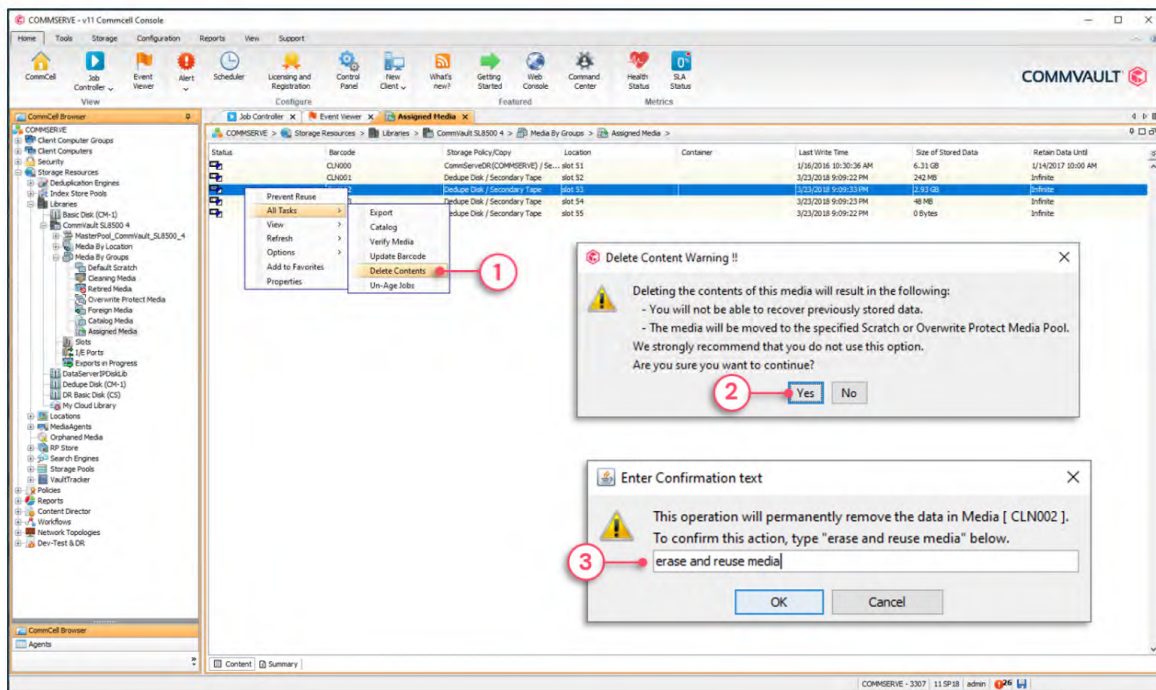
- A tape has been lost or destroyed.
- A tape was marked bad and has been physically removed from the environment. Note that until the tape is physically disposed, it is recommended NOT to delete the tape. A tape marked bad will not be reused, but a deleted tape that has been rediscovered will be treated as a new tape.
- Duplicate barcodes exist in a library where an old tape has been replaced by a new tape. Note the same caveats apply in this situation as the previous one.
- A tape was improperly marked bad. This can happen when a drive causes a tape to be marked bad.

Delete a Tape

This cannot be performed on active tapes.

1. Right-click **tape** > **All Tasks** > **Delete**
2. Click **Yes** to confirm the operation

Delete Tape Contents



Delete Contents is a logical operation that automatically marks all jobs on a tape as aged and moved to a scratch group. The data can still be recovered to the point where the tape is mounted and the OML header is overwritten.

Delete Contents is most often when there is not enough spare media to run scheduled jobs. This typically happens when storage policies are improperly configured, or retention expectations are unrealistic. If an administrator frequently uses the Delete Contents option to free-up tapes for jobs, consider readdressing environment configurations or purchasing more tapes. Manual operations such as Delete Contents can accidentally delete critical data.

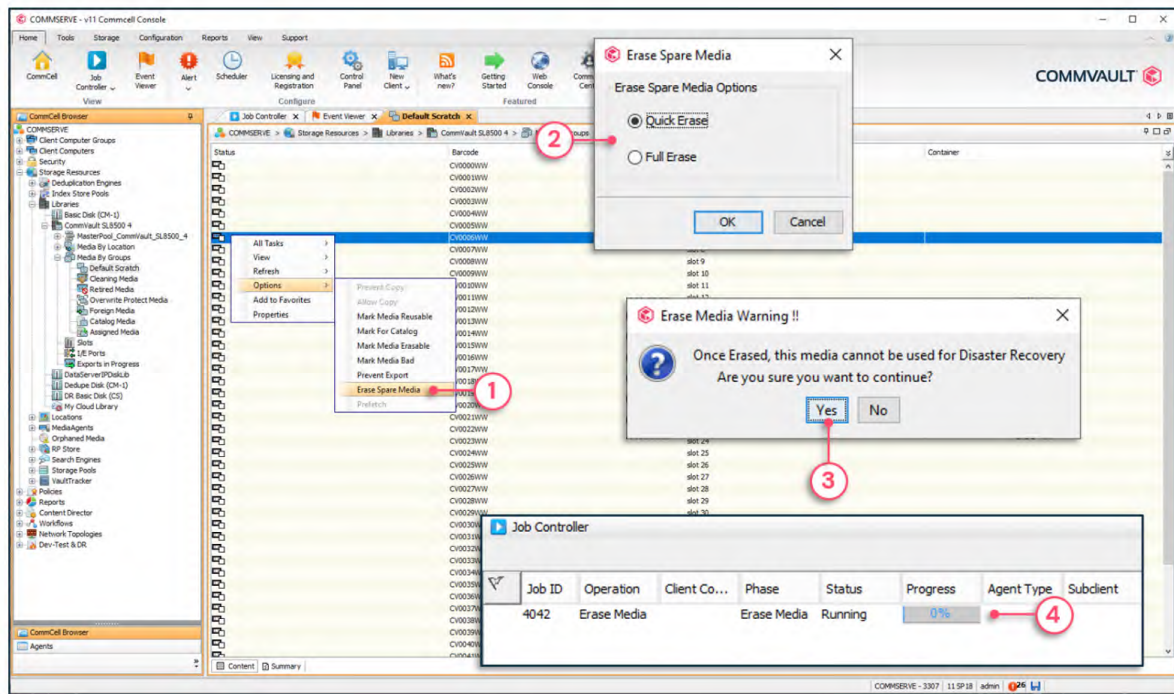
1. **Assigned Media Group** tab > right-click a **tape** > **All Tasks** > **Delete Contents**
2. Review the delete contents warning message and click **Yes** to continue
3. Additional confirmation is required. You must enter **erase and reuse media** in the text box

Delete a tape to allow for a full recovery operation when the tape is missing

An interesting use for the Delete Contents option is when a tape containing an incremental backup job is bad or missing at the time of a recovery operation. In the case of a restore, the full backup and all incremental backup jobs will need to be available for the restore to complete.

Commvault software prompts the administrator to put the tape in the library. If it is known that the tape is permanently unavailable, use the Delete Contents option to erase the records of the jobs. When the restore runs, the jobs on the unavailable tape are skipped and the restore operation completes successfully.

Erase Media



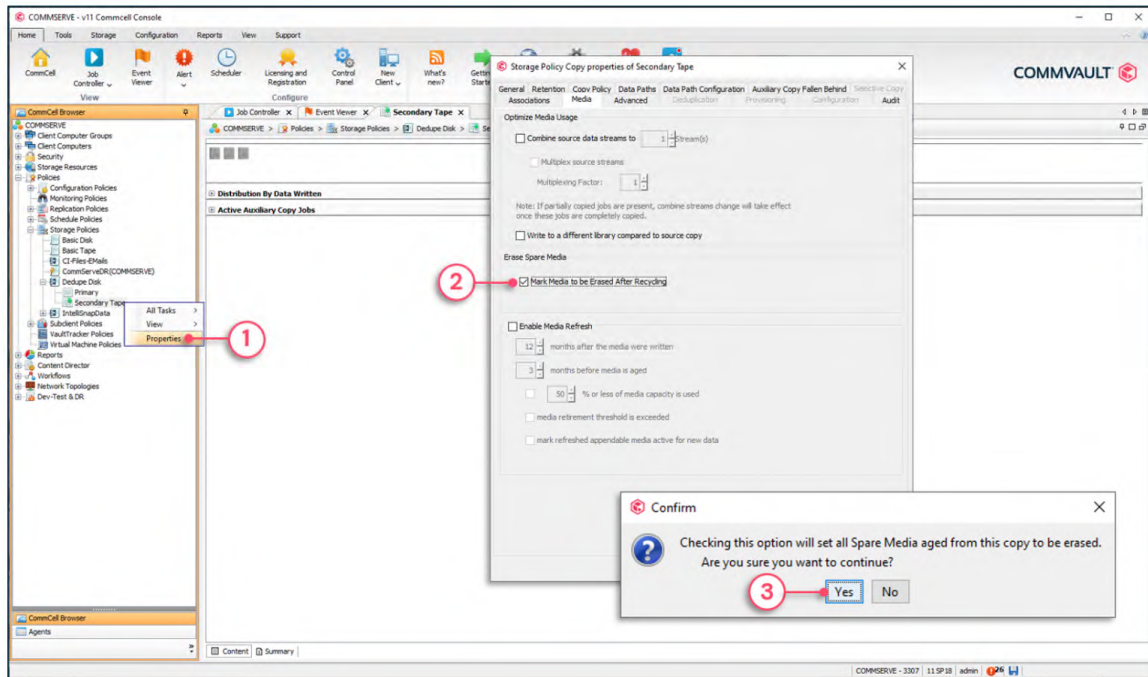
Erase Media is a physical operation that mounts the tape and overwrites the OML header. Once the header is overwritten data cannot be recovered using any method Commvault software provides. This is considered a destructive operation, so it cannot be performed on any tapes where jobs are actively being retained. The option to erase media is available in all logical groups except the Assigned Media group.

Manual Erase Operations

A tape can be erased on demand by right-clicking on the tape and selecting Erase Media. This action then loads the tape in a drive and overwrites the header physically preventing data to be recovered through Commvault software.

1. Right-click **tape** > **Options** > **Erase Spare Media**
2. Select **Quick Erase** or **Full Erase**
3. Click **Yes** to confirm the operation
4. Erase media operations will appear in the Job Controller

Automatic Erase Operations

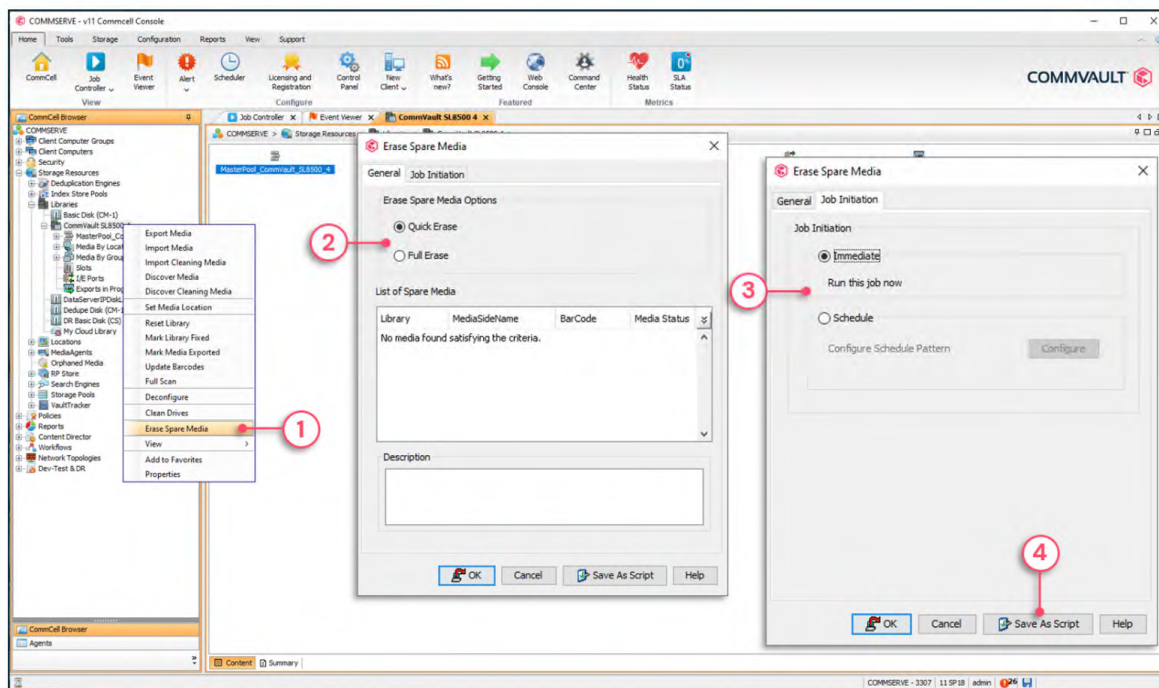


When tapes are recycled, they can automatically be marked to be erased. You can mark all tapes managed by the storage policy to be erased once all jobs have aged. An erase media operation must be scheduled for the library, which will physically load each marked tape and overwrite the OML header. The erase media operation mounts the tape and writes a new OML header to the tape. This makes the data completely unrecoverable, although the data on the tape is not actually erased.

Note: The only guarantee to prevent data from being recovered is to physically destroy the media. Even with the erase media and data encryption, there is no 100% guarantee that data cannot be recovered.

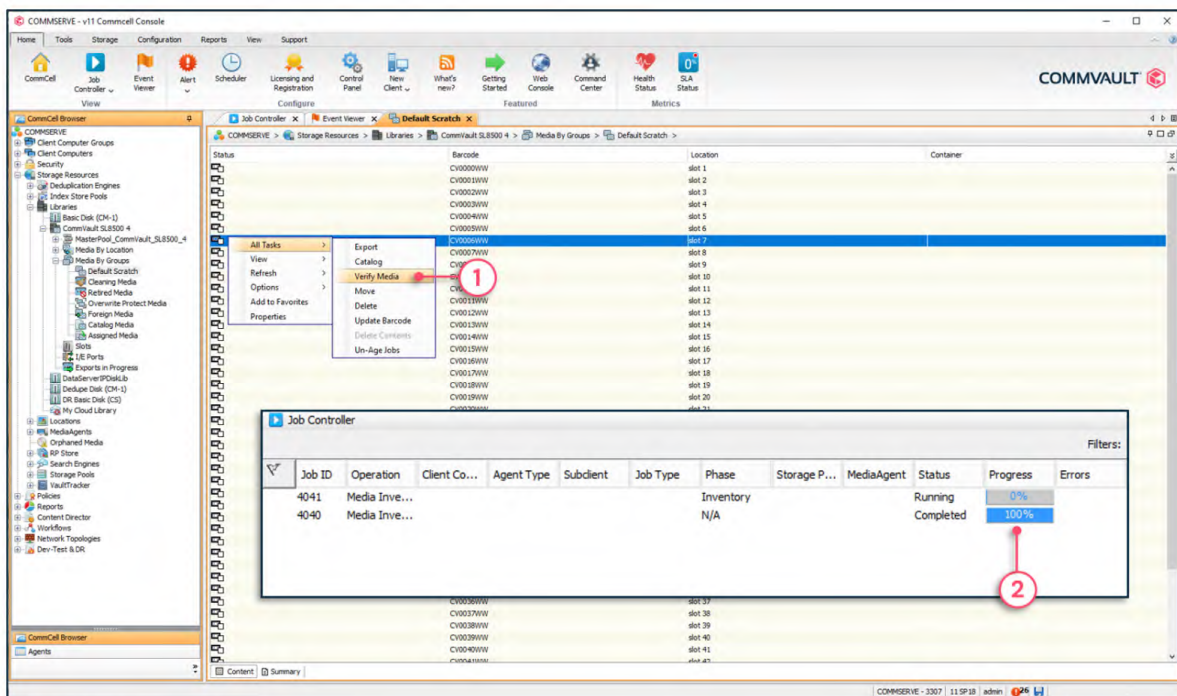
1. Right-click a **storage policy copy** > **Properties**
2. From the *Media* tab select 'Mark Media to be Erased After Recycling'
3. Click **Yes** to confirm

Schedule Erase Spare Media Operations



1. Right-click **Library > Erase Spare Media**
2. Set the erase operation to be quick (overwrite OML header) or full (overwrite entire tape)
3. The erase operation can be run immediately or scheduled for later
4. Alternatively, click **Save As Script** to save as a script

Verify Media



A Media verification operation is conducted to physically verify the barcode label, verify that the OML header matches, and that the tape belongs to the CommCell environment. Since this is a physical operation, the tape is mounted in a drive and a job appears in the job controller. When the job is initiated, the operation type is Media Inventory.

The Media verification operation is commonly used to:

- Ensure that the barcode label and the OML header properly match
- Identify damaged tapes
- Identify tapes that belong to a different CommCell environment
- Identify tapes that have an OML header written by different backup software

Verification Steps

1. Right-click a **tape > All Tasks > Verify Media**
2. Verification is a physical operation which will load the tape to conduct the verification operation

Mark Media Full

A tape in the assigned media pool can manually be marked full to prevent additional data from writing to the tape.

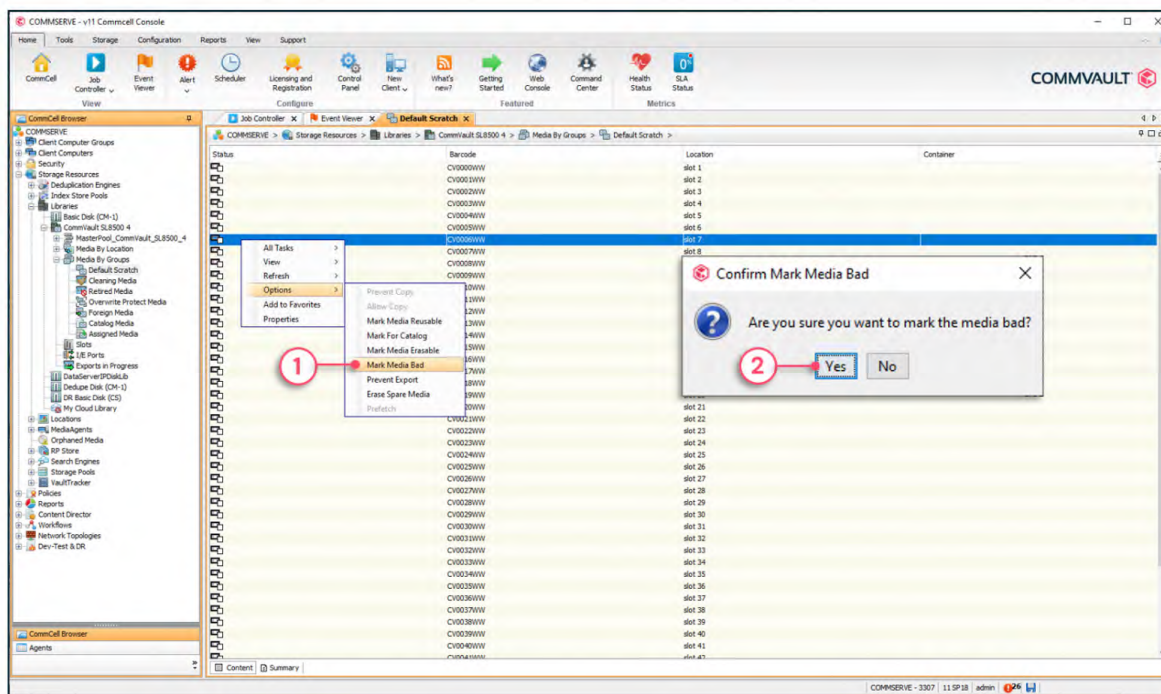
Mark media full is most used for the following situations:

- Jobs on the tape should be isolated for preservation purposes.
- The tape is marked full so the jobs on the tape age based on existing job data only (not additional jobs that may be written). This will allow the tape to potentially recycle sooner.
- Data on a tape needs to be refreshed to new media. Only tapes that have been marked full will be eligible for media refresh operations.

Various methods for marking tapes full

The Mark Media Full action presents a manual method to mark a tape full. Tapes can also be set to be automatically marked full. This is done through the Advanced backup or auxiliary copy options. This automatically marks the tape full after the job completes. Using the mark media full option is best used when jobs need to be isolated on tapes, such as when extended retention rules are applied.

Mark Media Bad



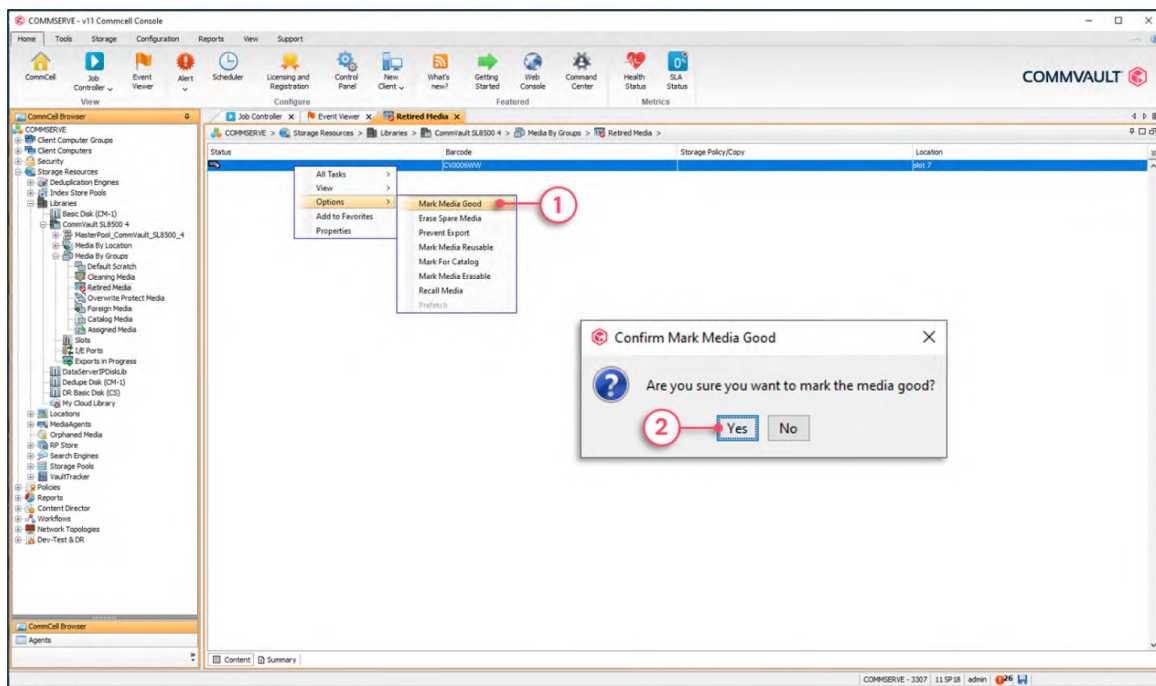
A tape can be marked bad from any group within the 'Media by Groups' tree. Once a tape is marked bad it cannot be used for future backup operations.

1. Right-click the **tape** > **Options** > **Mark Media Bad**
2. Click **Yes** to confirm the operation

Tape State	Description
Bad tape with jobs currently being retained	The tape is marked bad and no future jobs are written to the media. The tape remains in the assigned media group displaying a box icon with a red outline.
Bad tape with no jobs currently being retained (no jobs on media or all jobs exceeded retention)	The tape is marked bad and no future jobs are written to the media. The tape is moved into the Retired Media group.

It is important to note that depending on the state of a tape when it is marked bad, it appears differently within the CommCell console.

Fix a Tape that was Incorrectly Marked 'Bad'



1. *Retired Media* tab > right-click the **tape** > **Options** > **Mark Media Good**
2. Click **Yes** to confirm the operation

View Contents

The screenshot shows the Commvault console interface. On the left is the 'CommCell Browser' tree. The main pane shows 'Assigned Media' with a table of tapes. A right-click context menu is open over a tape, and the 'View' > 'View Contents' path is highlighted. A dialog box titled 'Data on Media [CLN001]' is displayed, showing the following information:

Size on Media: 245.01 MB
Total Data Written: 245.01 MB
Required by Auxiliary Copy: No
Retain Data Until: Infinite

Job ID	Status	Client	iDataAgent	Instance	Backup Set	Subclient	Backup Lev
3021	Completed	SHAREPOINT	SharePoint ...	defaultinsta...	Databases	default	Full
3021	Completed	SHAREPOINT	SharePoint ...	defaultinsta...	Databases	default	Full
3022	Completed	ORACLEXE	Oracle Data...	XE	default	default	Full
3022	Completed	ORACLEXE	Oracle Data...	XE	default	default	Full
3022	Completed	ORACLEXE	Oracle Data...	XE	default	default	Full
7123	Completed	SHAREPOINT	SharePoint ...	defaultinsta...	Documents	default	Full
7123	Completed	SHAREPOINT	SharePoint ...	defaultinsta...	Documents	default	Full

Job contents and details can be viewed for any tape any tape that has active or aged jobs. The view contents option only appears if jobs are on the tape.

1. Right-click **tape** > **View** > **View Contents**
2. Jobs on the tape will be listed. Jobs in black font are active and jobs in gray font are aged

Determine Why Tapes are not Recycling

One of the most common issues that may arise regarding media management is when tapes are not properly recycling. Using the View Contents of a tape lets the administrator view which jobs are causing the tape not to recycle. This information can then be used to track down problems that are causing the jobs to remain active within the environment.

Another method to assist in predicting when tapes will recycle is using the Data Retention Forecast and Compliance report. This report lists all tapes, their expected aging date, and specific reasons why the tapes have not aged. Each of the reasons are hyperlinks to the Commvault documentation site which will provide more details on the explanations.

Media Refresh

The Media Refresh tool writes job data from existing tapes to a new destination tape. Tapes can be manually selected, or automatic refresh operations can be triggered based on predefined criteria. Media refresh is a physical operation and requires a source and destination drive. It is best to run media refresh operations when backup and auxiliary copy jobs are not running. If multiple storage policy copies require refresh operations, separate schedules must be configured for each policy.

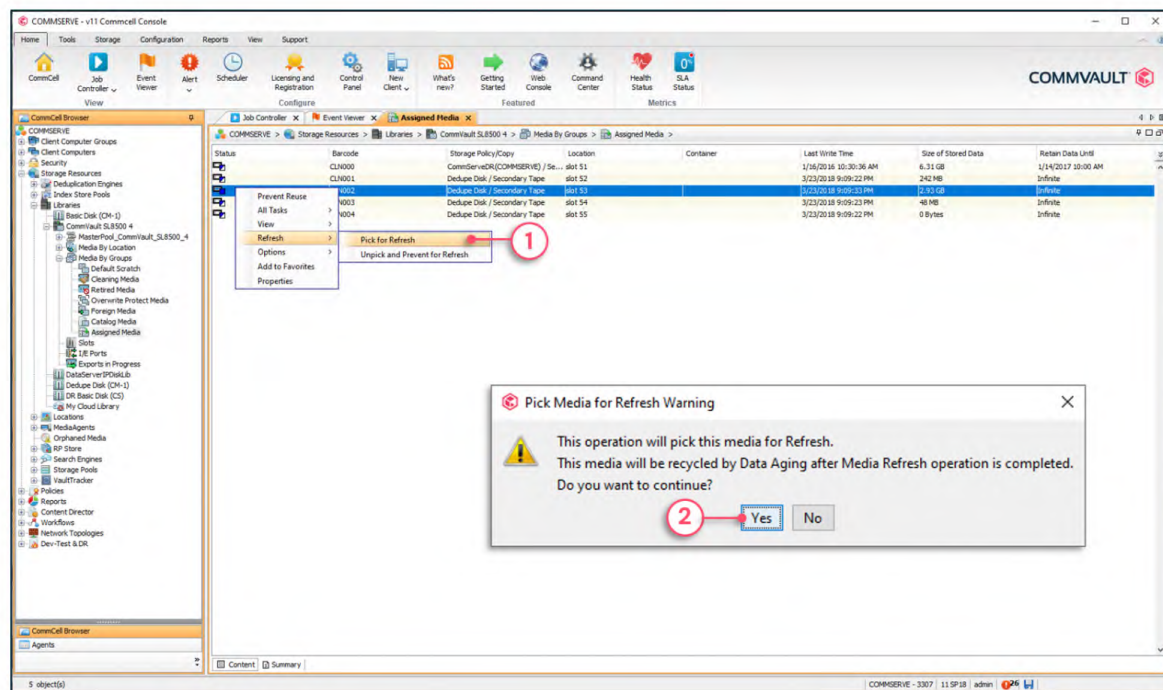
Media refresh is commonly used for:

- When specific jobs are currently being retained on the tape, but all other jobs have aged. This situation results in inefficient use of tape resources since a tape does not recycle until all jobs are marked aged. The media refresh operation copies all active jobs to a new tape and the old tapes are recycled and moved back into a scratch pool.
- When upgrading to new tape libraries, the media refresh can copy existing jobs from an old tape format to a new format such as LTO4 to LTO5. For this method to be used, the tape library must be capable of reading the old tapes.
- When data is required to be periodically rewritten to new tapes to better ensure the integrity of the data. Data integrity on tape is especially important when long retention periods are required.

Media refresh must be enabled for the storage policy copy, the media must be marked as full, and you must define criteria through a storage policy copy. Thresholds can be set for:

- Number of months after the media was written (best for critical data and data with long retention needs)
- Number of months before the media is aged
 - o Tape usage drop below a specified percentage – can reduce the total amount of data written to new media and maximize tape efficiency by writing jobs on tapes not fully utilized to new tapes, so old tapes can recycle.
- Media retirement threshold is exceeded

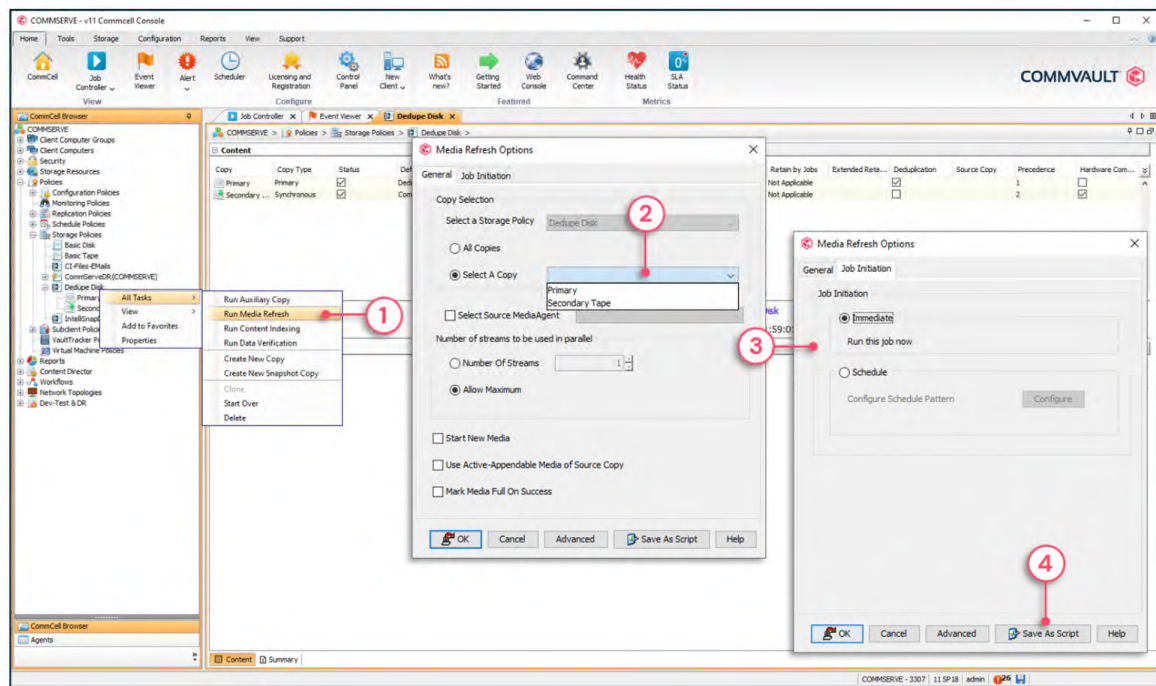
Manual Selection for Refresh



Manually selecting tapes for refresh is useful in situations where tapes need to be recycled for new jobs, and only a few jobs on tape have not aged.

1. Right-click a **tape in the Assigned Media group** > **Refresh** > **Pick for Refresh**
2. Click **Yes** to confirm

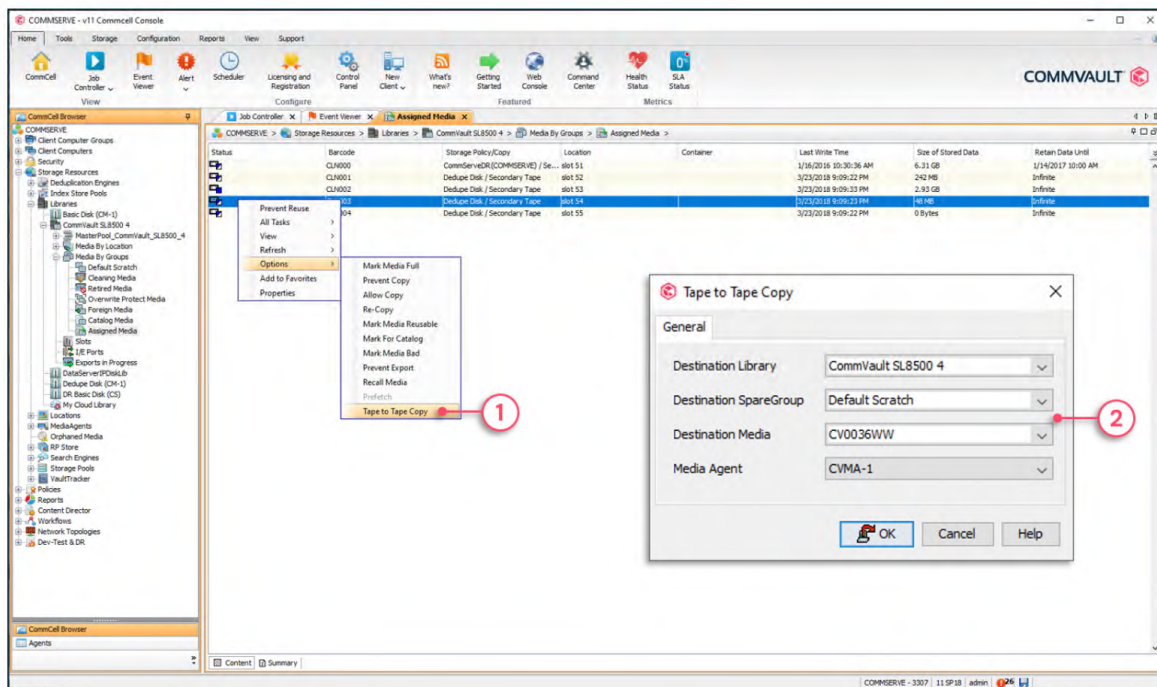
Run a Media Refresh Job



The media refresh feature follows the rule that a storage policy copy manages all data associated with it. This means that the scheduling for refresh operations is conducted at the storage policy level. The media refresh schedule is based on the auxiliary copy schedule, so the same options found in auxiliary copies are available for media refresh.

1. Right-click on the **storage policy** > **All Tasks** > **Run Media Refresh**
2. Configure media refresh job options
3. The media refresh job can be run immediately or scheduled for a later time
4. Alternatively, click **Save As Script** to save as a script

Tape to Tape Copy



The Tape to Tape Copy is a Windows-based option that manually copies the content of a tape onto a new tape from the scratch media pool. This is useful when a tape is suspected to be defective you want to copy it onto a new tape. The advantage of using the Tape to Tape Copy option over a traditional media refresh is that there is you do not need to enable media refresh setting on the storage policy copy.

1. *Assigned Media* tab > right-click the **media** > **Options** > **Tape to Tape Copy**
2. Select the MediaAgent, spare tape pool, and the media

- Locations
- Containers
- Virtual mail slots
- VaultTracker® policies
- VaultTracker® reports

VaultTracker is used to track media that are removed from a library and stored in offsite locations. This technology identifies media associated with storage policy copies. VaultTracker uses the retention and media management settings to identify when tapes are required for export or due back from export location.

VaultTracker provides the following capabilities:

- Identify media that must be sent off-site for storage or brought back from off-site locations
- Automatically move the media in sequence in the library and provides a pick-up list for the operators
- Use virtual slots for easier removal of media from a library with a limited number of I/E ports
- Identify and track media during transit
- Track record and track containers when the media is stored and moved using containers
- Record and track the movement of non-CommCell or foreign media

VaultTracker jobs are initiated in three ways:

- Tracking policy initiated on-demand or scheduled
- Advanced options of a data protection job
- VaultTracker options of an auxiliary copy job

A VaultTracker job that monitors exported media is reported in one of three states:

- At Source – Media is in a static location or library
- At VMS – Media is in the Virtual Media Slots awaiting removal
- In Transit – Media has been removed from a library or has been marked as 'picked up' from a location other than a library

Departure and Arrival of Media

Departure and arrival of media for a library is automatically recorded by the library's slot inventory mechanism. Therefore, media removed from a library source when the 'Auto Acknowledge' option is selected is automatically assumed to be in transit (if a transit location is specified) or at the specified export location.

Media Departing an Export Location

Media departing an export location can be automatically or manually flagged as removed from that location. It can be marked 'picked up' to put it 'in transit,' or at its new destination 'In transit' media or exported media that has not been marked 'picked up' because you want to confirm its arrival at its intended new export location, can be marked as 'reached destination.'

Individual media in a transit state between two export locations that require the return of the media to the originating source can be marked 'return to source.' Alternatively, a VaultTracker job that moved media between two export locations can be 'rolled back' to indicate return of its entire media to the source location.

Exported Media Reused as Spare Media

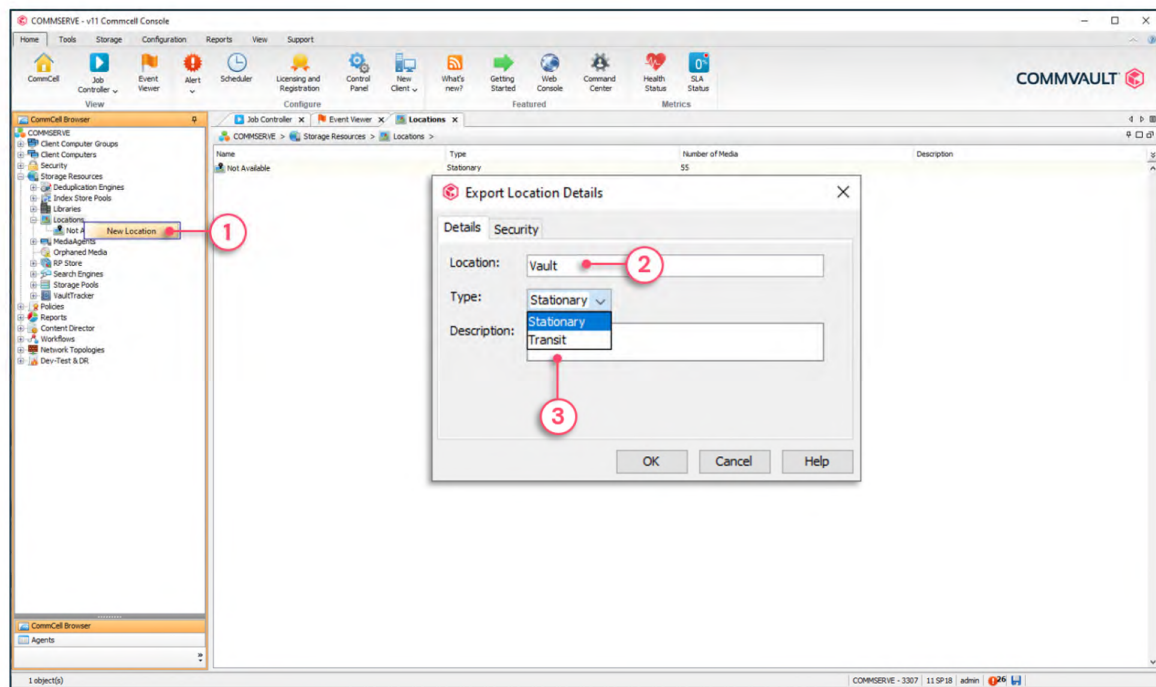
Exported media that can be reused as spare media can be tracked and scheduled with a 'due back' tracking policy. This creates a VaultTracker job consisting of media at an export location that are in a 'spare' state.

If a data protection operation requests an exported media (depending on the options selected for the library in the Media tab of the Library Properties dialog box), one of the following operations will be performed:

- Automatically mark the exported media as 'full' and use a new media for the data protection operation.
- The job will remain in the 'waiting' state, with the 'reason for job delay' stating that the media is outside the library.

However, if a restore or auxiliary copy job requires data from an exported media, the system prompts you to import the media to complete the operation. Information about exported media is retained in the system, which do not have to be rediscovered.

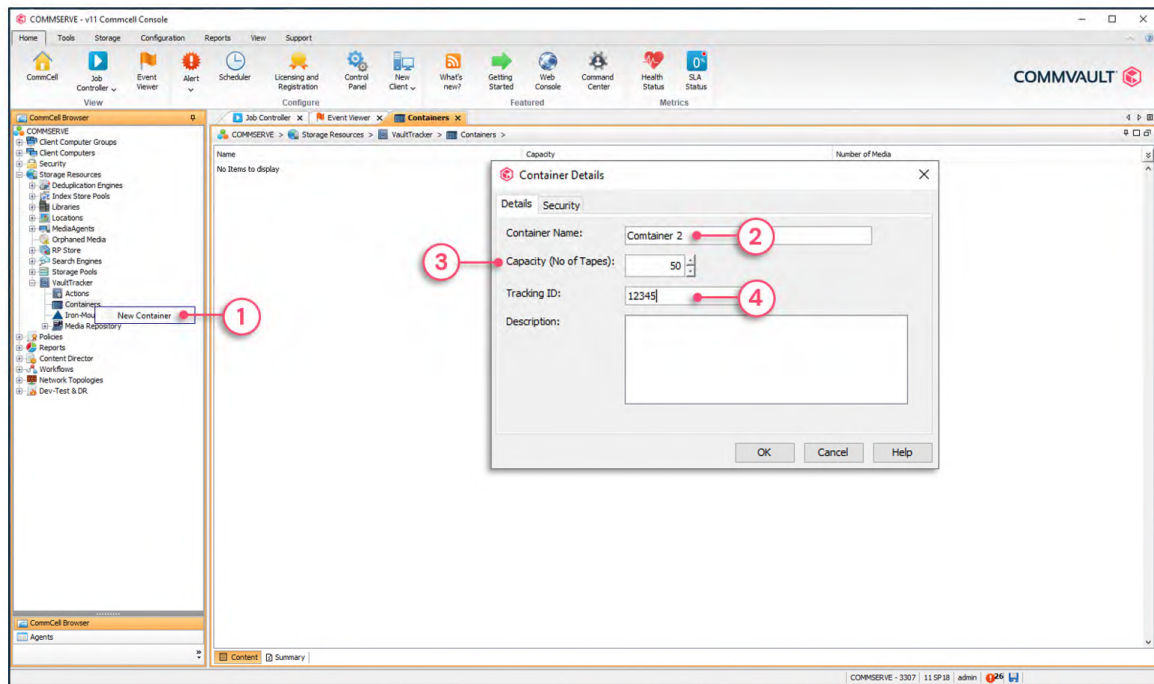
Define Locations



Locations are defined to track media throughout its lifecycle. Multiple transit and stationary locations are used to provide flexibility and accuracy in tracking tape media. Knowing where your data is and when it's time to recall the media for recycling is key for data retention and destruction policies.

1. Expand Storage Resources > right-click **Locations** > **New Location**
2. Stationary or transit locations can be defined

Containers



Containers are used to logically group exported media into a single addressable unit. This reduces the number of objects to track and facilitates movement of a set of media into, or out of, third party off-site storage. For example, ten media placed in a single container for storage is a single unit cost in both directions.

Containers can be pre-defined by the user or dynamically defined by a VaultTracker job. Each container reports the volume of data it contains.

Containers can be automatically created from a tracking policy, or manually created in the CommCell browser.

1. Expand VaultTracker > right-click **Containers** > **New Container**
2. Give a name to the container
3. Define the maximum number of media that fits in the container
4. Set the ID number of the container, if applicable

Right-click the container in the CommCell browser to view or edit properties, and to move, remove, delete, or recall all media associated with that container by right-clicking on the container in the CommCell browser and selecting the option.

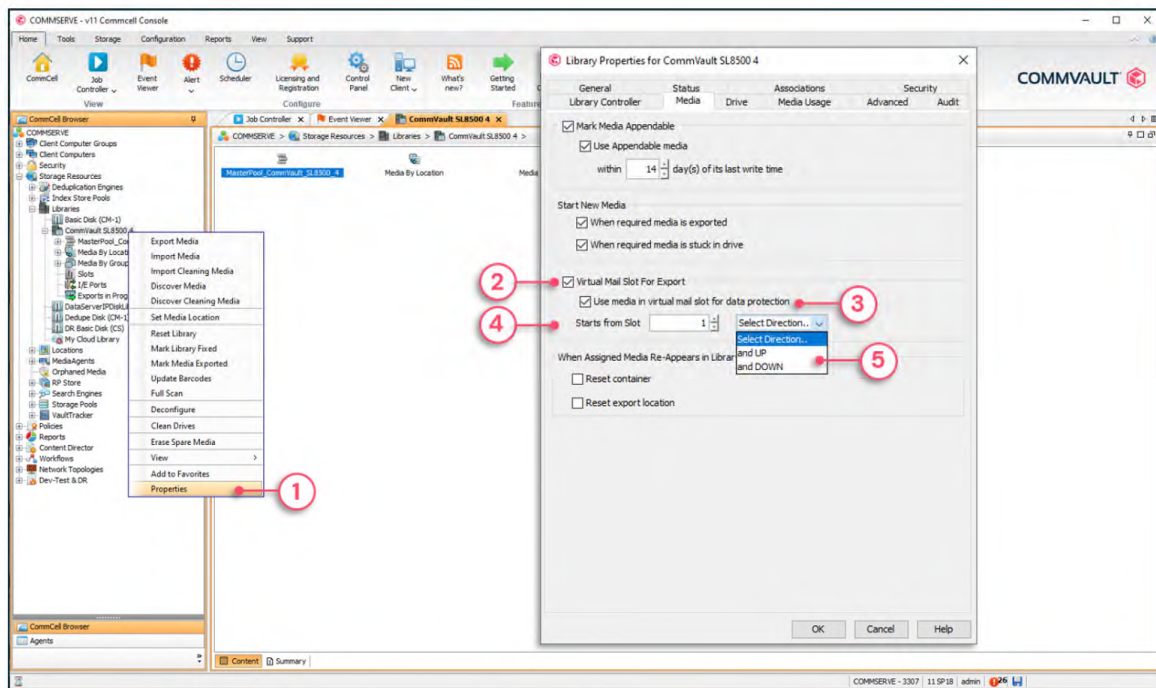
Media not Associated with a Container

For media that are not automatically associated with a container, there are several ways of associating the container information. When you record the 'Pending Action as Reached Destination,' you will be provided with the option to set a container if necessary. You can set the container for any 'Pending Action' that are displayed in the CommCell browser, when you click the 'Actions' icon under the VaultTracker entity.

Containers can also be associated with media displayed in the 'Exported Media' pool.

- Individually associate a container for a specific media by opening 'Media Properties' and selecting a container as the location for the media.
- Set the container for all media in the 'Exported Media pool,' by right clicking the 'Exported Media' icon displayed under 'Media By Location' and then clicking the 'Set Container' option.

Virtual Mail Slots



Exporting multiple media from a library with a single mail slot can be time consuming. Each media moved to the mail slot must be manually acknowledged and removed individually.

Opening the library to remove multiple media manually may be quicker, but it requires a visual search of all the slots to find the appropriate media. Virtual mail slots alleviate the need to visually search the library for each media by designating a static block of slots within the library to hold the media awaiting export.

1. Right-click the **tape library > Properties**
2. Check the box to use virtual mail slots for export
3. Check the box to specify if a media in a virtual mail slot can be used for data protection job when space is available
4. Define the starting port to use as virtual mail slot
5. Define the order to use for additional virtual mail slots

VaultTracker Policies

Tracking policies are used to track regular media movements, such as when media is shipped from a library to off-site storage, media repository, or shipment of media between two locations.

There are two types of VaultTracker policies:

- Export Media Policies
- Due Back Policies

Export Media Policies

An Export Media policy is created to track the movement of media between any source and any destination as listed below. If the tracking policy is used to export media from a tape library, it also moves the media in the I/E ports or virtual mail slots.

- Library to location
- Location to location
- Location to library
- Library to library

Auto Acknowledge

When creating a tracking policy, you can enable the Auto Acknowledge option within the policy. When the policy is run, the system automatically acknowledges all pending actions associated with the media movement if both source and destination is a location. If the source is a library, the pending actions is automatically acknowledged when the media is removed from the library.

Delay Export of Media

It is also possible to delay the export of the media, which creates a time gap between when the policy is run and when the media movement actions start. This provides time for you to create and review a Vault Tracking Report with all the policy's actions detailed prior to any actions taking place. If any changes to the policy are required, you can perform these changes without aborting or rerunning an action.

Set Export into 'Pending' State

Temporarily stopping an action by placing the export operation in a pending state is useful if you need to perform library maintenance or urgent data protection and recovery operations, but do not wish to abort or restart what has already been completed. You can manually restart the policy from the point it left off when the library is available, or you can set the policy to resume automatically at a specific time. When you use the 'Auto Acknowledge' option to export tape media from a library, use the 'Delay the export by *nn* minutes' option to delay moving the media to the mail slots. If the tapes are physically removed from the library after the Export Media tracking policy completes, but before the tracking report is generated and sent to the tape operator, the Auto acknowledge feature marks tapes as 'in vault' resulting in an empty report. For instance, the export can be delayed by ten minutes to ensure that the report is generated before tapes are moved to the mail slots.

Example: The Export Media Policy is scheduled to start at 9:00 AM with a delay of 10 minutes to export media. The tracking report is scheduled to start at 9:05 AM. This setup results in the following:

- 9:00 AM – Export Media tracking policy starts, which creates a list of all tapes required to be sent to vault, but since the export is delayed, tapes are not moved in I/E ports yet.
- 9:05 AM – The Tracking report is generated listing all media required to go to vault and is sent to tape operators by email.
- 9:10 AM – The delay for exporting media is reached and media are physically moved in the I/E ports.

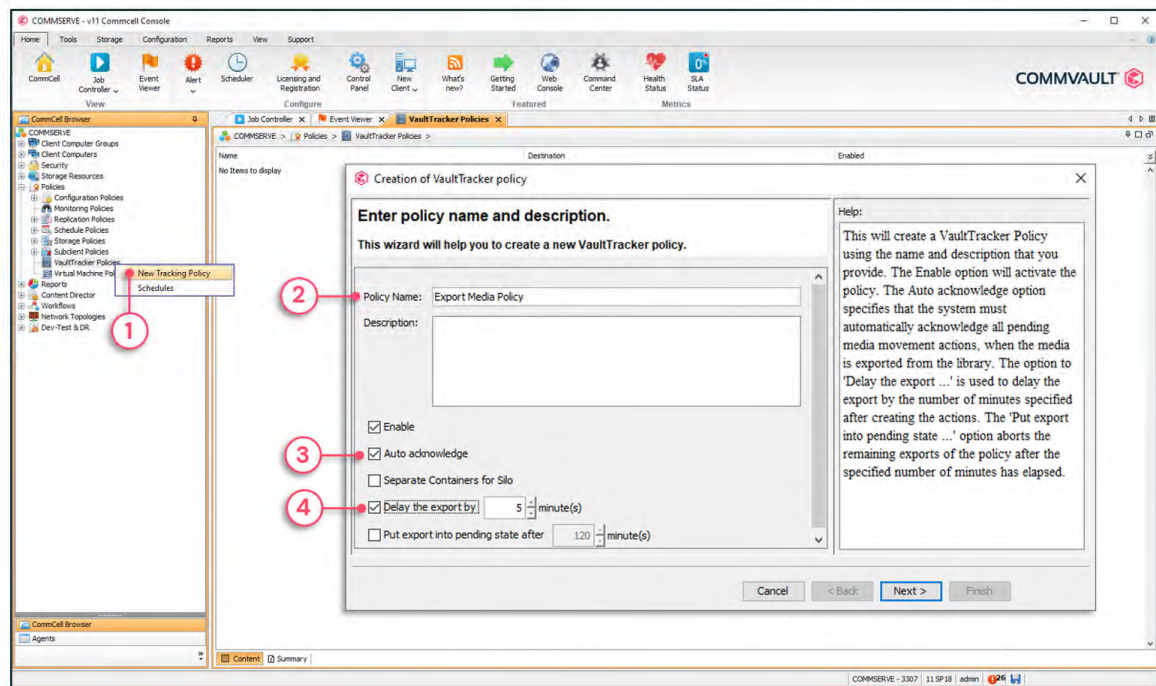
Additional screens include several other options, such as storage policy copies. Storage policy copies must be selected to be included and managed by the policy. This can be useful for scheduling purposes. A good example could be to create a tracking policy to export tapes containing production data daily, and another tracking policy to export tapes containing development data once a week.

Media Status to Export

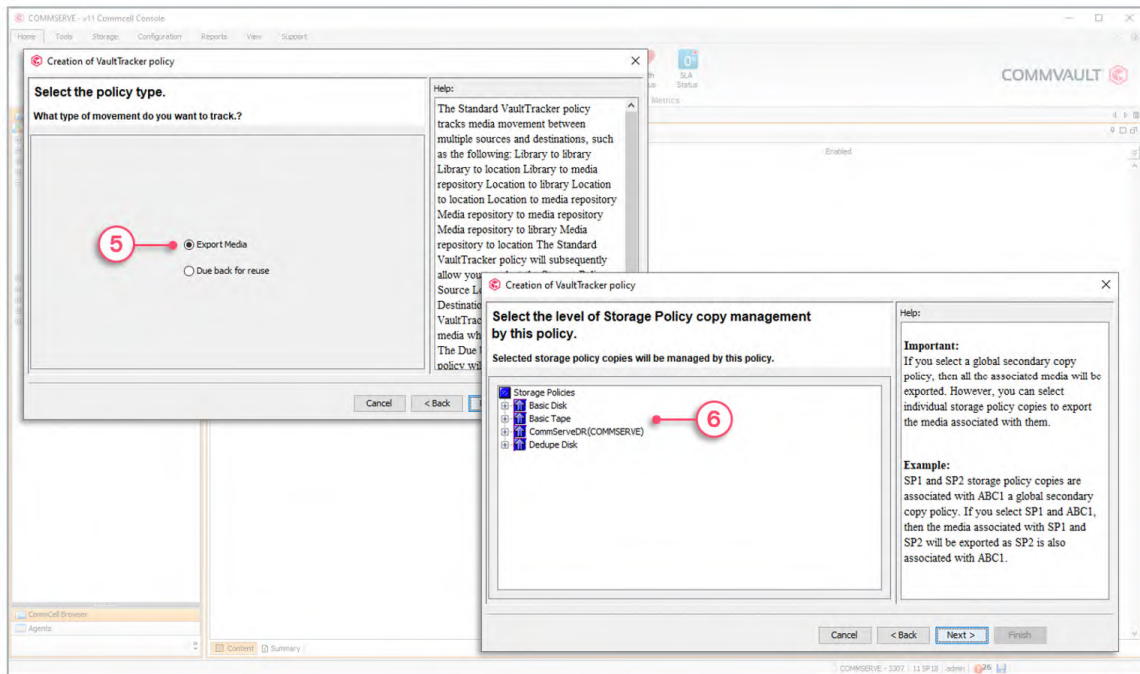
By default, only full media are exported. If tape media is your primary recovery method, a disaster could potentially lead to data loss if the library is destroyed with tapes partially filled still located in it. In this case, Media Status to Export can also be configured to include Active media, which means that partially filled media are also exported when the job runs.

By default, when exporting media, a limit of 10 media is defined for the export process. If you want to export all media meeting your criteria and avoid any data loss, it is recommended to uncheck the limit box.

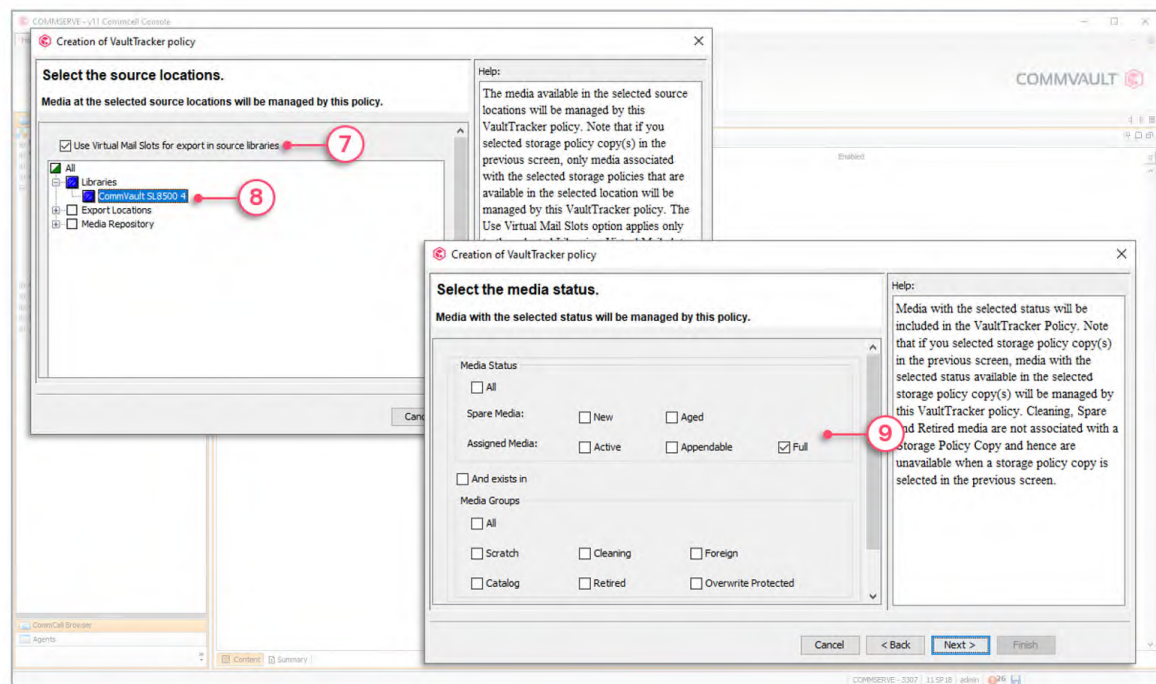
Create an Export Media VaultTracker® Policy



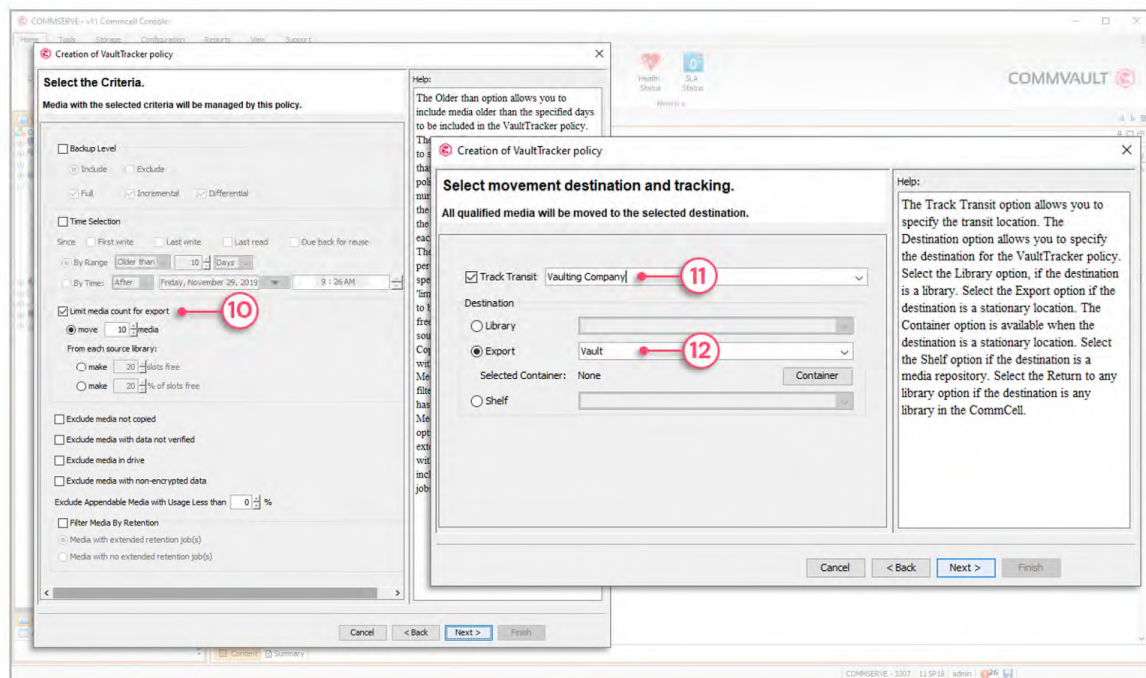
1. Expand Policies > right-click **VaultTracker Policies** > **New Tracking Policy**
2. Enter a name for the policy
3. Check the box for **Auto Acknowledge** to auto acknowledge pending action and avoid human intervention
4. Check the box and set a number in minutes to delay the physical export of media



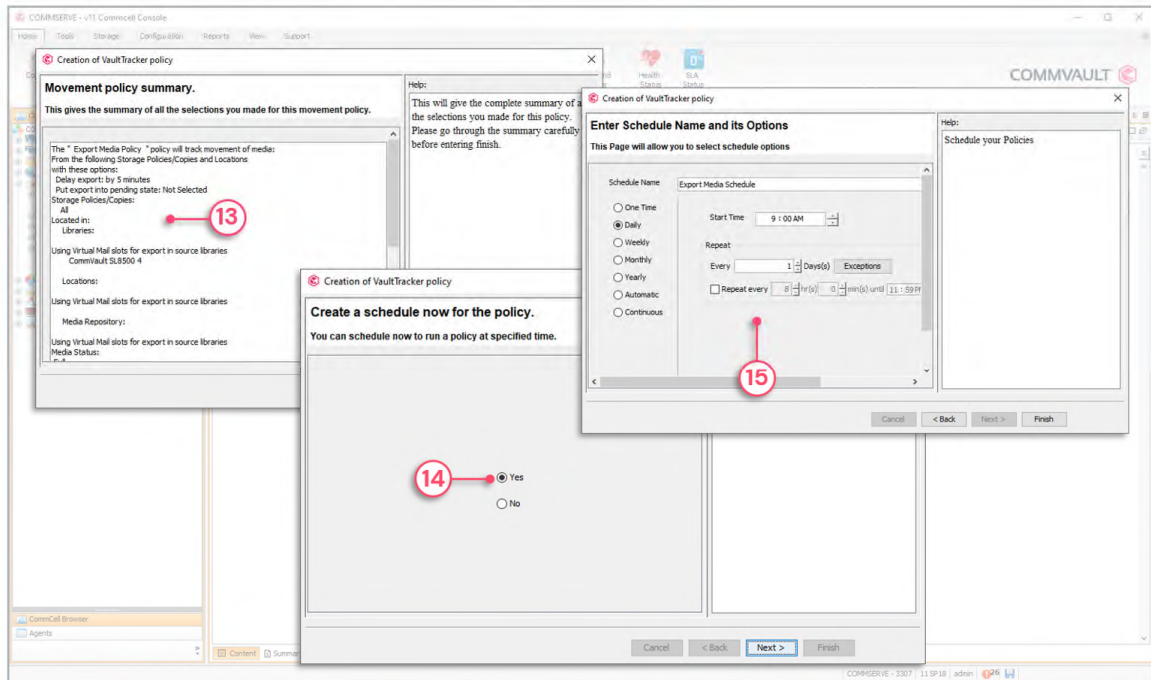
5. Select option to define an Export Media policy
6. Select the storage policy copies to be managed by the policy



7. Check the box to use tape library virtual mail slots for export
8. Select the source location for the export
9. Select the media to exported

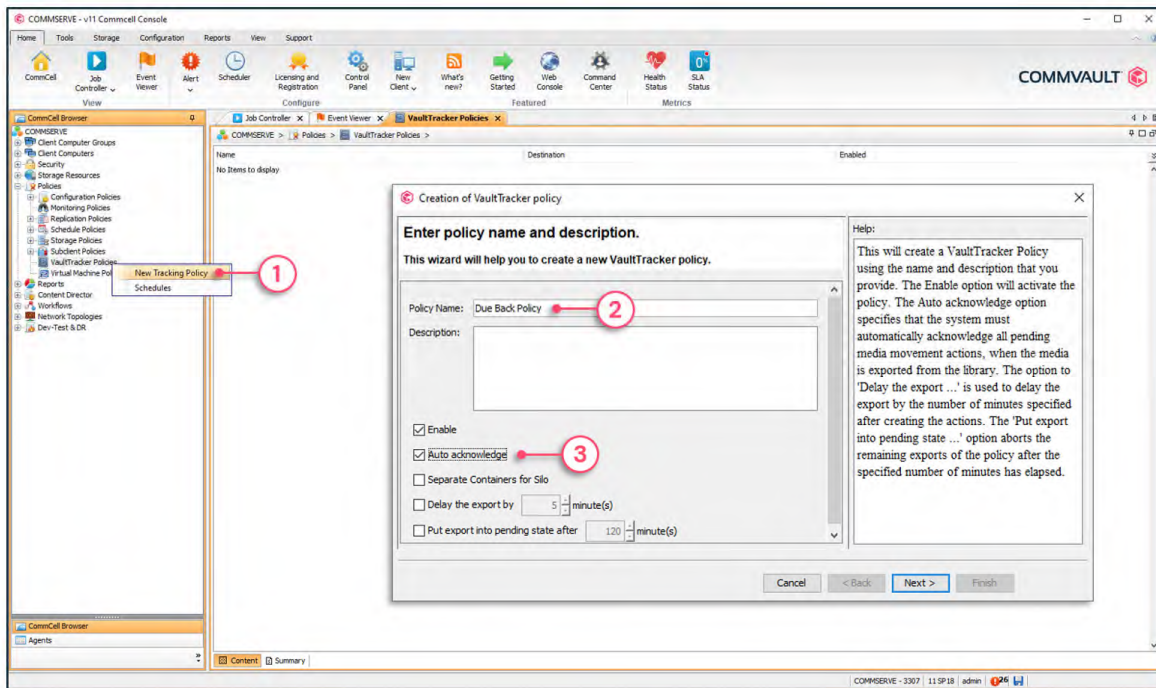


10. Uncheck the box to export all media meeting criteria
11. If using a transit location, check the box and select the transit
12. Select the destination



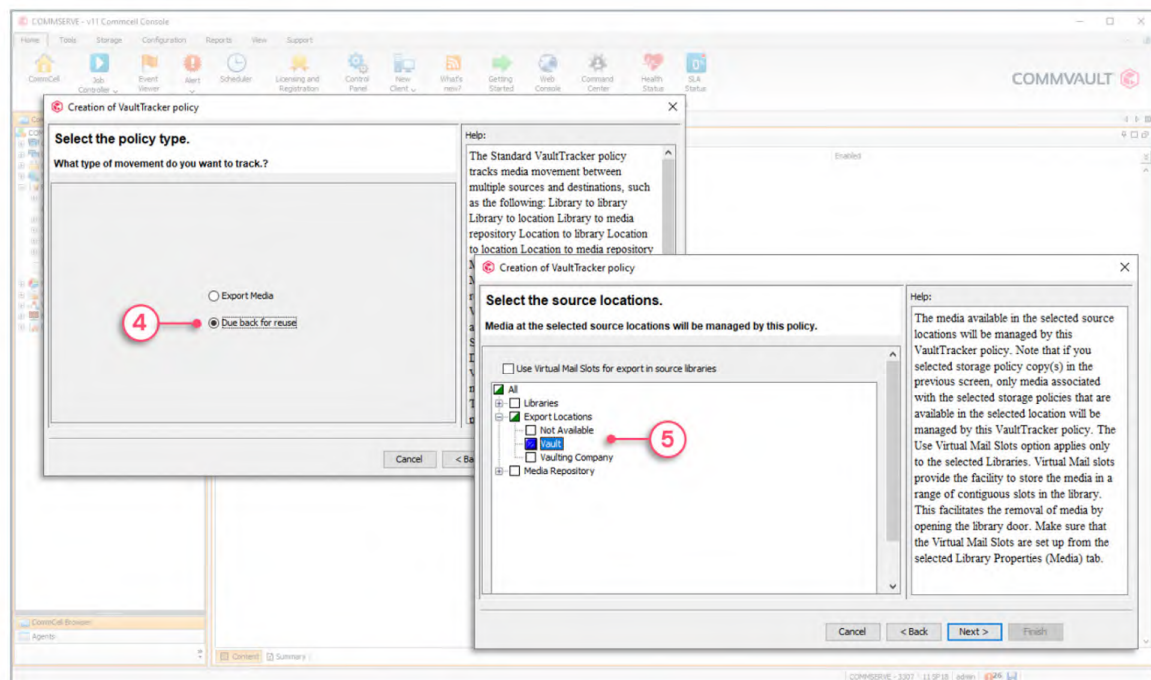
13. Review the summary of selected options and click **Next**
14. Select **Yes** to schedule the VaultTracker policy
15. Enter a name and define the scheduling options

Due Back Policies

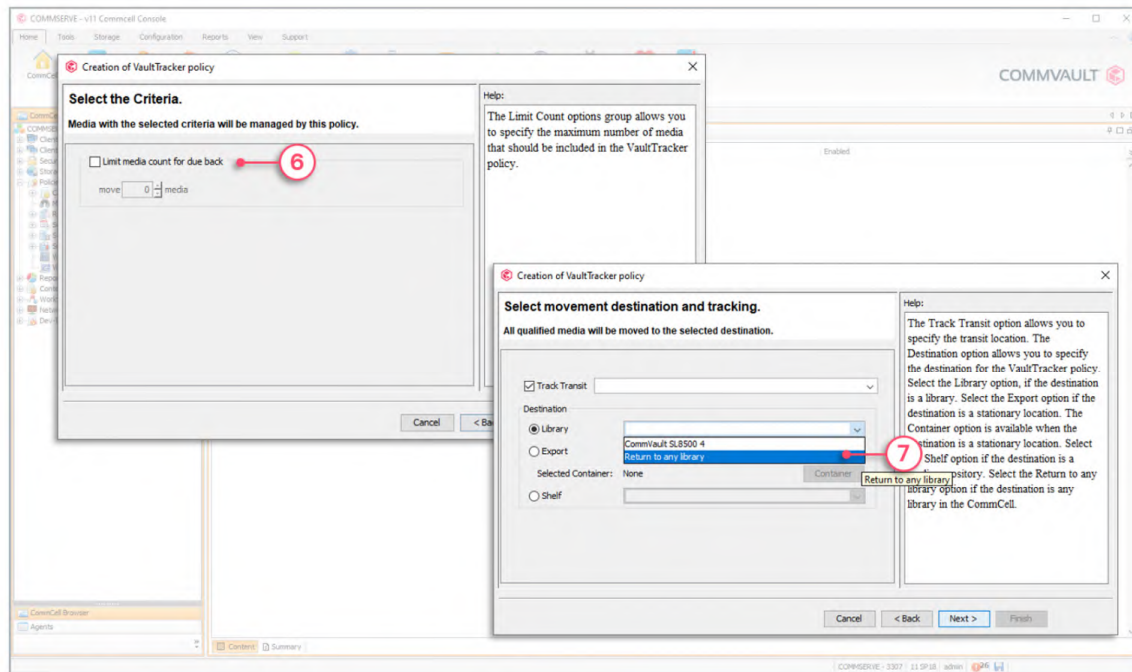


A due back policy tracks the movement of spare media stored in an off-site location. When all jobs stored on an off-site media expire, the media becomes a spare media that can be recalled and reused. The due back policy creates the list of anticipated media to recall. If the 'auto acknowledge' option is selected, as soon as the media is inserted in the I/E ports, it is imported in the library and the status is automatically changed from 'In Vault' to 'In Library.' When creating a policy, you can schedule the policy to run later if you anticipate changes being made to the policy prior to media movement taking place.

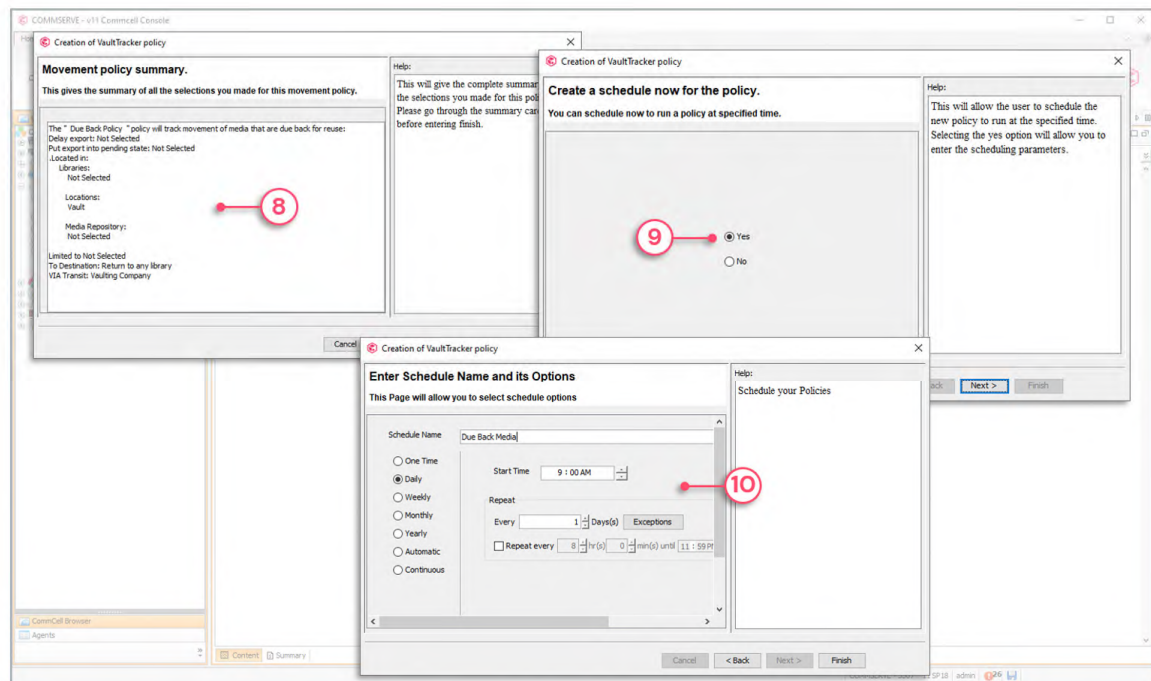
1. Expand Policies > right-click **VaultTracker Policies** > **New Tracking Policy**
2. Name the policy
3. Check the box for **Auto acknowledge** to auto acknowledge pending actions



4. Select **Due back for reuse**
5. Select the source location from which media are recalled

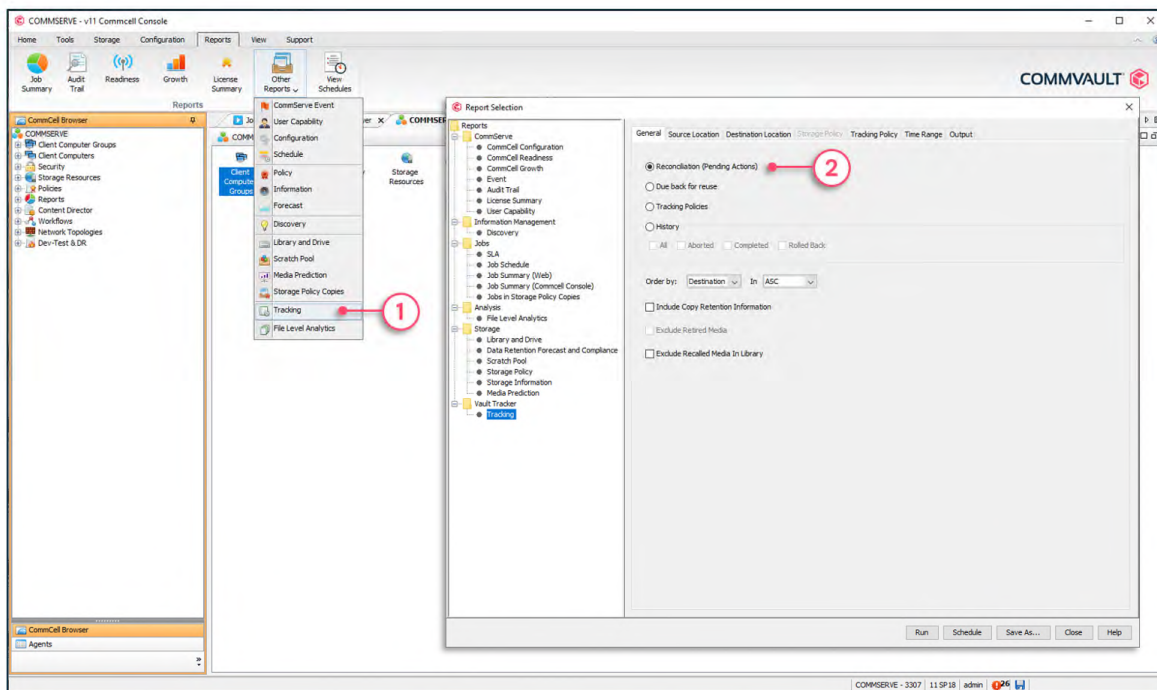


6. Check the box to set a limit for the number of tapes to recall, or uncheck the box recall all spare media
7. Define the destination where recalled tapes will be stored



8. Review selected options and click **Next**
9. Select **Yes** to schedule the VaultTracker policy
10. Enter a name and define the schedule options

VaultTracker® Reports

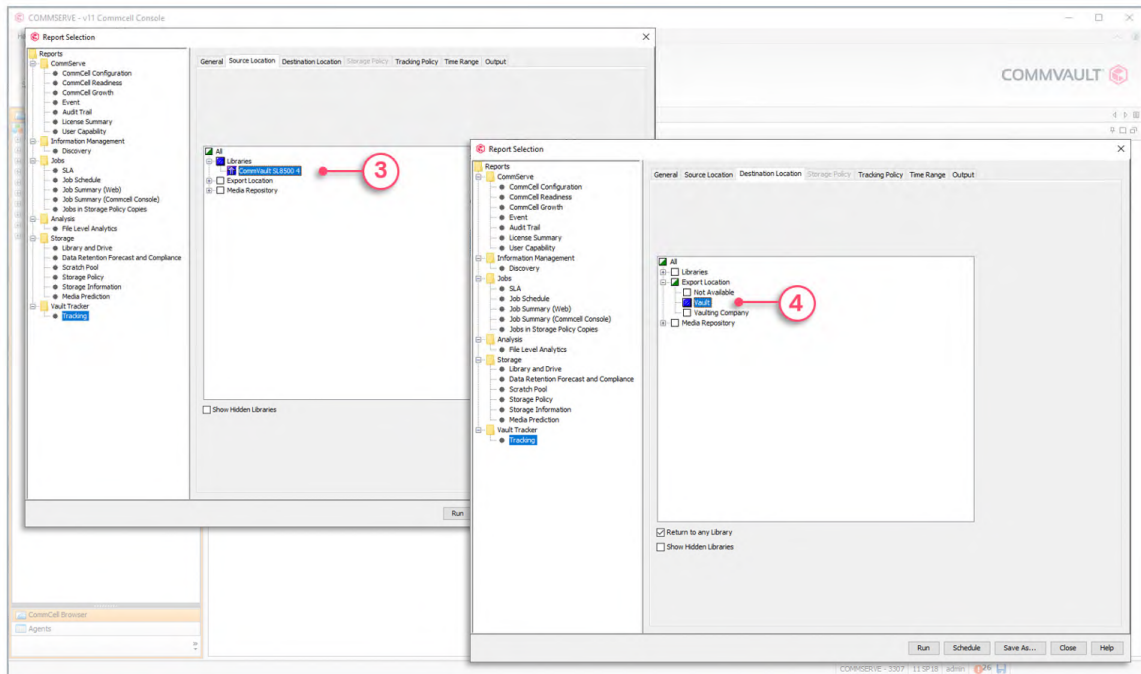


VaultTracker reports can be sent to tape operators or backup administrators to provide a list of tapes that are required to be exported or recalled. The list is automatically populated by the software based on the tracking policies media movement and media status.

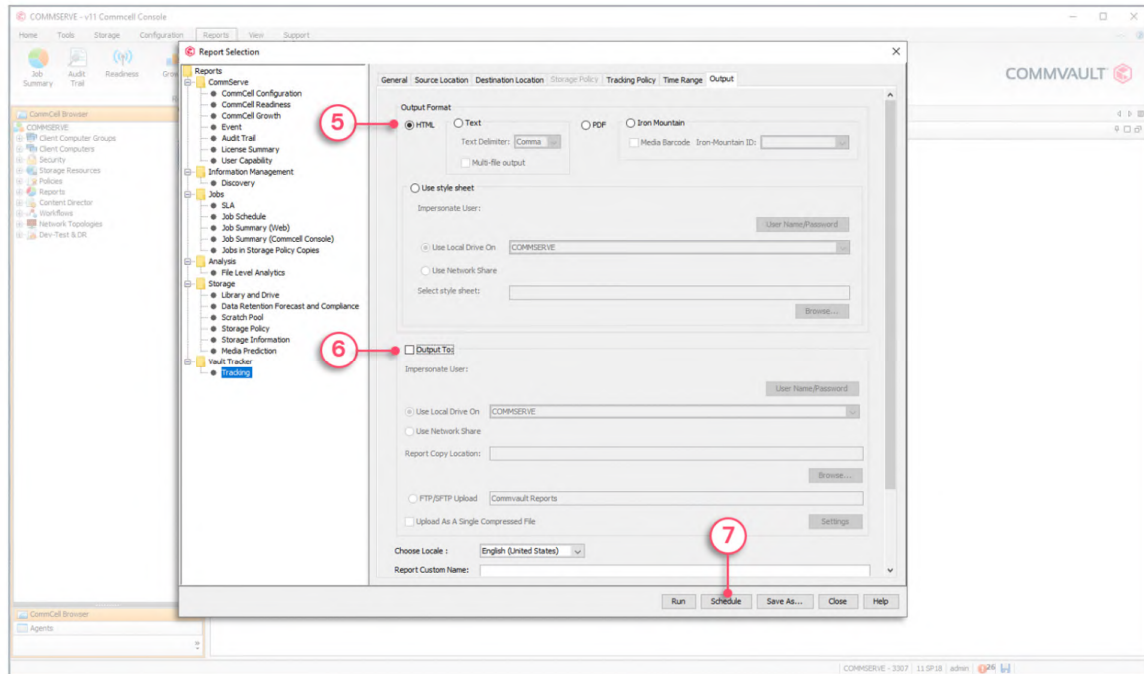
Export Media Report

The Export Media report is used to validate that all media required for vaulting has been moved to the I/E ports. It can also be used to prepare the paperwork for the vaulting company if required. Reports can be formatted specifically for Iron Mountain sent to the vaulting company's FTP site.

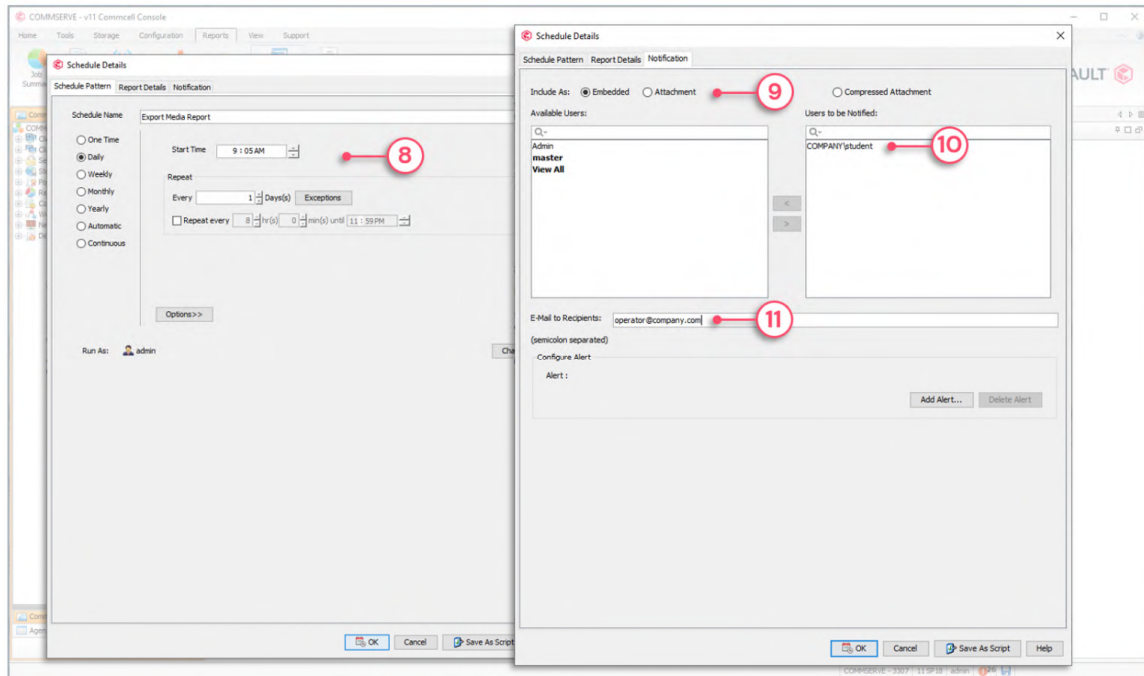
1. **Reports** tab > **Other Reports** > **Tracking**
2. Select **Reconciliation** to generate an export media report



3. From the *Source Location* tab, select the source location to include in the report
4. From the *Destination Location* tab, select the destination where media are sent

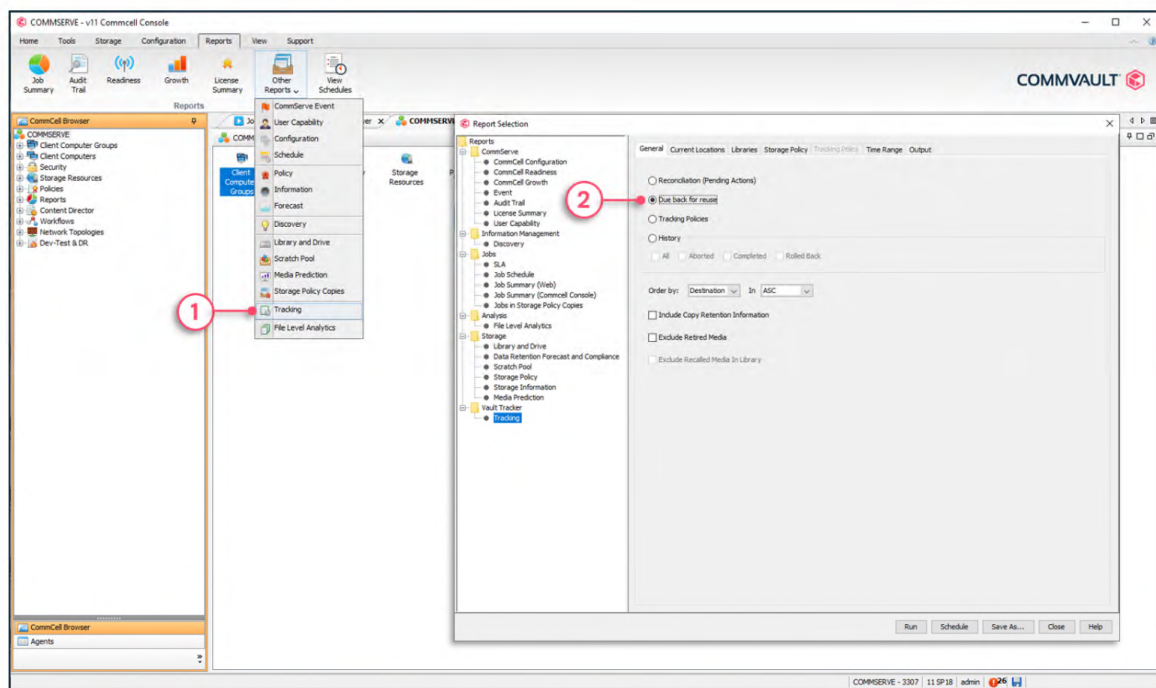


5. From the *Output* tab, select the report format
6. A copy of the report can be saved on local disks, network share, or sent to an FTP site
7. Click **Schedule** to schedule the report to be automatically generated



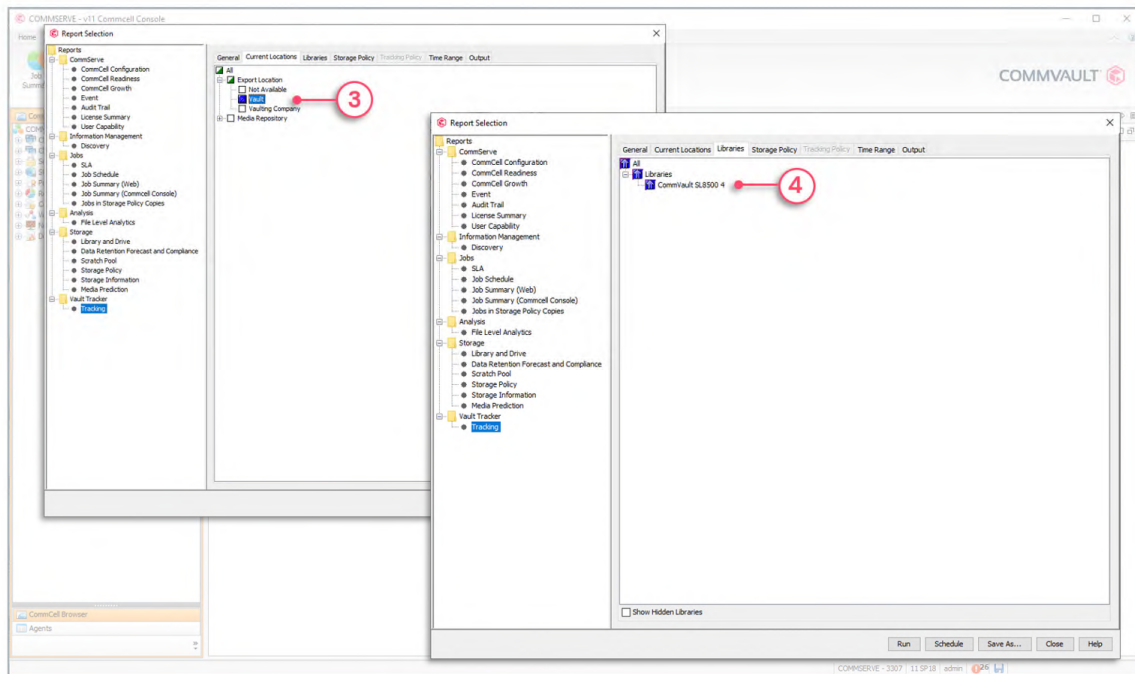
8. Name the report schedule and define the schedule pattern
9. Choose to embed or attach the report to the email
10. Select users to whom the report should be sent
11. Enter additional email addresses or distribution lists

Due Back Media Report

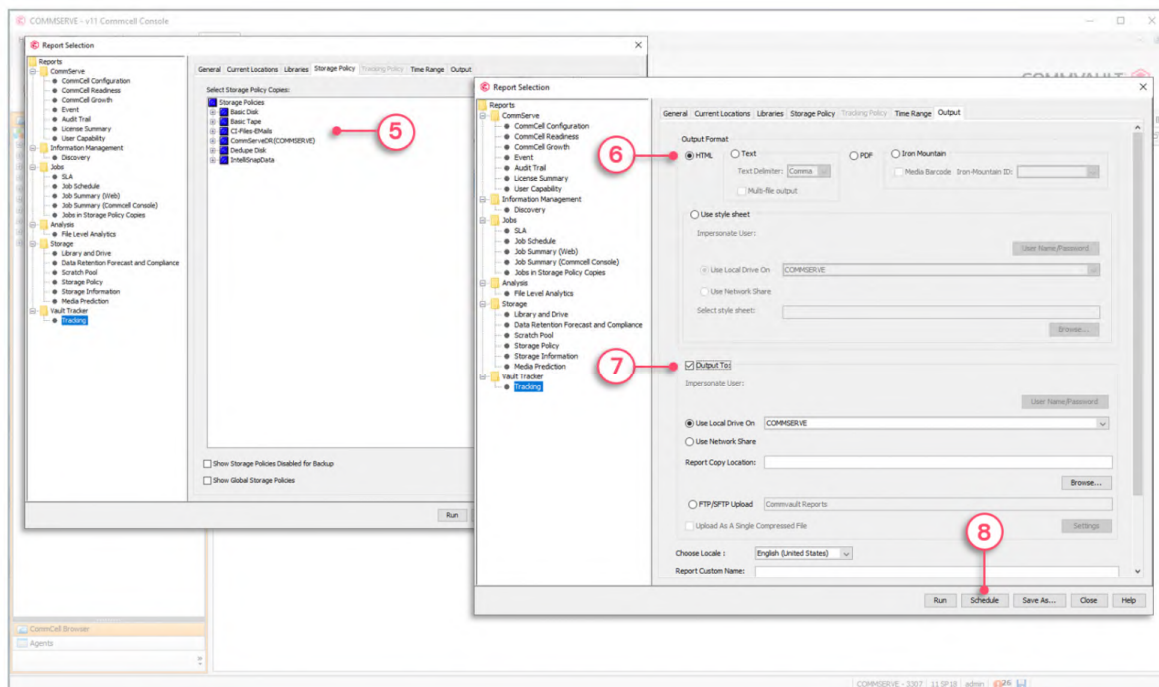


The Due Back Media report lists all media stationed in vault that have all jobs expired and can be recalled and used as spare media. The Due Back Media reports can be formatted specifically for Iron Mountain and sent to the vaulting company's FTP site. Running a Due Back Media report requires a license for VaultTracker Enterprise. If you are licensed for a different version, skip to the next section for instructions.

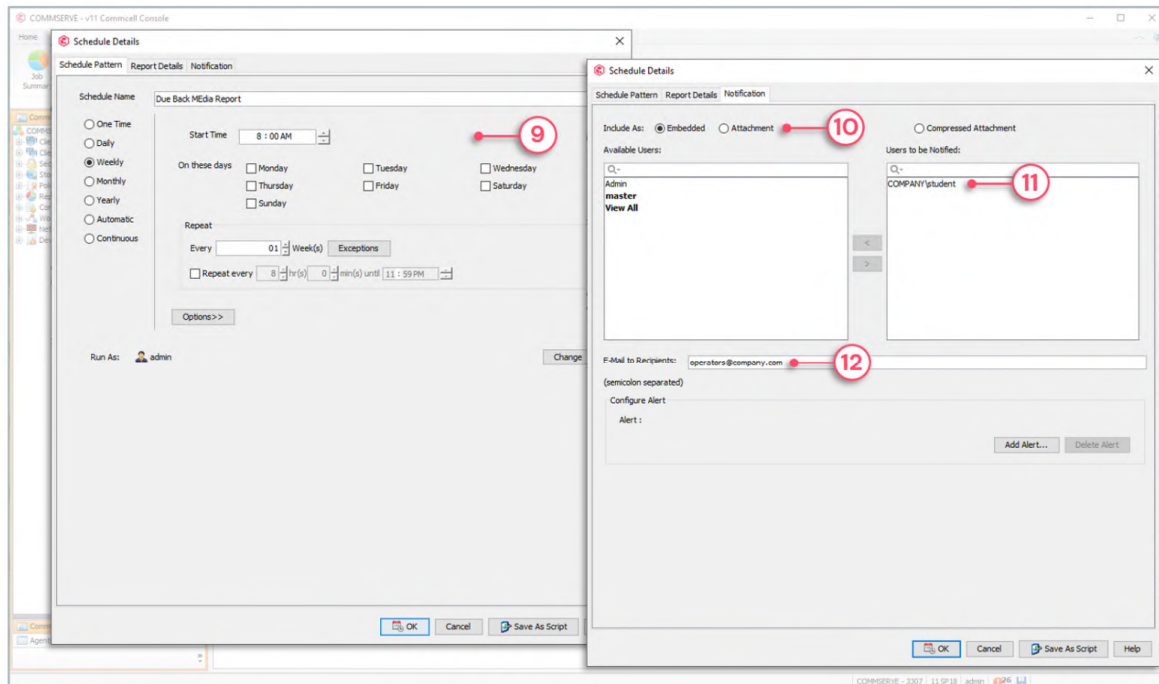
1. **Reports** tab > **Other Reports** > **Tracking**
2. Select **Due back for reuse** to generate a report listing all tapes that expired and that can be recalled



3. Select the current location from which media are recalled
4. Select the destination location that will receive the media

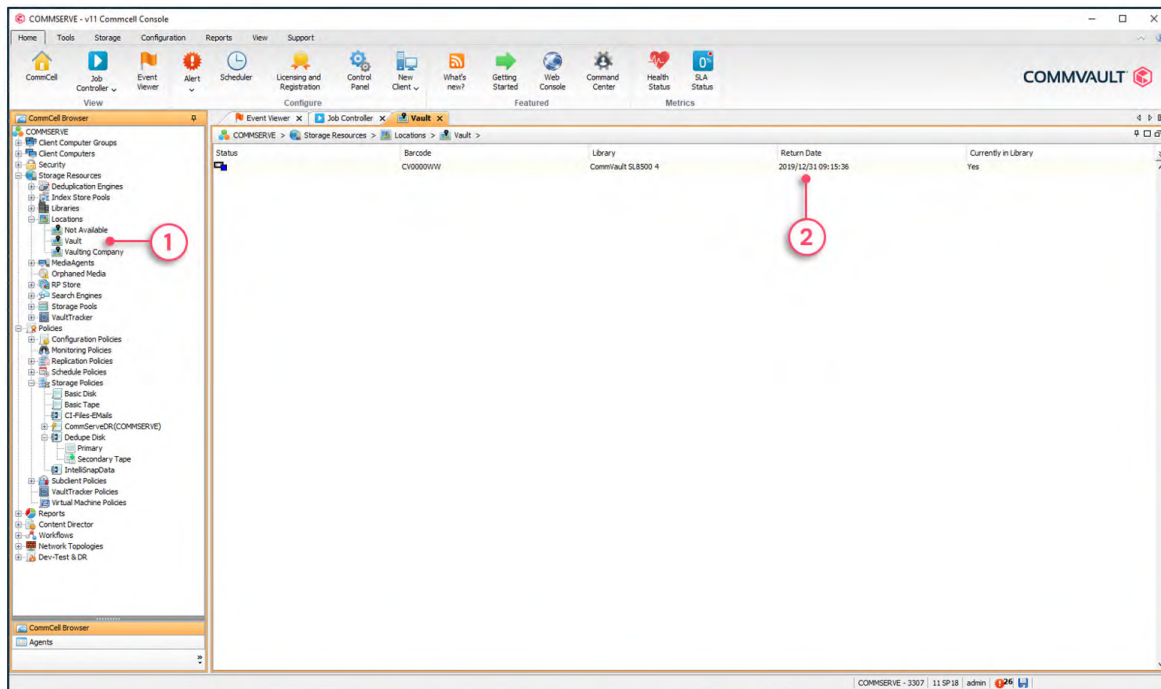


5. Select the storage policy copies to include in the report
6. Select the report format
7. A copy of the report can be saved on local disks, network share or sent to an FTP site
8. Click **Schedule** to schedule the report



9. Name the report schedule and define schedule pattern
10. Choose to embed or attach the report to the email
11. Select users to whom the report should be sent
12. Enter additional email addresses or distribution lists

Manual Recall



Running a Due Back Media report requires a license for VaultTracker Enterprise. If you are licensed for another VaultTracker edition, click the location in the CommCell browser and sort the media by 'return date'. All media having a return date listed as 'now' can be recalled.

1. Expand Location > click the vaulting location
2. All media information is listed including the return date; any tape listing 'Now' as a return date can be recalled